

Principles of bionics and biorobotics engineering

Bioelectronics and neuroengineering in
neurology and neurorehabilitation – Part 1

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Outline

- 1. Background on nervous system injuries and disorders**
- 2. Technologies for neurorehabilitation**
- 3. Technologies for functional assessment**

1. Background on nervous system injuries and disorders

1.1 Overview of the nervous system

- Central and peripheral nervous system
- Neuronal pathways: sensory and motor circuits

1.2 Principles of motor control

- Motor control of human movement
- Motor control of speech

1.3 Injuries and disorders of the nervous system

- Types and causes of injuries
- Neurodegenerative diseases
- Overview of neurological deficits related to motor and speech control

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Overview of the nervous system

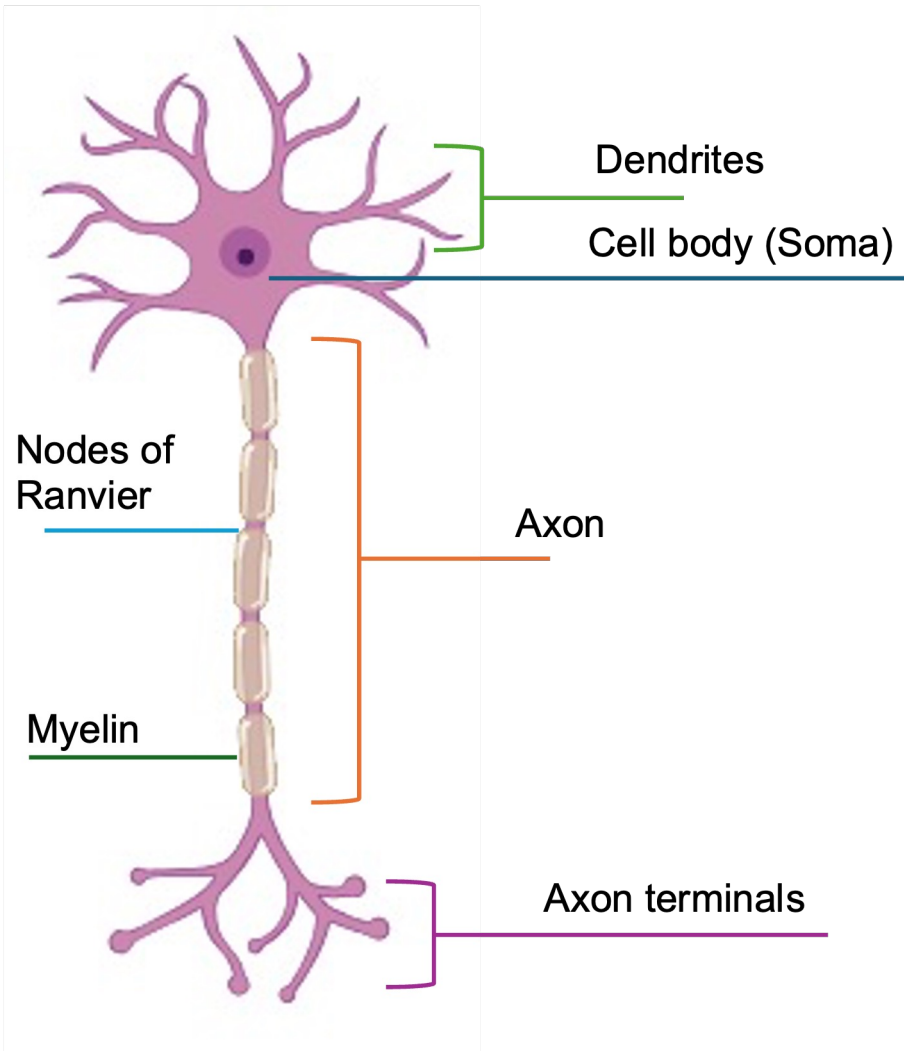
The nervous system is the body's control center, responsible for receiving, processing, and responding to information.

Key functions: sensation, integration, motor output, and regulation of homeostasis.

Major divisions of the nervous system

- **Central Nervous System (CNS):** Comprised of the brain and spinal cord, responsible for processing and interpreting sensory information and coordinating responses.
- **Peripheral Nervous System (PNS):** Includes all neural structures outside the CNS; facilitates communication between the CNS and the rest of the body.

Neuronal Cell Anatomy



Neurons are the functional units of the nervous system. They are excitable cells specialized in receiving, conducting, and transmitting signals

- **Cell Body (Soma):** Contains the nucleus and essential organelles; responsible for synthesizing proteins and processing signals.
- **Dendrites:** Branch-like structures that represent the receiving end of the neuron, gathering inputs from other neurons. The sum of inputs received by all the dendrites of a neuron determines whether or not the neuron will activate (i.e., generate an action potential).
- **Axon:** A long, slender projection that carries electrical impulses away from the cell body to other neurons, muscles, or glands.
 - **Axon Terminals:** The endings of the axon where neurotransmitters are released to communicate with neighboring cells.
 - **Myelin Sheath:** Insulating layer around the axon (formed by glial cells) that speeds up signal transmission.
 - **Nodes of Ranvier:** Gaps in the myelin sheath that allow for rapid signal propagation via saltatory conduction.

Neuronal Cell Functioning

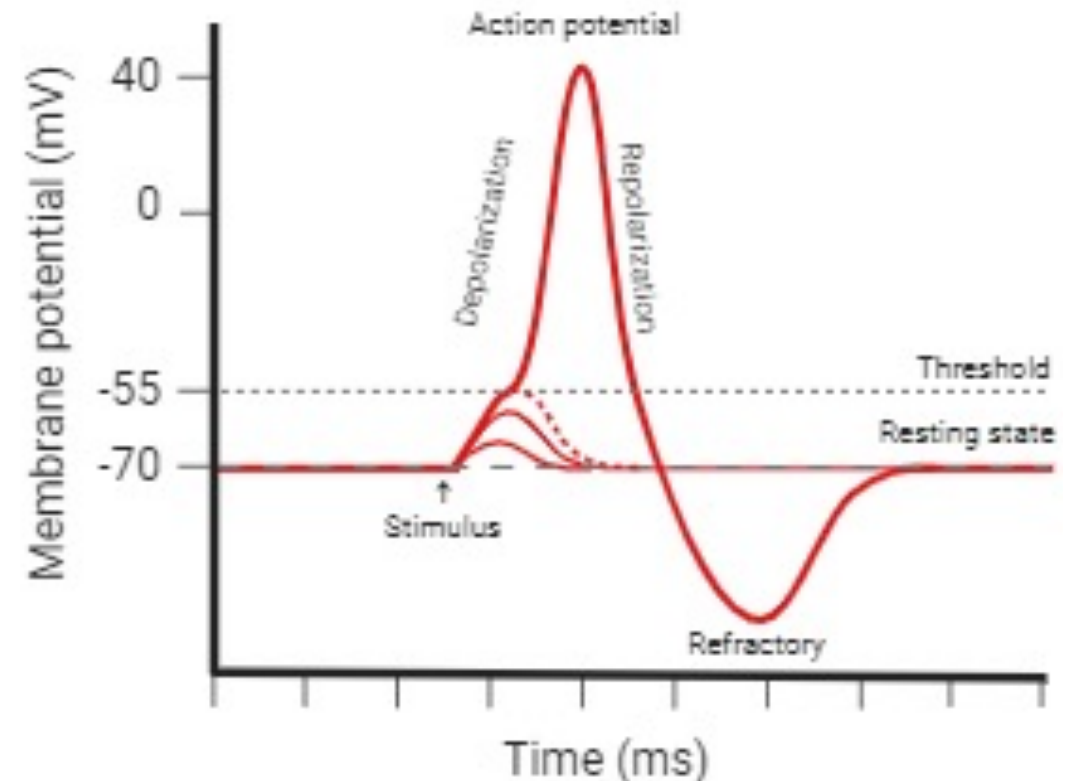
Resting membrane potential: Neurons maintain a resting membrane potential, with the inside of the cell negatively charged relative to the outside due to ion distribution.

Key ions: sodium (Na^+), potassium (K^+), and chloride (Cl^-), managed by ion pumps and channels.

Action Potential:

- **Depolarization:** When a neuron receives enough stimulation, Na^+ channels open, allowing sodium ions to rush in, making the inside of the cell more positive.
- **Repolarization:** K^+ channels open, allowing potassium ions to exit, restoring the negative charge inside.
- **Propagation:** The action potential travels down the axon to the axon terminals.

Action Potential Diagram



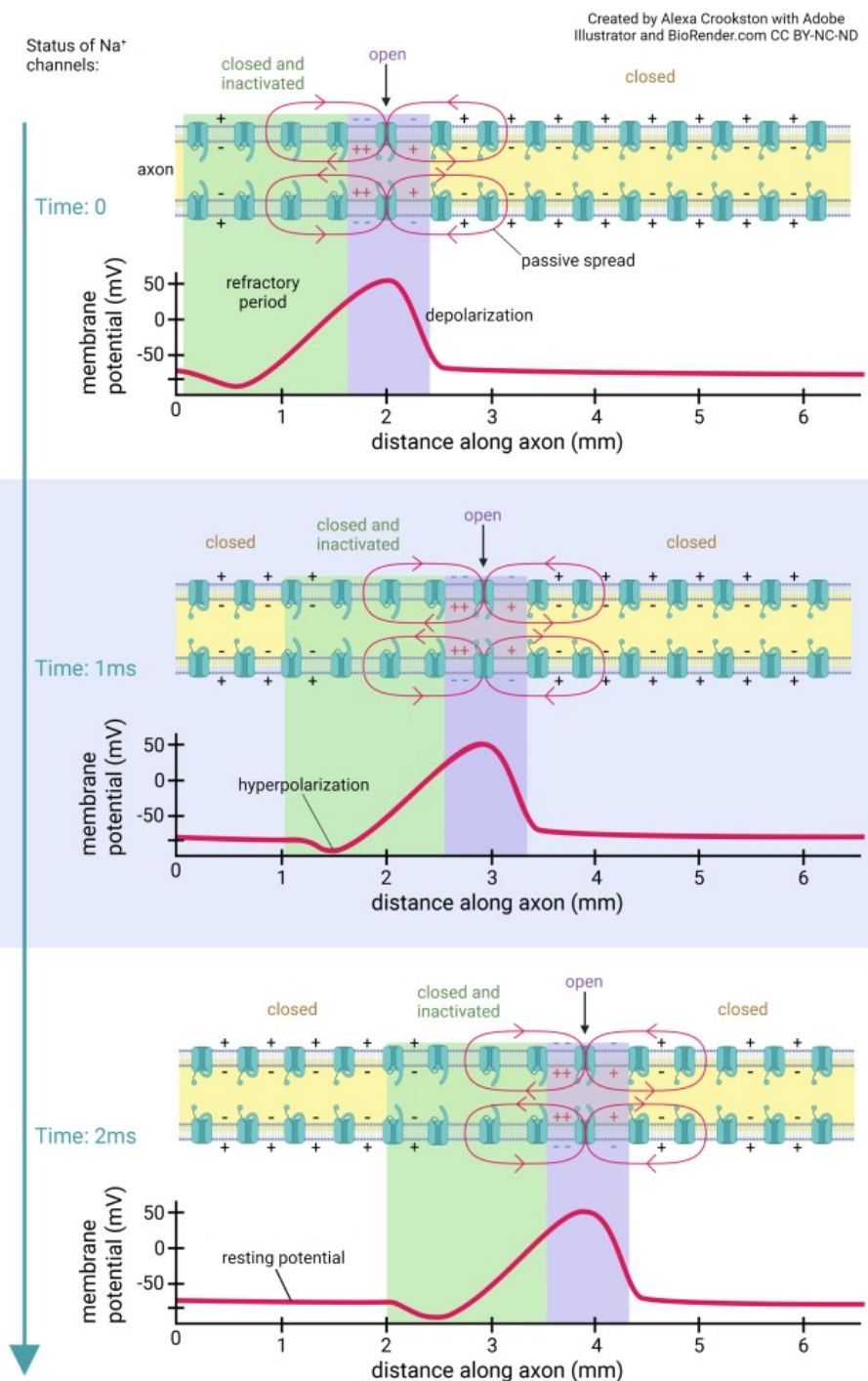
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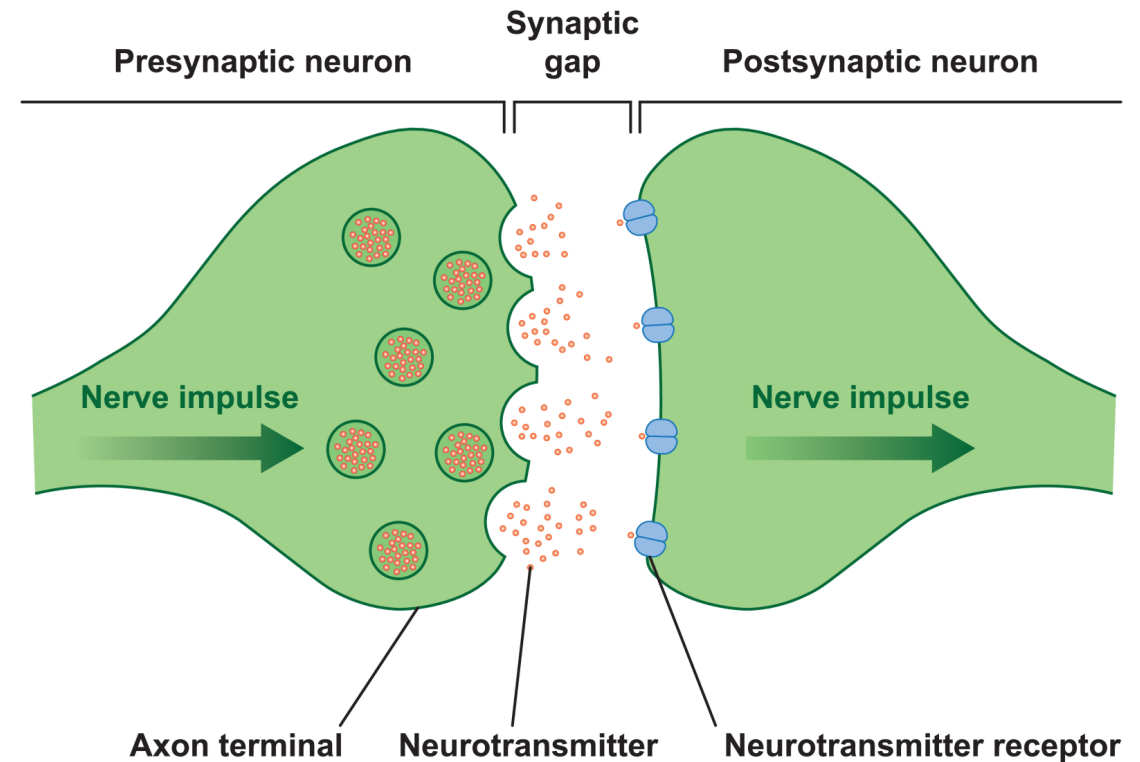
Neuronal Cell Functioning

When the action potential reaches the axon terminals, it triggers the release of neurotransmitters.

Synaptic transmission:

- **Neurotransmitter release:** Neurotransmitters are released into the synaptic cleft, the gap between neurons.
- **Binding to receptors:** Neurotransmitters bind to receptors on the postsynaptic neuron, triggering ion channels to open and starting a new electrical signal.

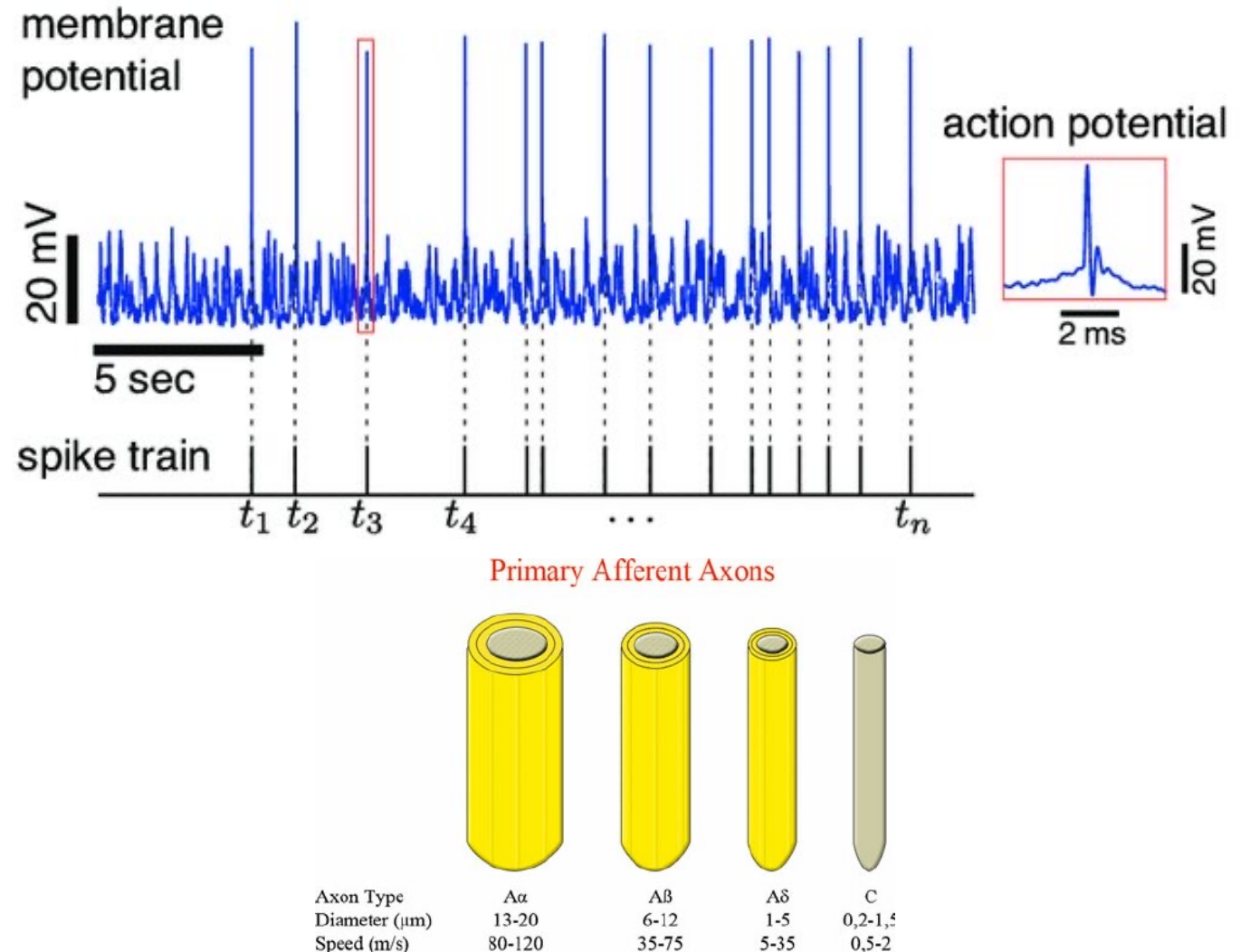
Neurotransmitters can excite or inhibit the postsynaptic neuron, influencing whether it generates an action potential.



Neuronal Cell Functioning

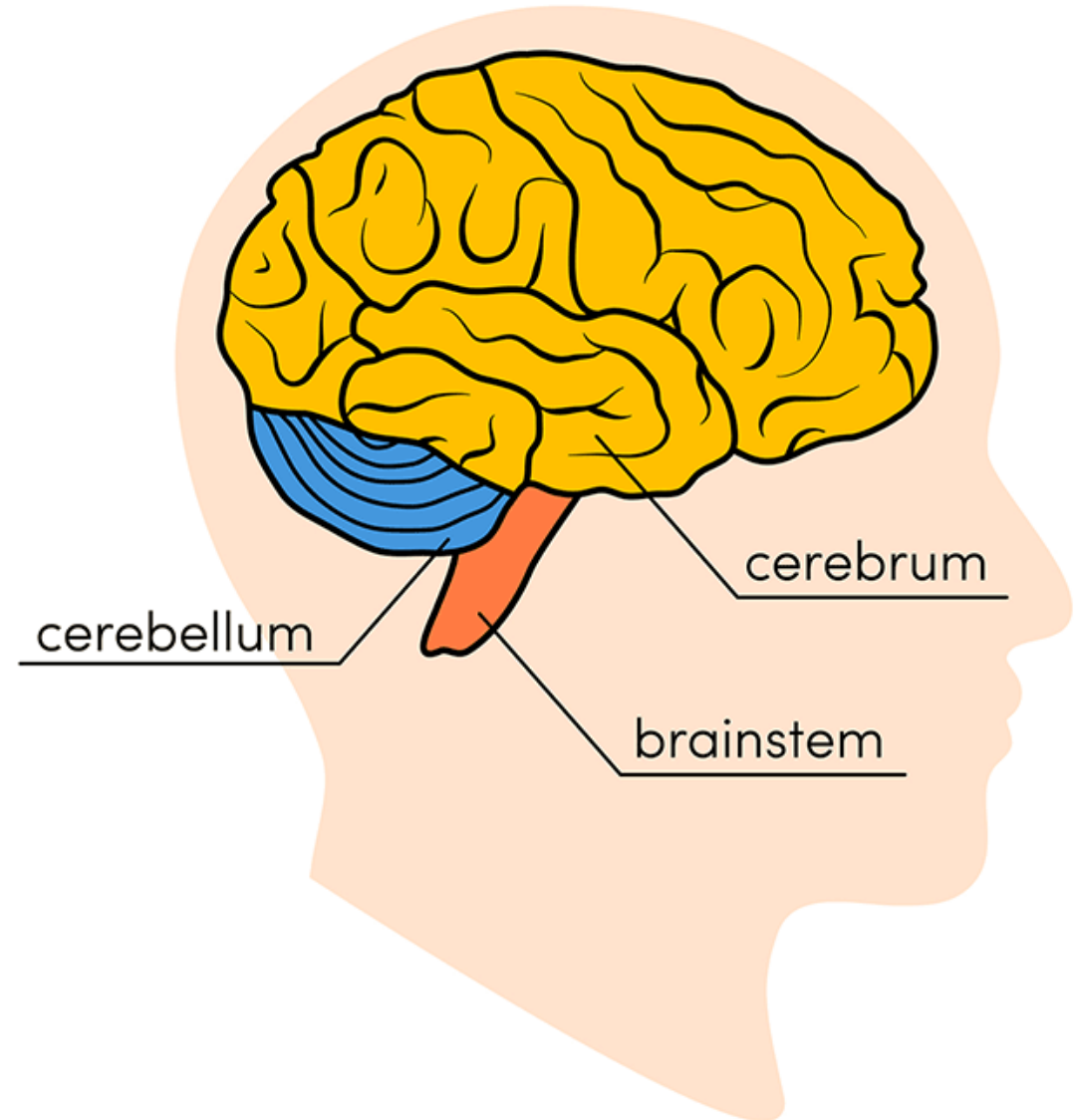
Impulse frequency: Neurons typically transmit trains of impulses at a frequency generally around 200-300 Hz. Higher frequencies can also be observed, but the maximum frequency is limited by the refractory period.

Conduction speed: Depends on the presence of myelin and the diameter of the axon. It is generally around 50 m/s, but large sensory fibers can reach speeds up to 120 m/s.



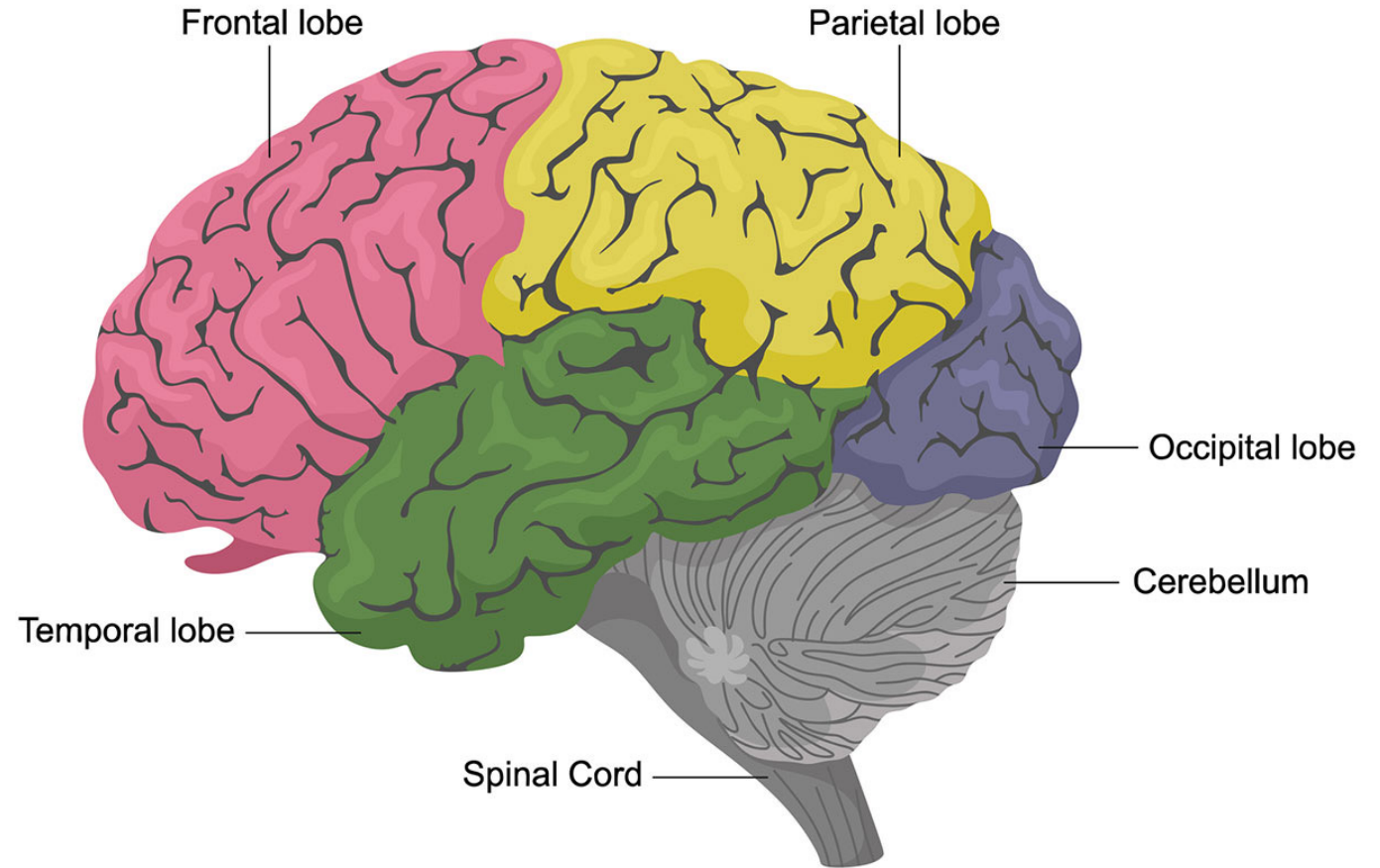
Central Nervous System

- **Brain:** Key regions for motor control, cognition, and autonomic functions
 - **Cerebrum:** Responsible for higher brain functions, including voluntary motor activity and sensory perception.
 - **Cerebellum:** Coordinates movement and balance.
 - **Brainstem:** Regulates basic life functions (e.g., breathing, heart rate).
- **Spinal Cord:** Pathway for neural signals between the brain and the body; plays a role in reflex actions and motor control.



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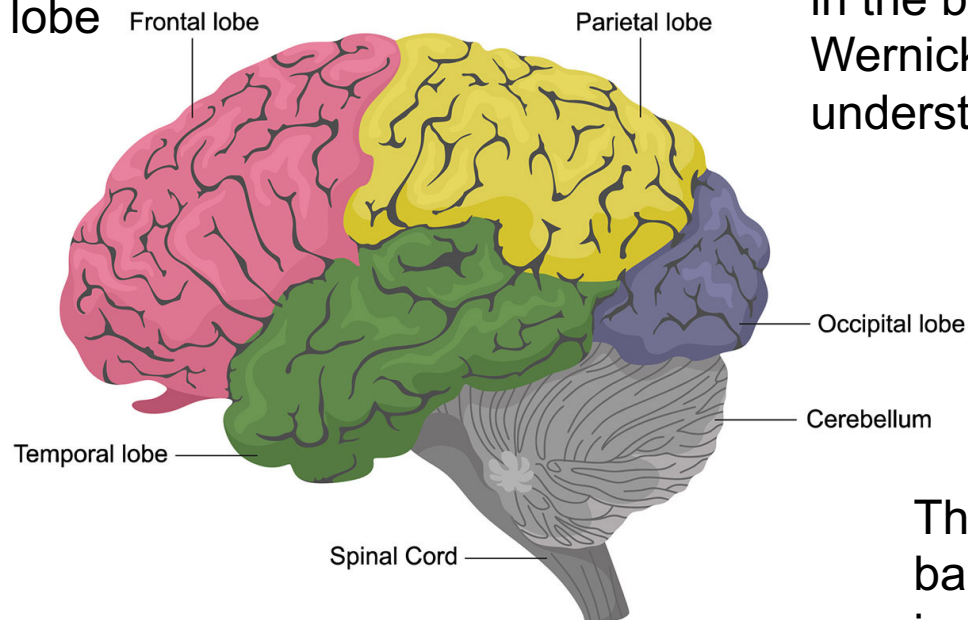


Cerebral cortex

The largest lobe of the brain, located in the front of the head, the **frontal lobe** is involved in personality characteristics, decision-making and movement.

Recognition of smell usually involves parts of the frontal lobe. The frontal lobe contains Broca's area, which is associated with speech ability.

The sides of the brain, **temporal lobes** are involved in short-term memory, speech, musical rhythm and some degree of smell recognition.

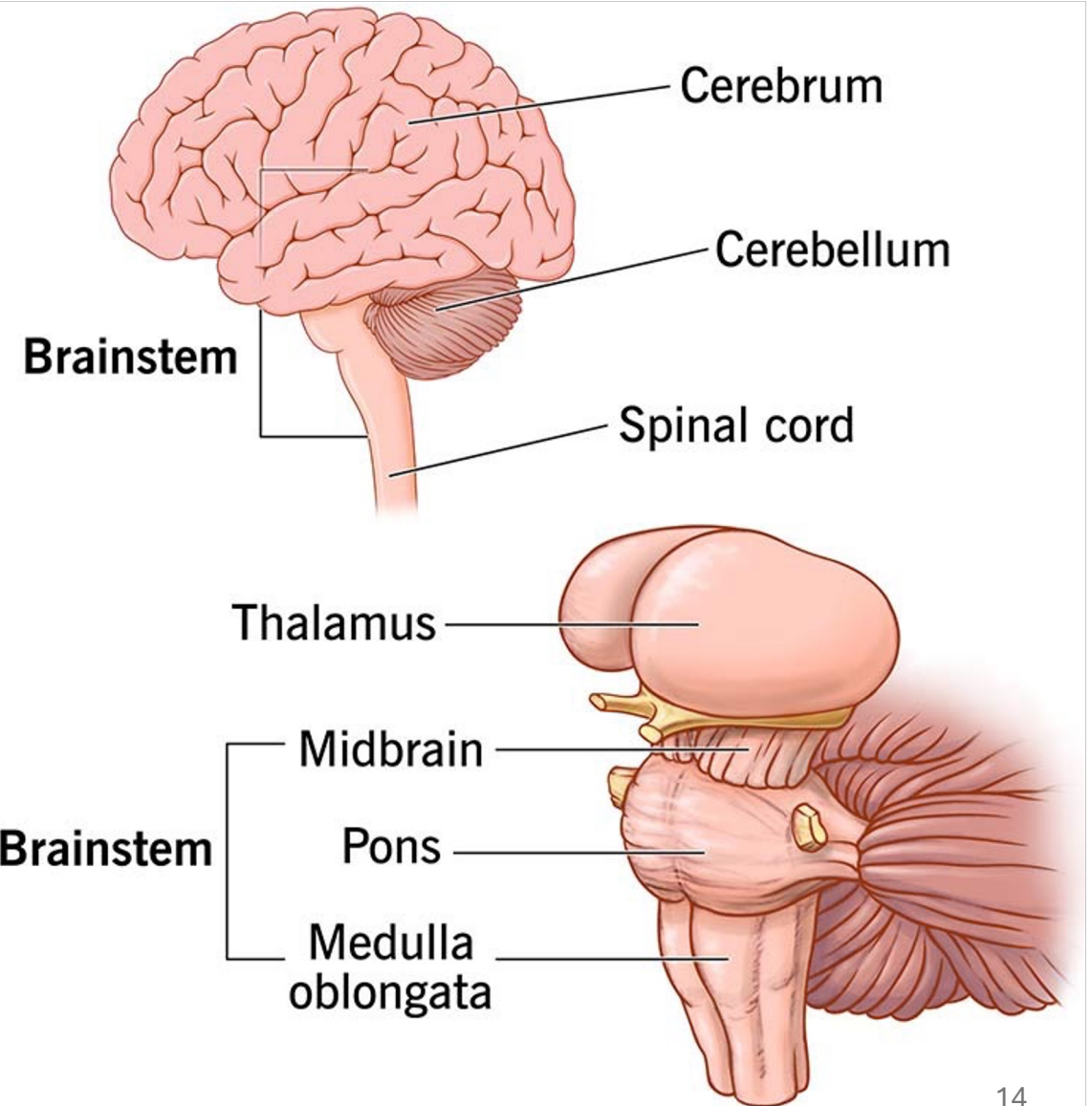


The middle part of the brain, the **parietal lobe** helps a person identify objects and understand spatial relationships (where one's body is compared with objects around the person). The parietal lobe is also involved in interpreting pain and touch in the body. The parietal lobe houses Wernicke's area, which helps the brain understand spoken language.

The **occipital lobe** is the back part of the brain that is involved with vision

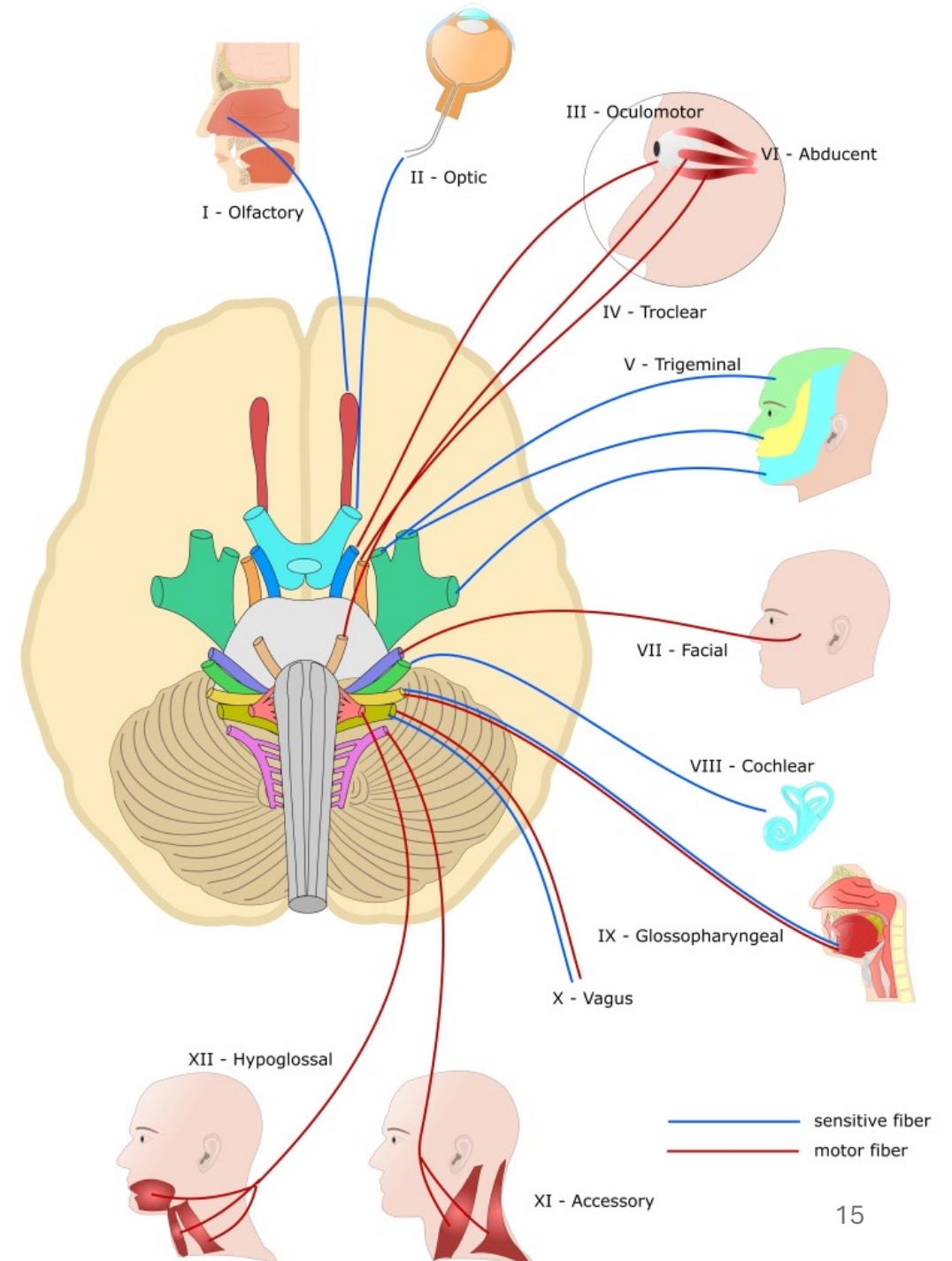
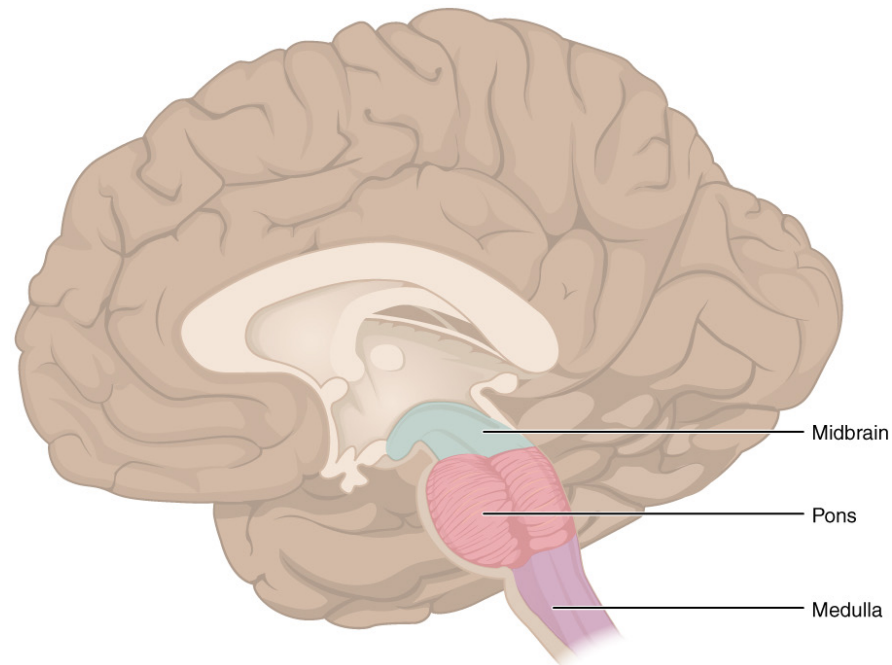
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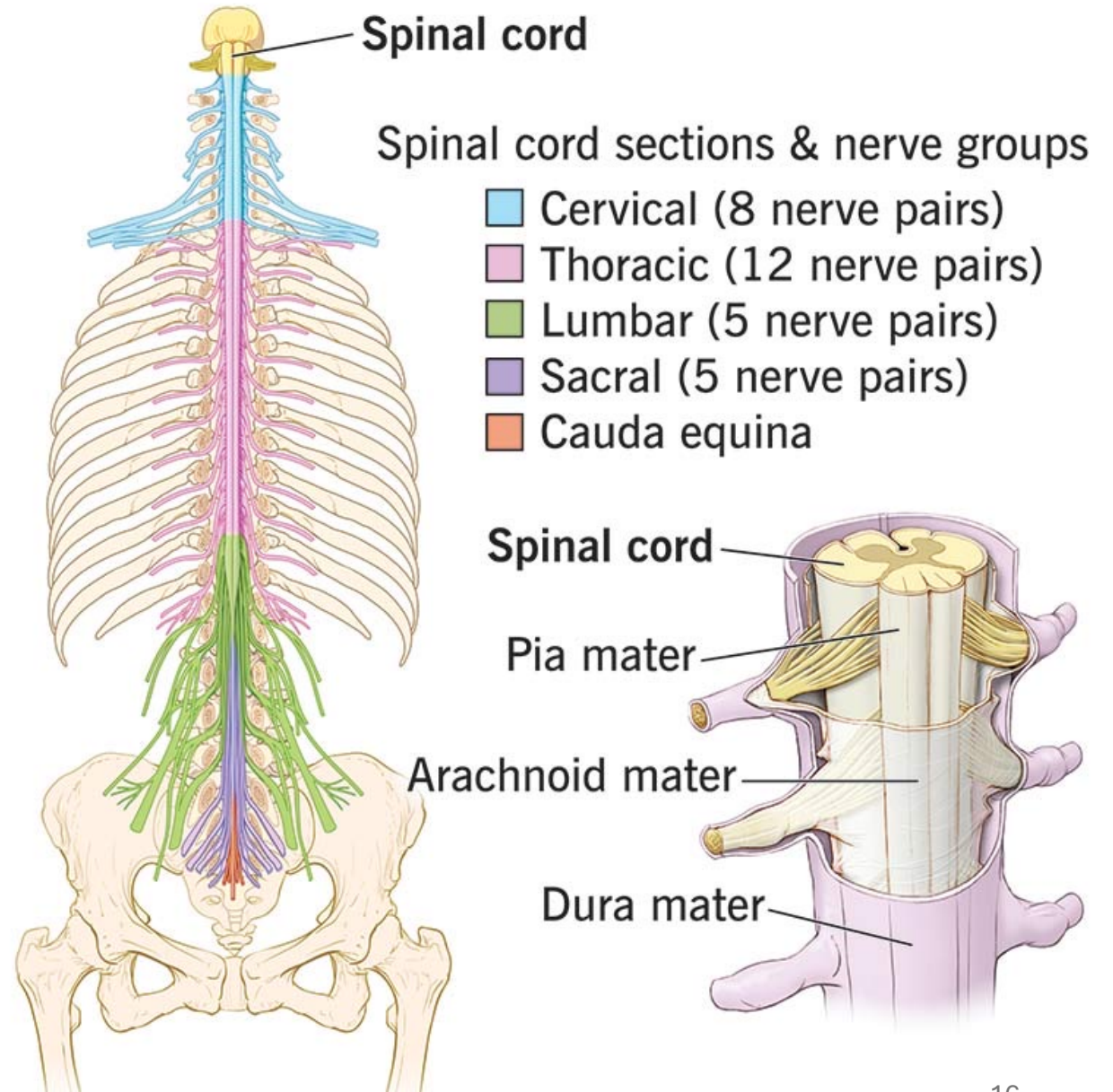
Brainstem and cranial nerves

The brainstem is the lower part of the brain that connects to the spinal cord, responsible for controlling essential life-sustaining functions. It regulates heart rate, breathing, blood pressure, and reflexes like swallowing and blinking. Comprising the midbrain, pons, and medulla oblongata, the brainstem also acts as a relay center, transmitting signals between the brain and the rest of the body.



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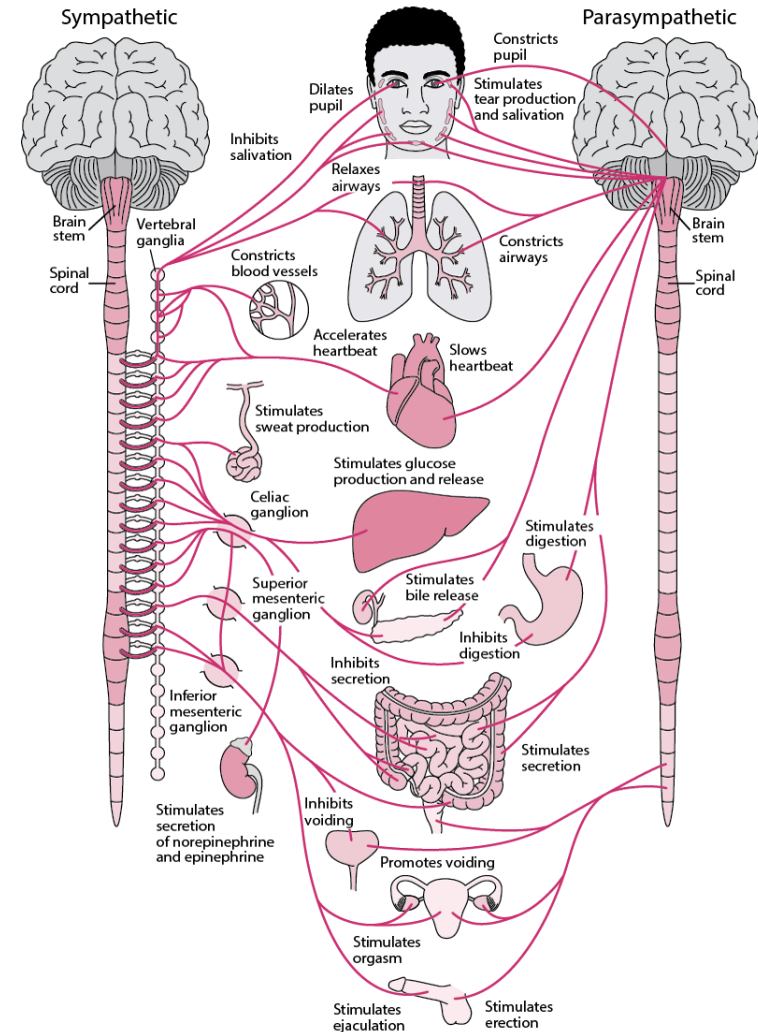
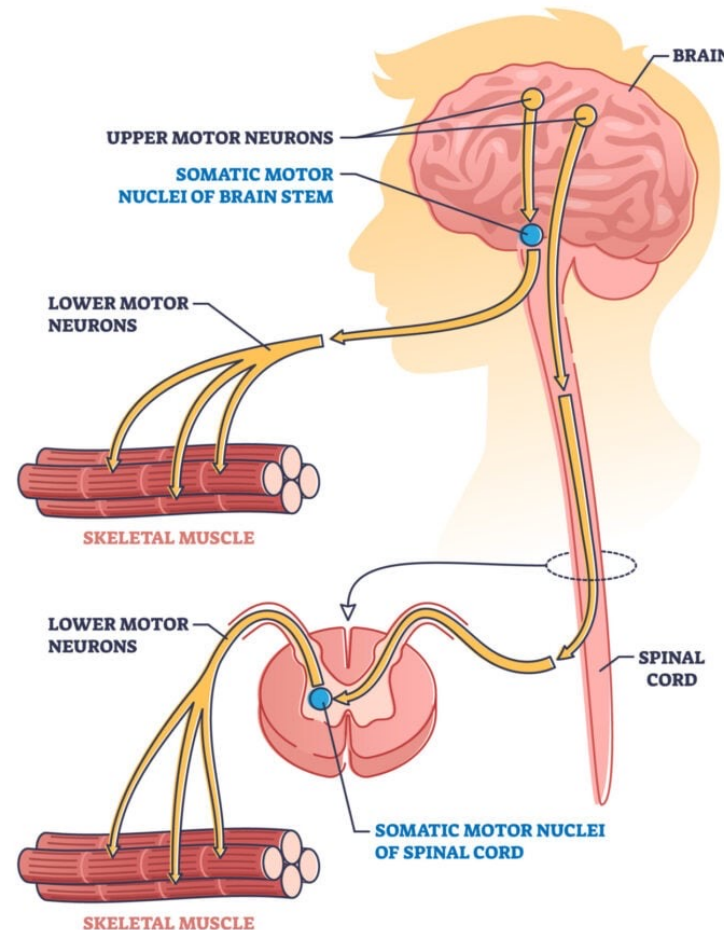
The PNS connects the CNS to limbs and organs, facilitating communication between the brain/spinal cord and the rest of the body.

It is divided into two main components:

- **Somatic Nervous System:** Controls voluntary movements by innervating skeletal muscles.
- **Autonomic Nervous System (ANS):** Regulates involuntary functions, including heart rate, digestion, and respiratory rate, with **sympathetic** (fight-or-flight) and **parasympathetic** (rest-and-digest) divisions.

Key Components:

- **Cranial nerves:** 12 pairs that emerge from the brain, involved in sensory and motor functions for the head and neck.
- **Spinal nerves:** 31 pairs that emerge from the spinal cord, connecting the CNS to different body regions.



Peripheral Nervous System

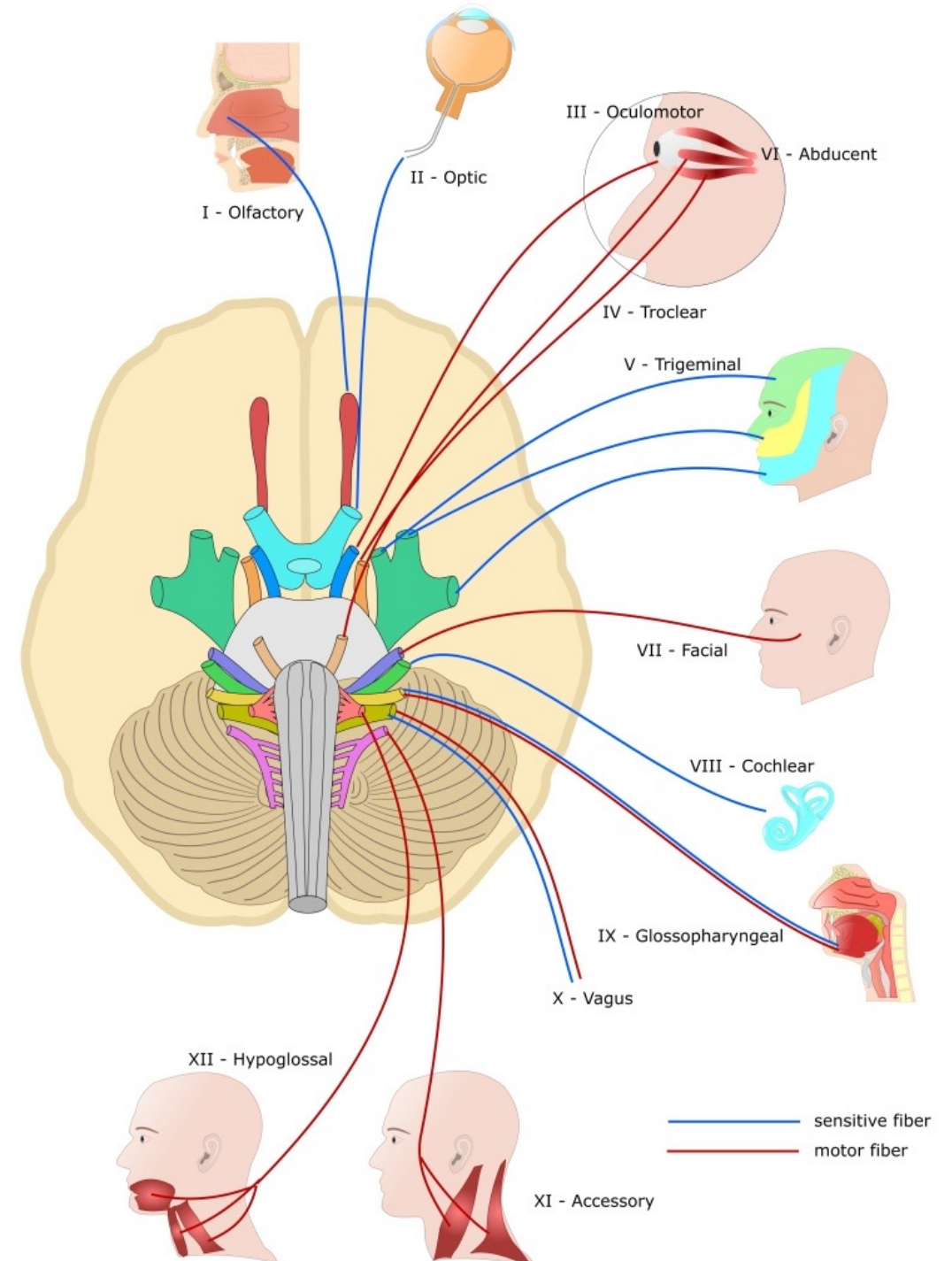
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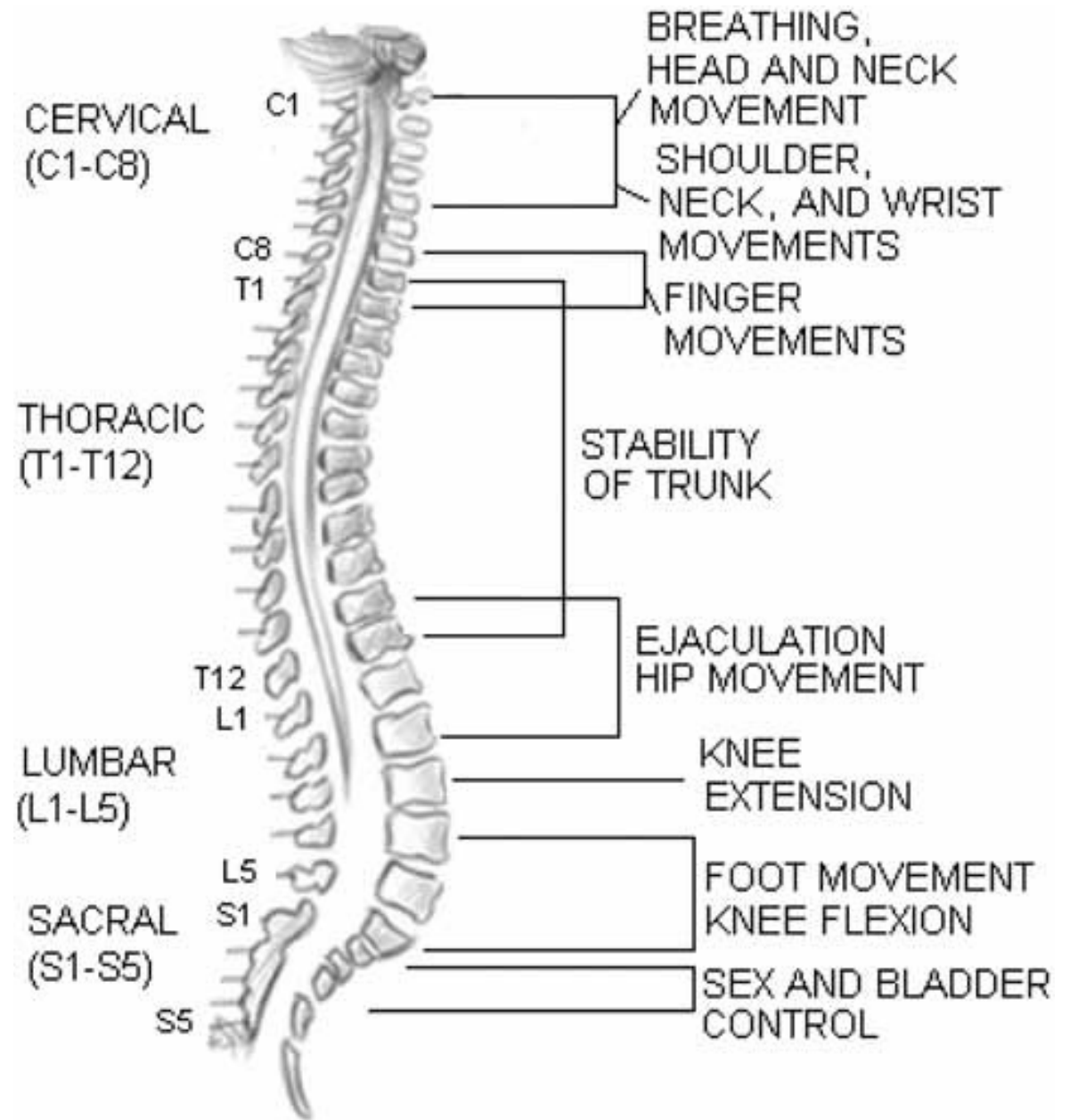
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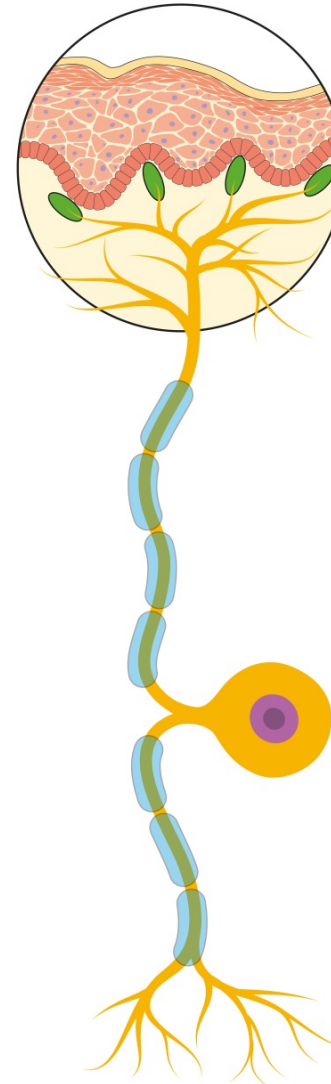
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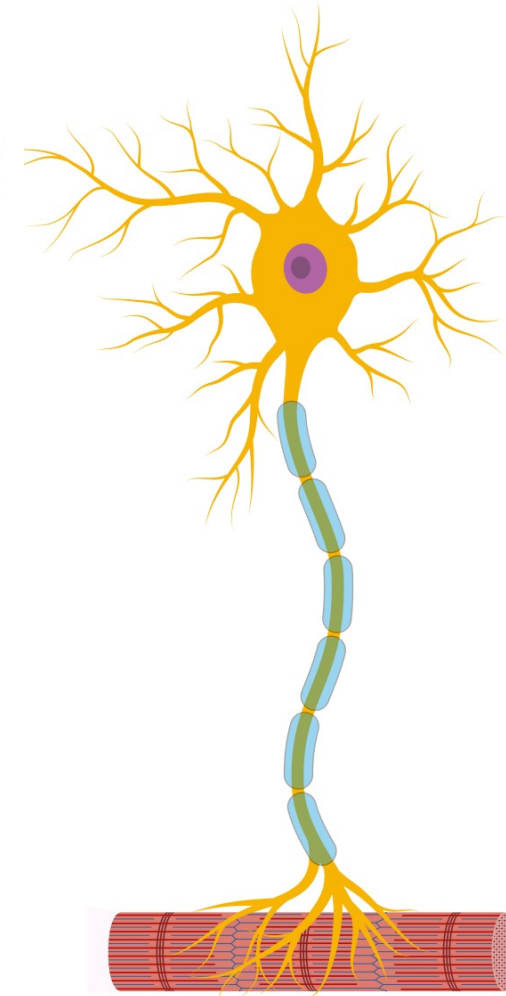
Types of neurons in CNS and PNS

Neurons in the human nervous system can be classified into three types based on their functions: **sensory neurons**, **motor neurons**, and **interneurons**.

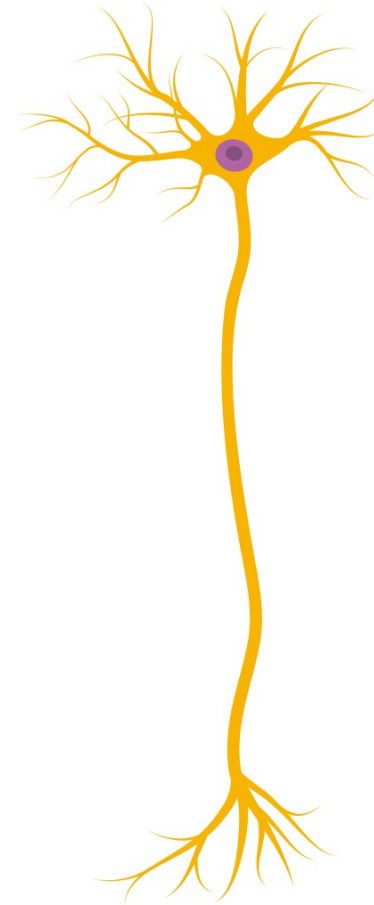
- **Sensory neurons** gather information from both the body and the external environment and transmit it to the CNS for processing. For example, if you touch a hot coal, sensory neurons in your fingertips will relay the sensation of heat to your CNS.
- **Motor neurons** receive signals from other neurons and send commands to muscles, organs, and glands. For instance, if you pick up a hot coal, motor neurons connected to the muscles in your fingers will prompt your hand to release it.
- **Interneurons** are found exclusively in the CNS, interneurons link one neuron to another. They receive signals from sensory neurons or other interneurons and relay information to motor neurons or additional interneurons.



Sensory neuron



Motor neuron



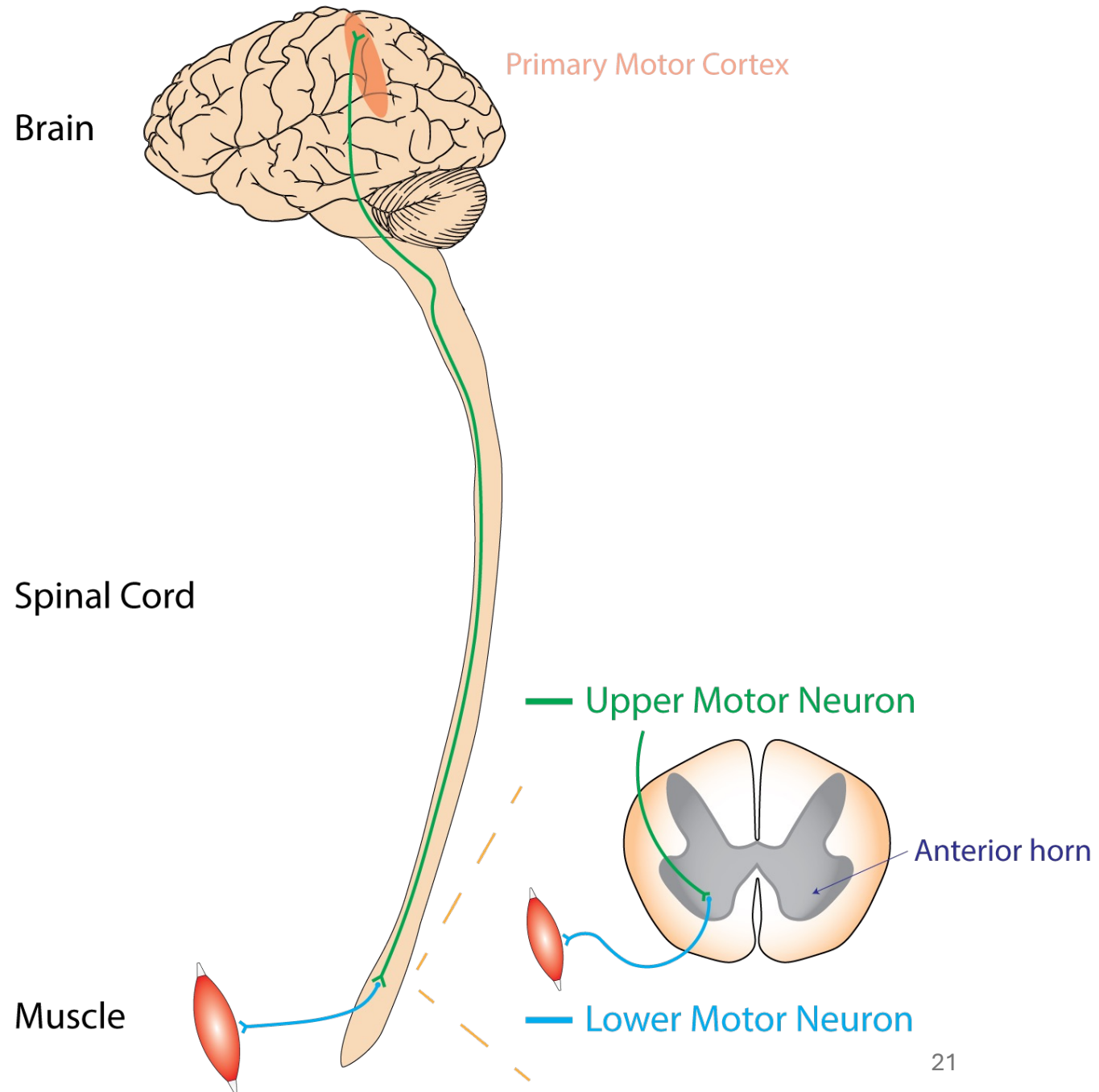
Interneuron

Motor neurons

Neurons that have their cell bodies in the motor cortex, brainstem, or spinal cord and whose axons extend to the spinal cord or beyond, allowing them to directly or indirectly control effector organs such as muscles and glands.

- **Upper motor neurons (UMNs):** they originate in the cerebral cortex and travel down to the brainstem or spinal cord
- **Lower motor neurons (LMNs):** they begin in the spinal cord and go to innervate the muscles and glands throughout the body

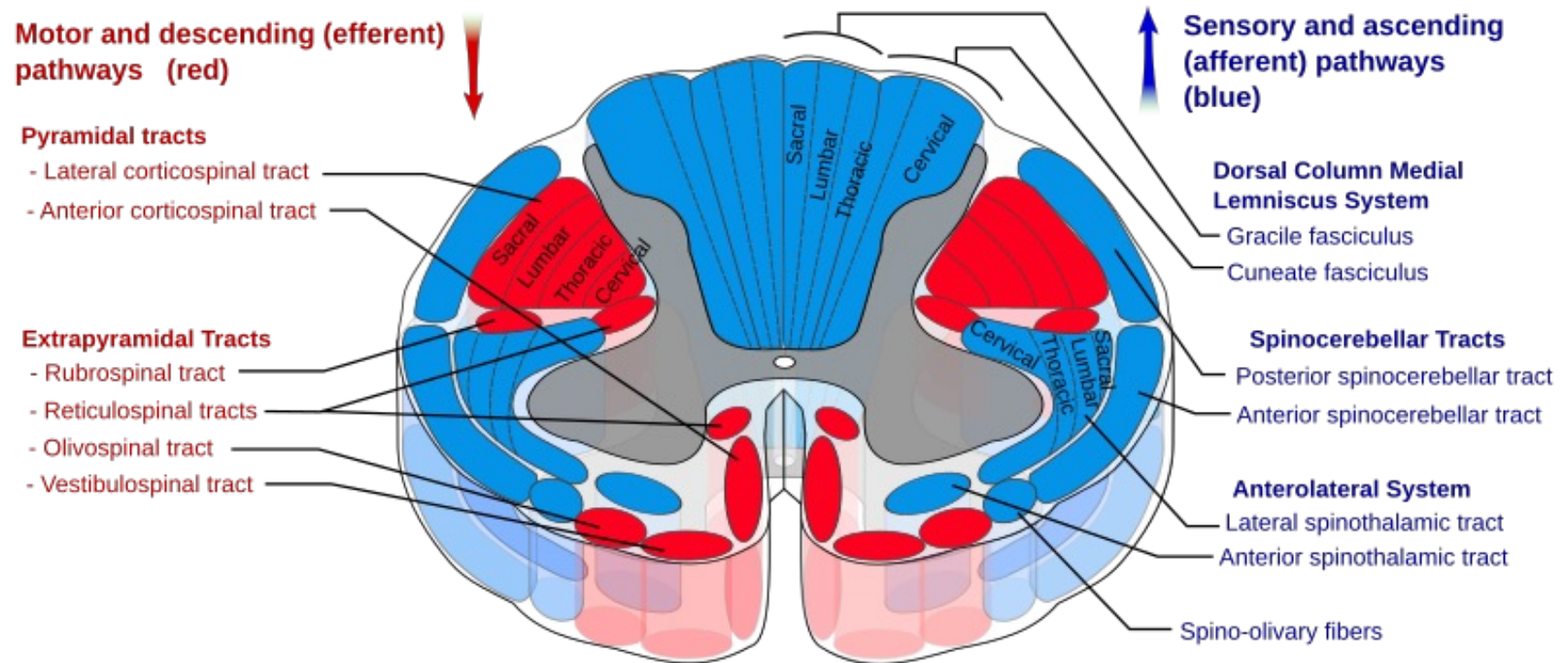
A **motor unit** is formed by one motor neuron and all the muscle fibers innervated by that motor neuron



Ascending and descending pathways

The central nervous system relies on ascending and descending pathways to interact with the external environment.

- **Ascending tracts** carry sensory information from the body, like pain, for example, up the spinal cord to the brain.
- **Descending tracts** carry motor information, like instructions to move the arm, from the brain down the spinal cord to the body.
- Both types of tracts are made up of neuronal axons that gather into long columns called funiculi, meaning long ropes, which are found inside the ventral, lateral and dorsal parts of the spinal cord.

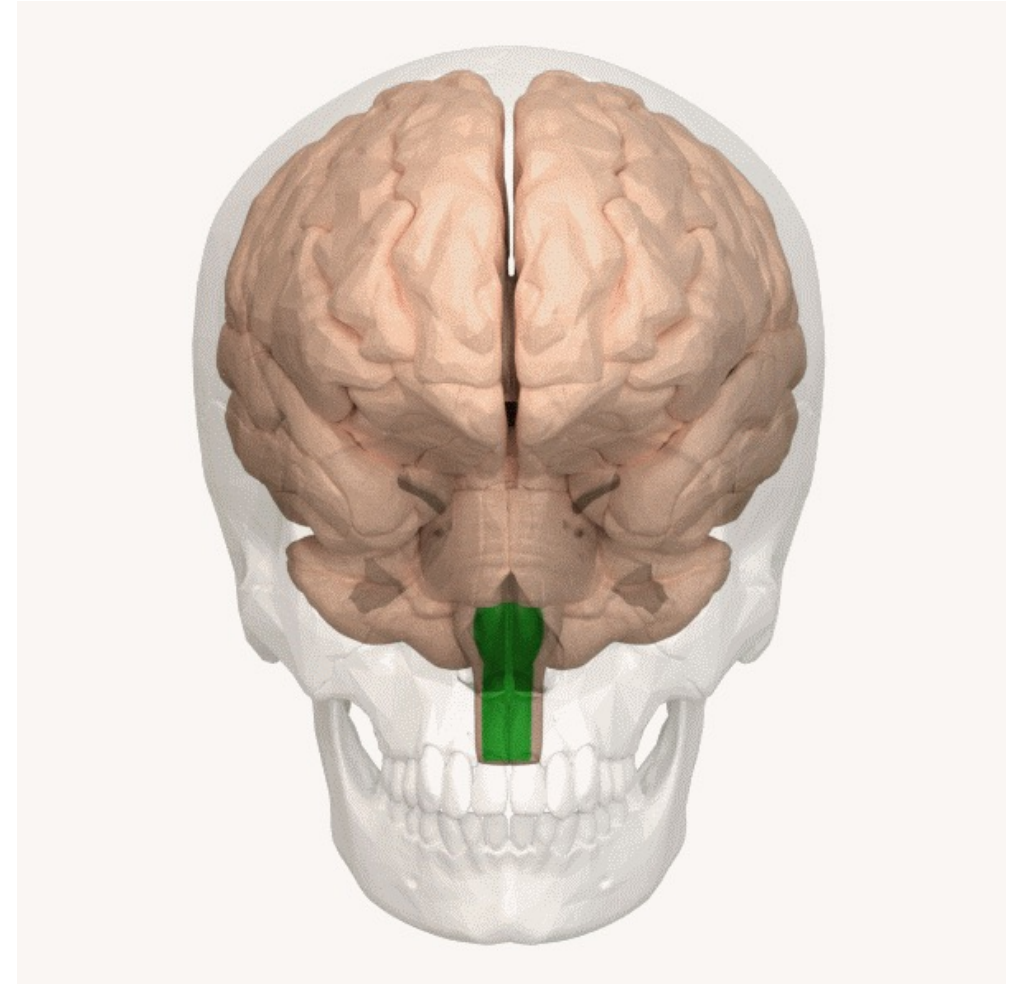


Descending motor pathways

The spinal cord contains multiple motor tracts, some of which operate under conscious control, while others function reflexively or responsively without conscious input. These motor tracts are functionally categorized into two main groups: the **pyramidal** and **extrapyramidal** tracts

- **Pyramidal tracts:** Responsible for conscious muscle control, transmitting signals from the cerebral cortex to the muscles of the body and face.
- **Extrapyramidal tracts:** Involved in unconscious, reflexive, or responsive muscle control, these tracts originate in various brainstem regions and primarily target postural and anti-gravity muscles.

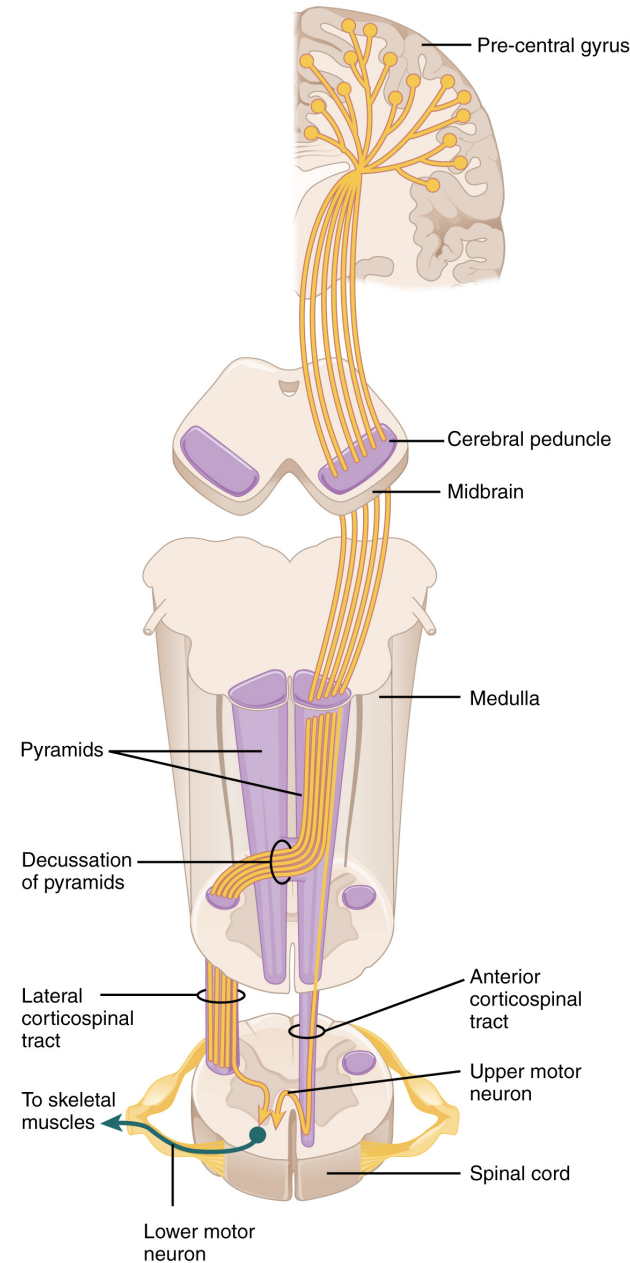
The **medullary pyramids** are paired white matter structures of the brainstem's medulla oblongata that contain motor fibers of the corticospinal and corticobulbar tracts (aka the pyramidal tracts)



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Key

Upper motor neuron

Lower motor neuron

Corticospinal tract

- It transmits motor signals from the cerebral cortex to the spinal cord.
- It affects the control of fine motor movements in the hands and feet.
- It primarily projects contralaterally, with about 85% of fibers crossing at the medullary pyramids in the brainstem to form the **lateral corticospinal tract**, while the remaining 15% cross within the spinal cord at their target level as part of the **medial corticospinal tract**.

Fiber organization within this tract results in cervical motor neurons being centrally located and lumbar/sacral neurons peripherally located, helping to explain certain patterns of functional loss in specific spinal injuries.

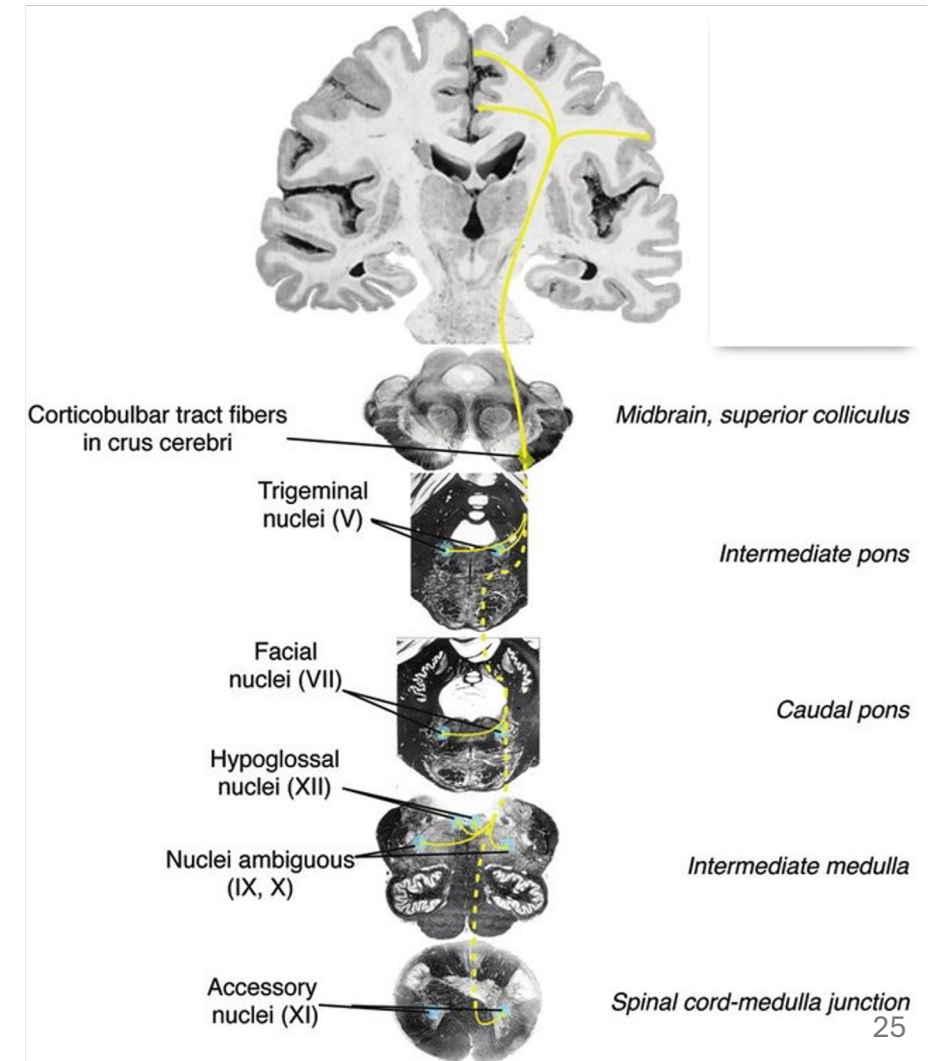
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Corticobulbar tract

- It provides upper motor neuron input to the cranial nerves controlling muscles of the head and face.
- It originates in the precentral gyrus of the primary motor cortex
- It also receives fibers from the premotor and supplementary motor areas, similar to the corticospinal tract.



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- 
- **Reticulospinal tract:** Divided into medial (pontine) and lateral (medullary) tracts, it primarily controls locomotion and posture.
 - **Vestibulospinal tract:** Increases tone in antigravity (extensor) muscles when the head tilts, helping maintain balance.
 - **Rubrospinal tract:** Regulates motor neuron activity and plays a key role in certain reflexive body movements.

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Motor control of human movement

Motor skills involve voluntary control of joint and body movements to achieve specific goals, such as riding a bicycle, walking, or weightlifting.

The study of motor skills encompasses **motor learning and control**, also known as skill acquisition. This area is crucial for both skill performance and rehabilitation, particularly in cases of neurological injuries and disorders.

Human movement involves a spectrum from simple reflex loops to complex neural networks that communicate across the central and peripheral nervous systems.

New motor patterns develop through movement experiences, rich sensory interactions, and problem-solving challenges. Understanding motor control and learning helps us comprehend how individuals progress from novice to skilled performers throughout their lives.

Motor control of human movement

Motor control is the process of initiating, directing, and regulating purposeful voluntary movement.

Motor control process

- **Task identification:** Recognize the task to be completed.
- **Sensory gathering:** Collect sensory information from the environment.
- **Perception:** Interpret the gathered information.
- **Movement planning:** Select an appropriate movement strategy to achieve the task goal.
- **Coordination:** Integrate the plan within the central nervous system (CNS).
- **Execution:** Implement the plan through motor neurons in the brainstem and spinal cord.
- **Communication:** Relay the outcome to muscles, coordinating postural and limb synergies.
- **Timing:** Ensure motor units activate in a specific sequence.

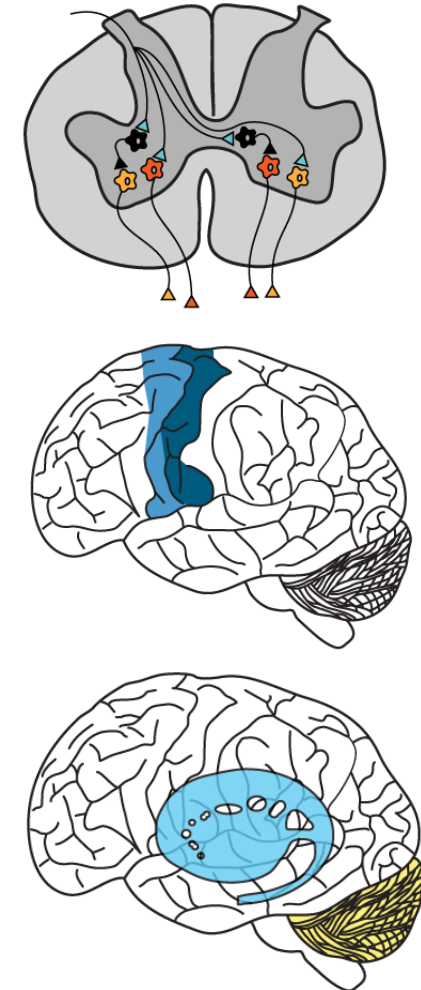
Feedback and adaptation

- Sensory feedback informs the CNS about the movement.
- Decisions made based on feedback may include:
 - Modifying the movement plan during execution.
 - Acknowledging the achievement of the task goal.
 - Storing information for future performance of similar tasks.

Motor control of human movement

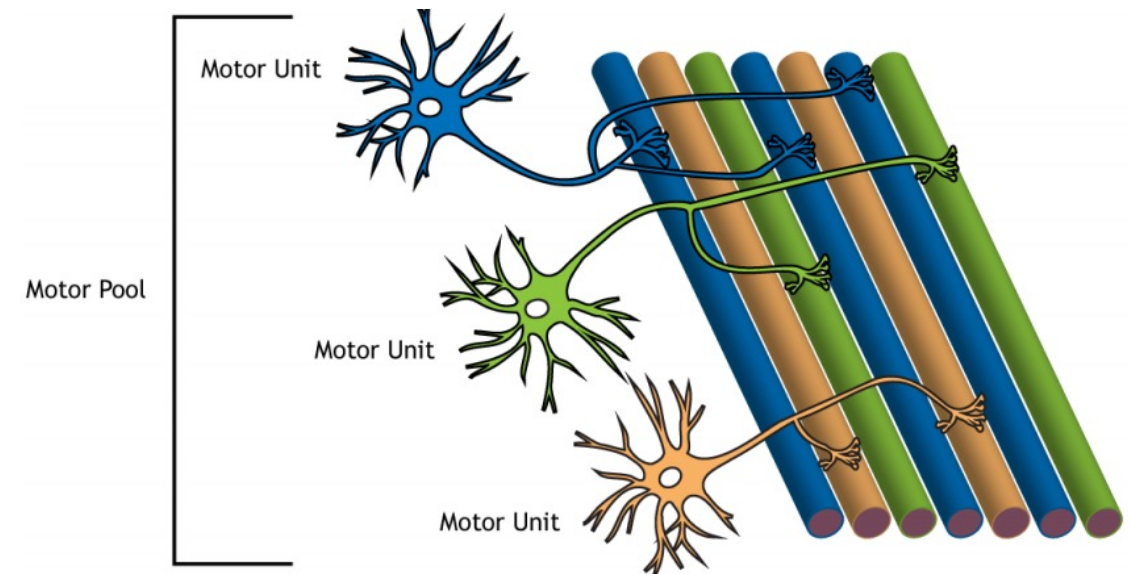
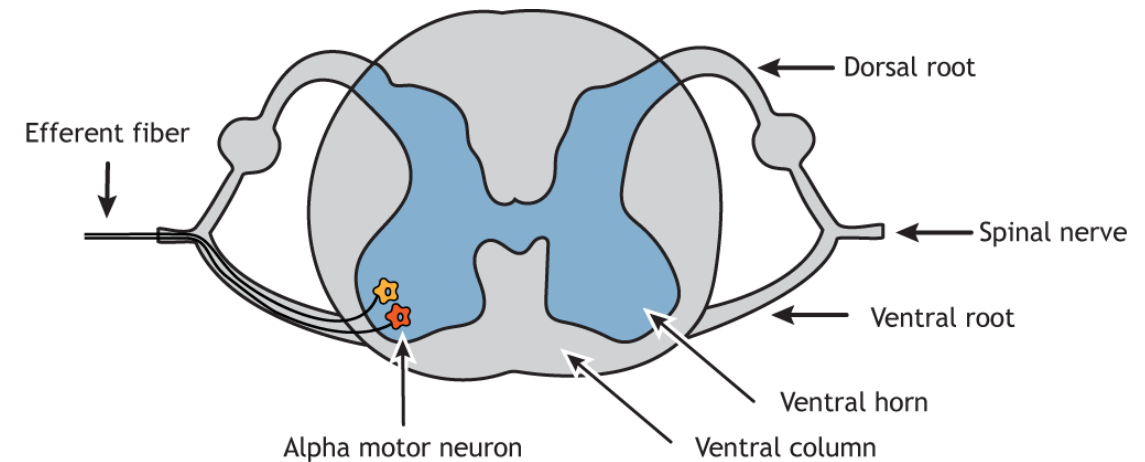
The motor system regulates all skeletal muscle movements and operates at multiple levels of control:

- **Spinal reflexes:** Simple reflexes can occur independently within the spinal cord without requiring input from the brain.
- **Central pattern generators:** These neural circuits facilitate more complex, repetitive movements, such as walking.
- **Motor and premotor cortices:** Located in the brain, these areas are responsible for planning and executing voluntary movements.
- **Basal ganglia and cerebellum:** These structures modulate the activity of neurons in the motor cortex, aiding in coordination, motor learning, and balance.



Spinal control of movement

- Muscle fibers are innervated by **alpha motor neurons**, whose cell bodies reside in the ventral horn of the spinal cord within the central nervous system.
- The alpha motor neurons' axons exit the spinal cord through the ventral roots and extend to the muscles via efferent peripheral spinal nerves.
- One alpha motor neuron can innervate several muscle fibers.
- Its axon branches to form synaptic connections with multiple fibers.
- A **motor unit** is the combination of a motor neuron and the fibers it innervates.
- The muscle fibers within a motor unit are typically distributed throughout the muscle to ensure a coordinated contraction across the entire muscle.
- The collective group of motor neurons that innervate all the fibers of a specific muscle is known as a **motor pool**.

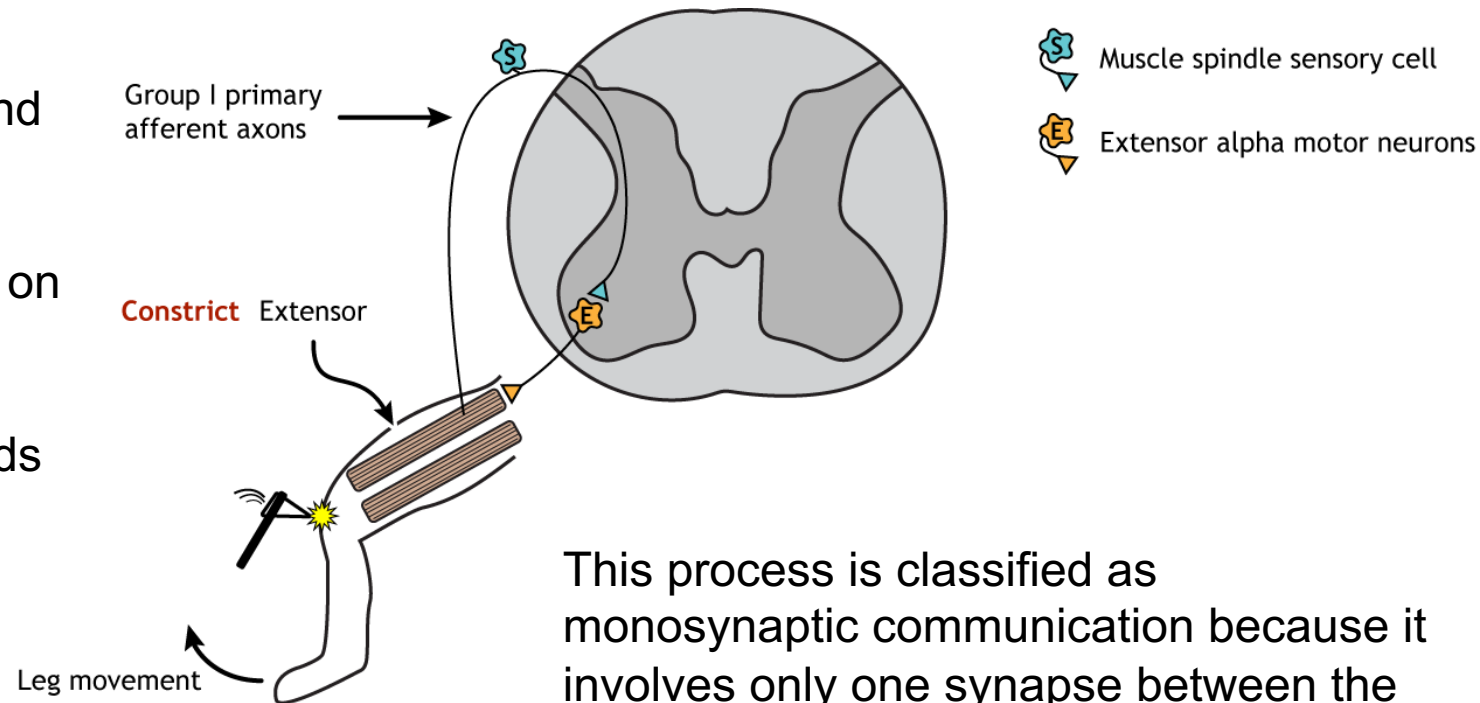


Reflexes

***Muscle spindles** are specialized sensory receptors located within skeletal muscles that monitor the stretch of muscle fibers and relay this information to the nervous system through large, myelinated Group I sensory axons. Muscle spindles are essential for facilitating spinal reflexes, helping to maintain muscle tone and coordinate movement.

The **stretch reflex** (aka myotatic, patellar, or knee-jerk reflex) is triggered by the activation of muscle spindle stretch receptors*. This reflex is entirely mediated within the spinal cord, requiring no input from the brain.

- The tendon connecting to the quadriceps muscle is tapped, stretching the muscle and activating stretch receptors.
- Sensory information is sent to the dorsal horn of the spinal cord, where it synapses on alpha motor neurons innervating the quadriceps.
- The activation of these motor neurons leads to the contraction of the quadriceps, extending the lower leg.



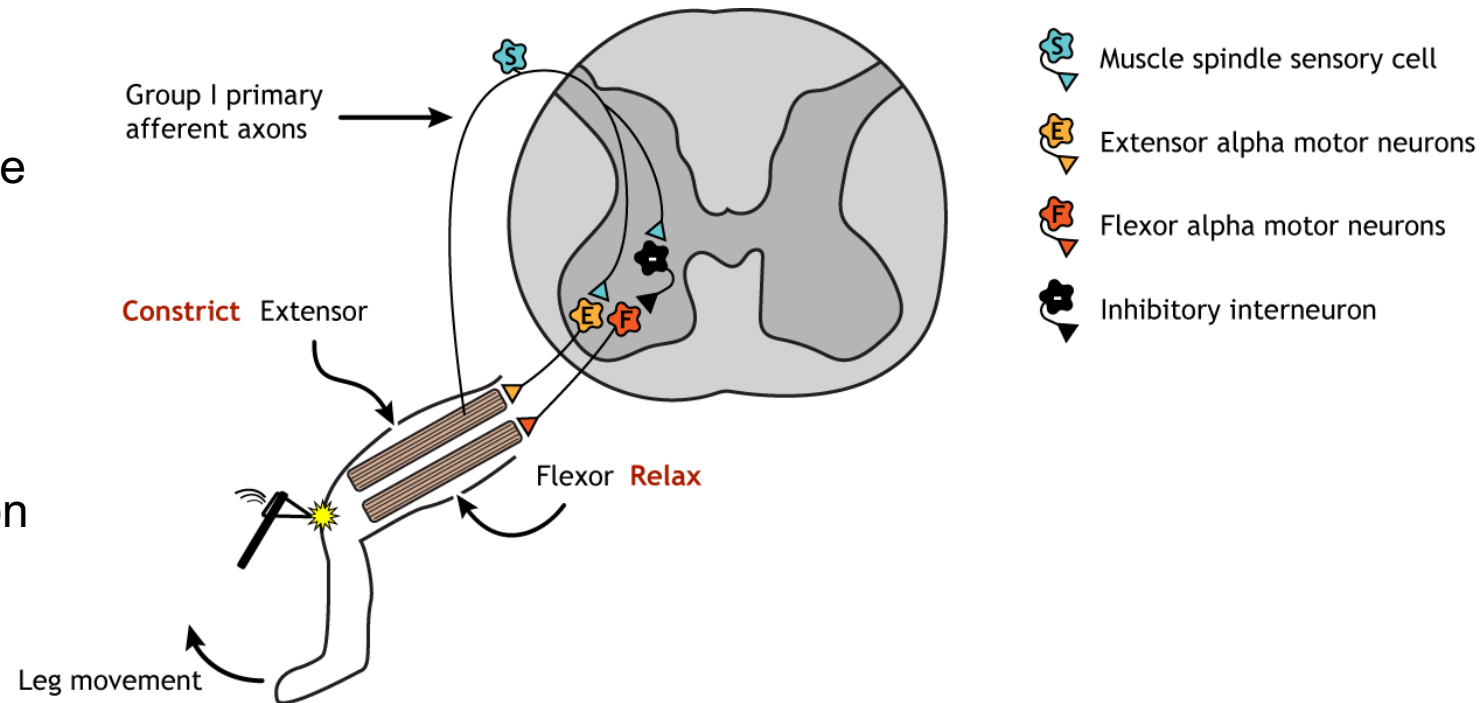
This process is classified as monosynaptic communication because it involves only one synapse between the sensory input and the motor output.

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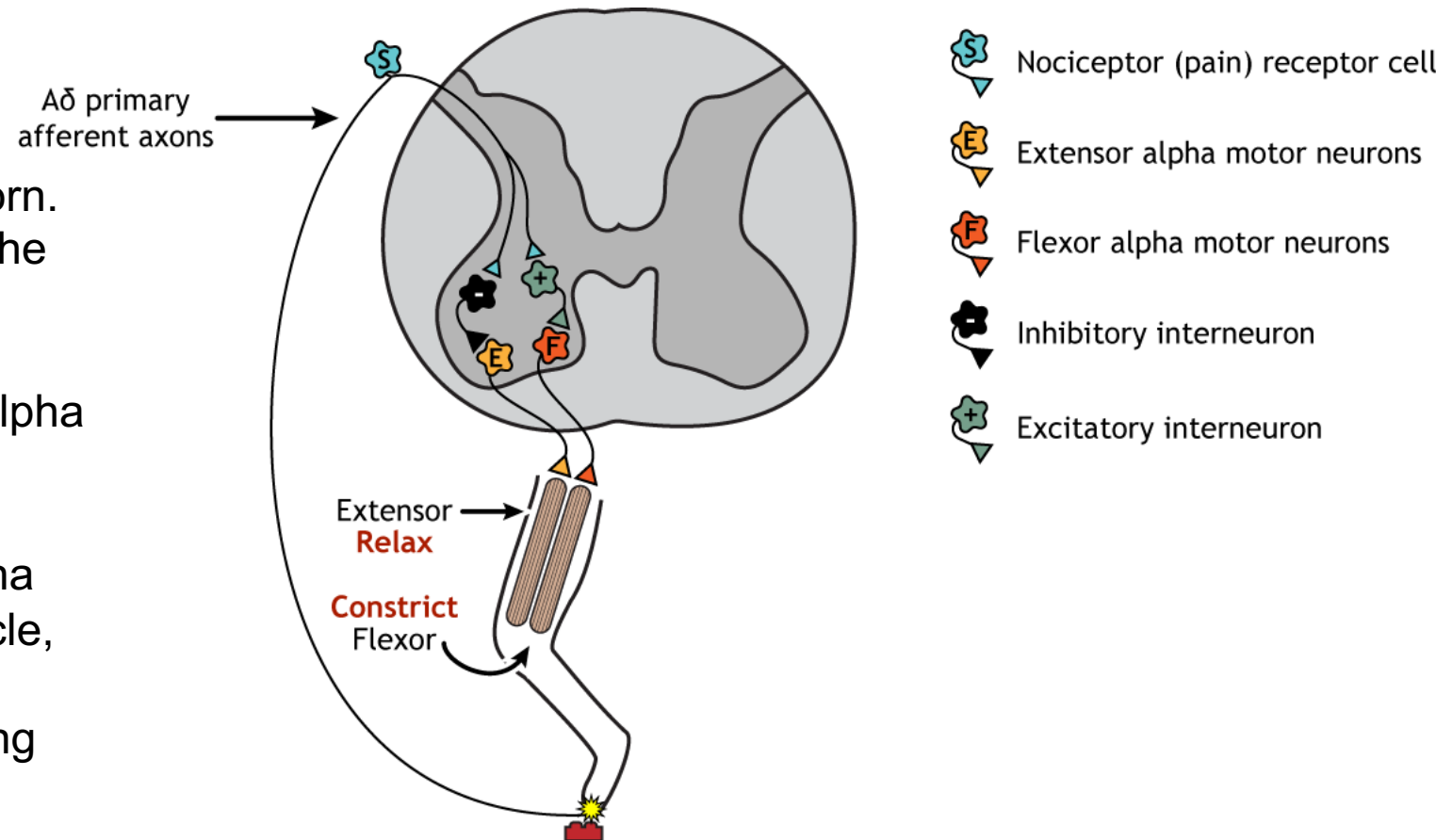
- Sensory neurons synapse on inhibitory interneurons in the spinal cord.
- These inhibitory interneurons then synapse on alpha motor neurons that innervate the hamstring, which is the antagonistic flexor muscle to the quadriceps.
- Inhibition of these motor neurons leads to relaxation of the hamstring muscle.
- This relaxation allows for easier contraction of the quadriceps.



Reflexes

The **withdrawal reflex** causes muscle flexion in response to a painful stimulus, such as stepping on something sharp. This reflex allows for quick, protective withdrawal from harm.

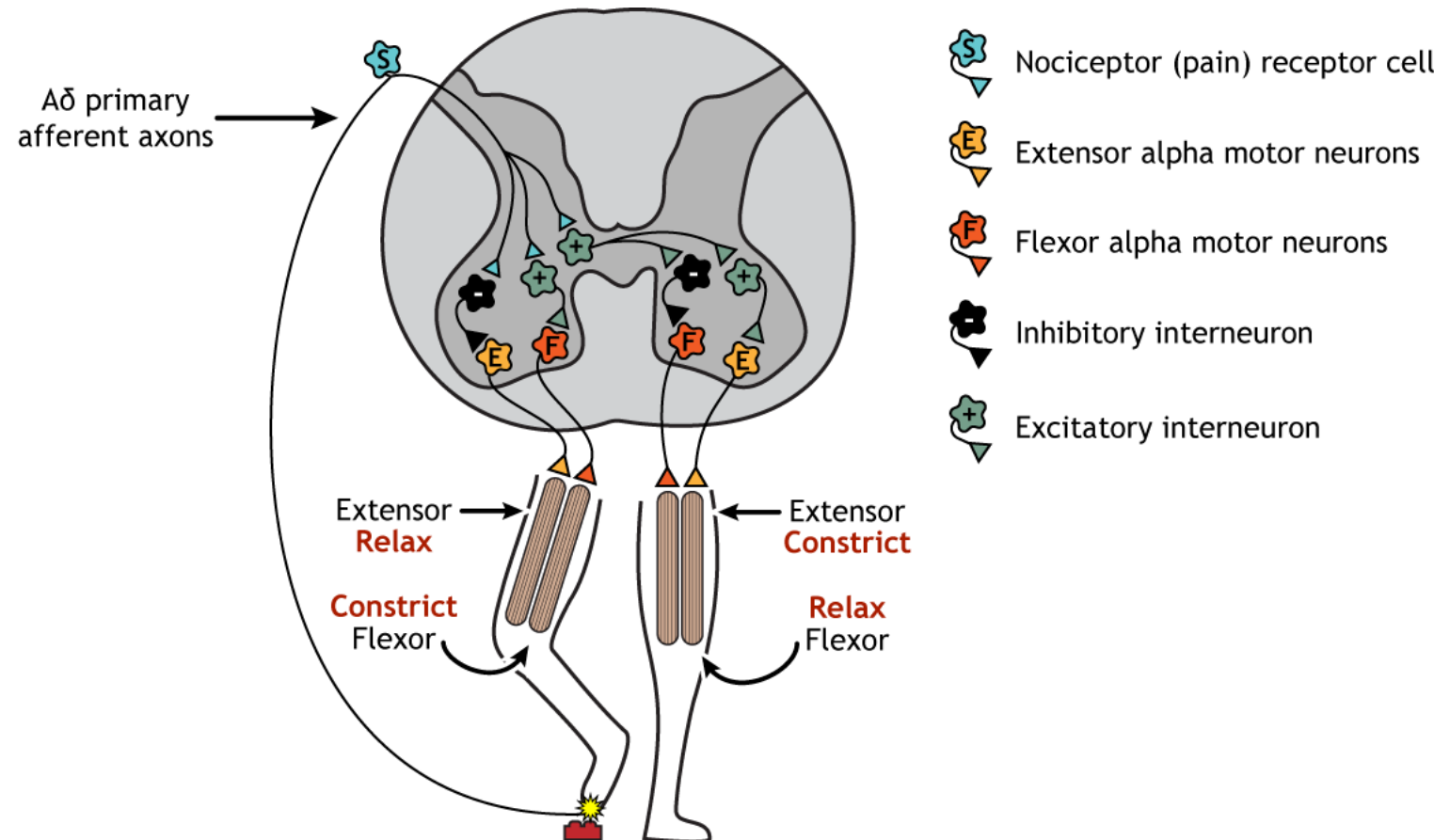
- Pain receptors (nociceptors) detect the painful stimulus, sending sensory information to the spinal cord's dorsal horn.
- Unlike the monosynaptic stretch reflex, the withdrawal reflex involves interneurons (polysynaptic pathway)
 - **Excitatory interneurons** activate alpha motor neurons in the flexor muscle, causing it to contract.
 - **Inhibitory interneurons** inhibit alpha motor neurons in the extensor muscle, allowing it to relax.
- Response: The leg flexes upward, moving away from the painful stimulus.



Reflexes

The **crossed-extensor** reflex works in tandem with the withdrawal reflex to maintain balance when responding to pain. When you lift one leg in response to stepping on something sharp, the opposite leg must support your weight shift. This reflex helps stabilize posture during rapid withdrawal from pain.

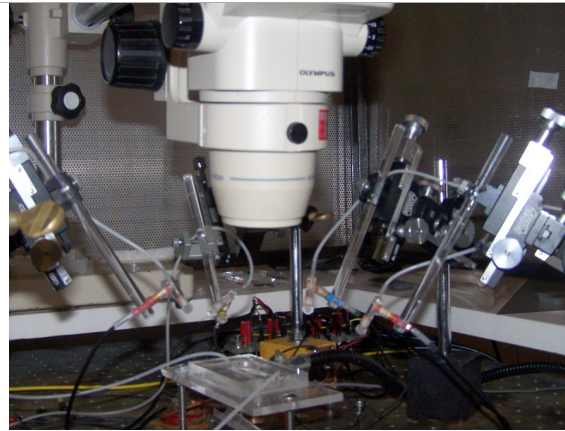
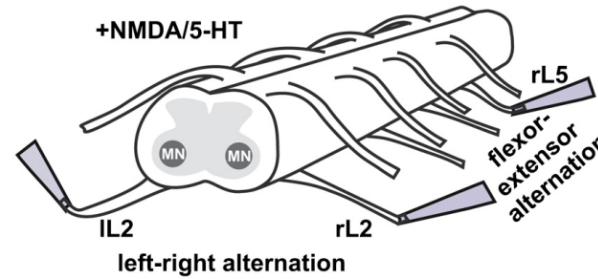
- **Initiation:** Painful sensory input triggers both the withdrawal and crossed-extensor reflexes.
- **Crossing the midline:** Interneurons cross to the contralateral side of the spinal cord to activate the opposite leg.
 - **Excitatory interneurons** activate alpha motor neurons of the extensor muscle on the opposite leg, allowing it to straighten and support the body.
 - **Inhibitory interneurons** inhibit alpha motor neurons of the flexor muscle on the opposite leg, keeping it relaxed.
- **Result:** The opposite leg extends to provide a stable base, allowing for balanced weight support while the injured leg withdraws.



Central pattern generators (CPGs)

Mike the Headless Chicken

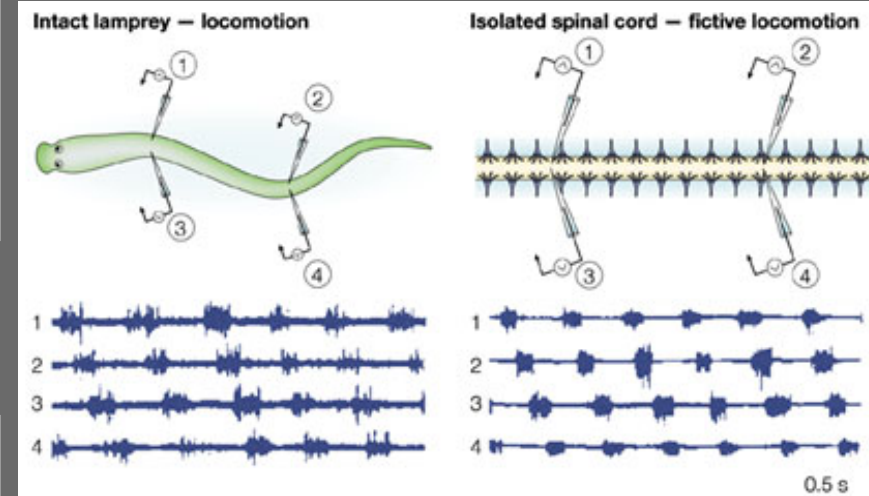
(April 20, 1945 – March 17, 1947)



Sten Grillner



THE SAME PATTERN OF LOCOMOTOR ACTIVITY IS OBSERVED WITH AND WITHOUT BRAIN INPUTS



CENTRAL PATTERN GENERATORS

Locomotor CPGs are operatively defined as neural networks that can endogenously produce rhythmic patterns of motor output, i.e. in the absence of phasic input from descending or peripheral projections

Central pattern generators (CPGs)

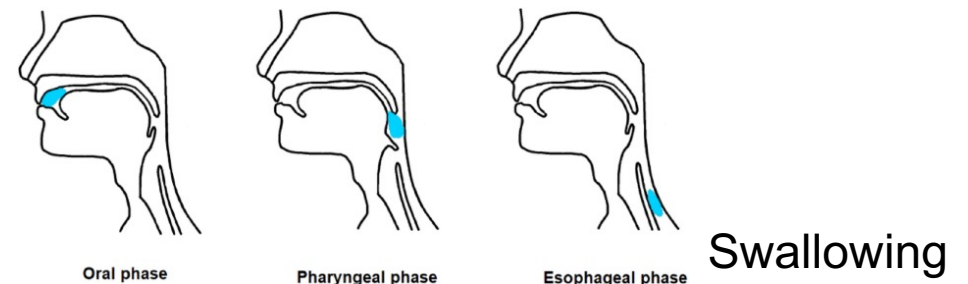
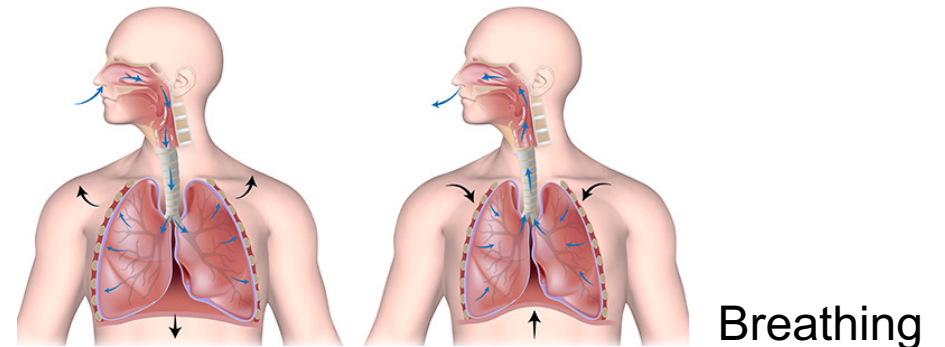
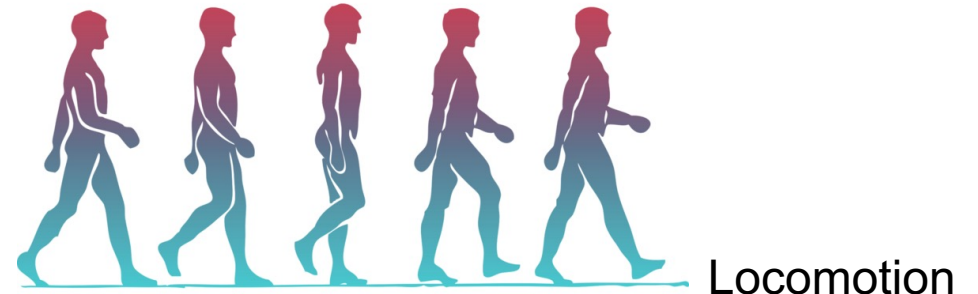
In vertebrates, rhythmic movements like walking are believed to be generated by spinal circuits called central pattern generators (CPGs).

CPGs create rhythmic motor patterns independently of sensory input or signals from the brain but can be adjusted by them.

These spinal circuits are thought to support a range of rhythmic human movements and can adapt based on sensory feedback and environmental demands. They integrate sensory information and brain signals, allowing smooth and coordinated movement.

Locomotion is an example of a fundamental rhythmic movement that relies on the coordinated action of multiple muscle groups (other examples include swimming, flying, breathing, and swallowing).

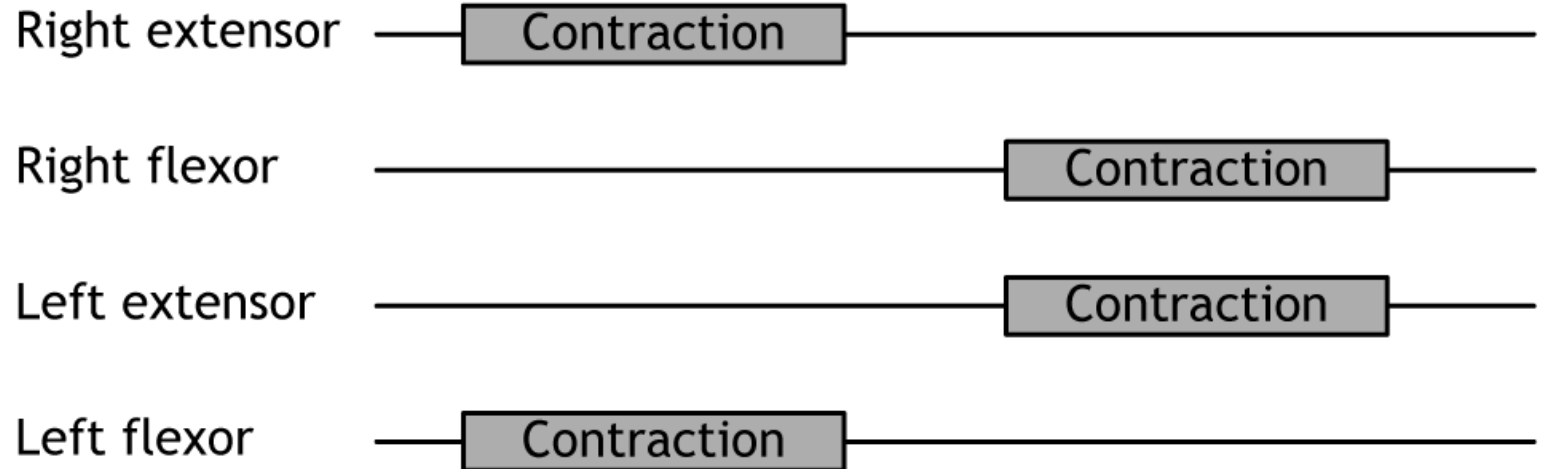
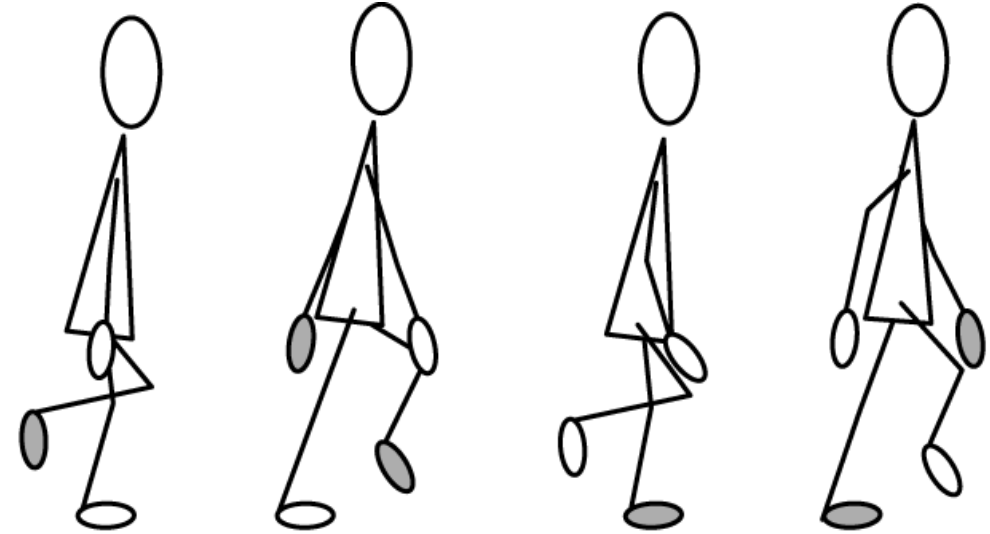
Examples of rhythmic movements regulated by CPGs:



CPGs for locomotion

For smooth locomotion, extensor and flexor muscles in both legs must be precisely coordinated to prevent falls. This **rhythmic movement is regulated at the spinal cord level by the CPGs.**

In walking, the spinal cord circuitry alternates leg movements in opposite patterns: as one leg lifts to step forward, the other remains stable, maintaining contact with the ground.

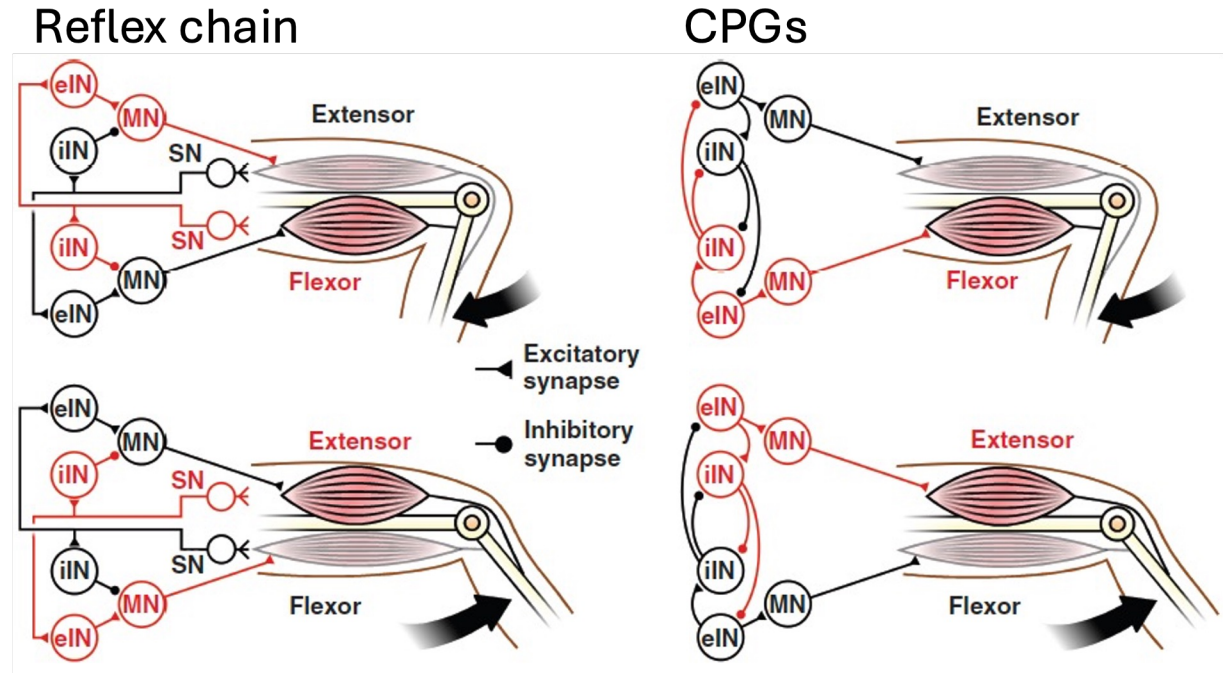


CPGs for locomotion

Originally, it was thought that rhythmic movements arose from reflex chains. For example, contraction of a muscle controlling flexion (bending) would activate sensory feedback that then triggers the opposite (extensor) muscle to contract. This reflexive alternation would repeat to create rhythmic movement.

We now know that CPGs are responsible for these rhythmic patterns, operating independently of continuous sensory feedback. CPGs use networks of neurons, including excitatory and inhibitory interneurons, that interact to generate rhythmic motor activity by alternately activating opposing muscle groups (agonists).

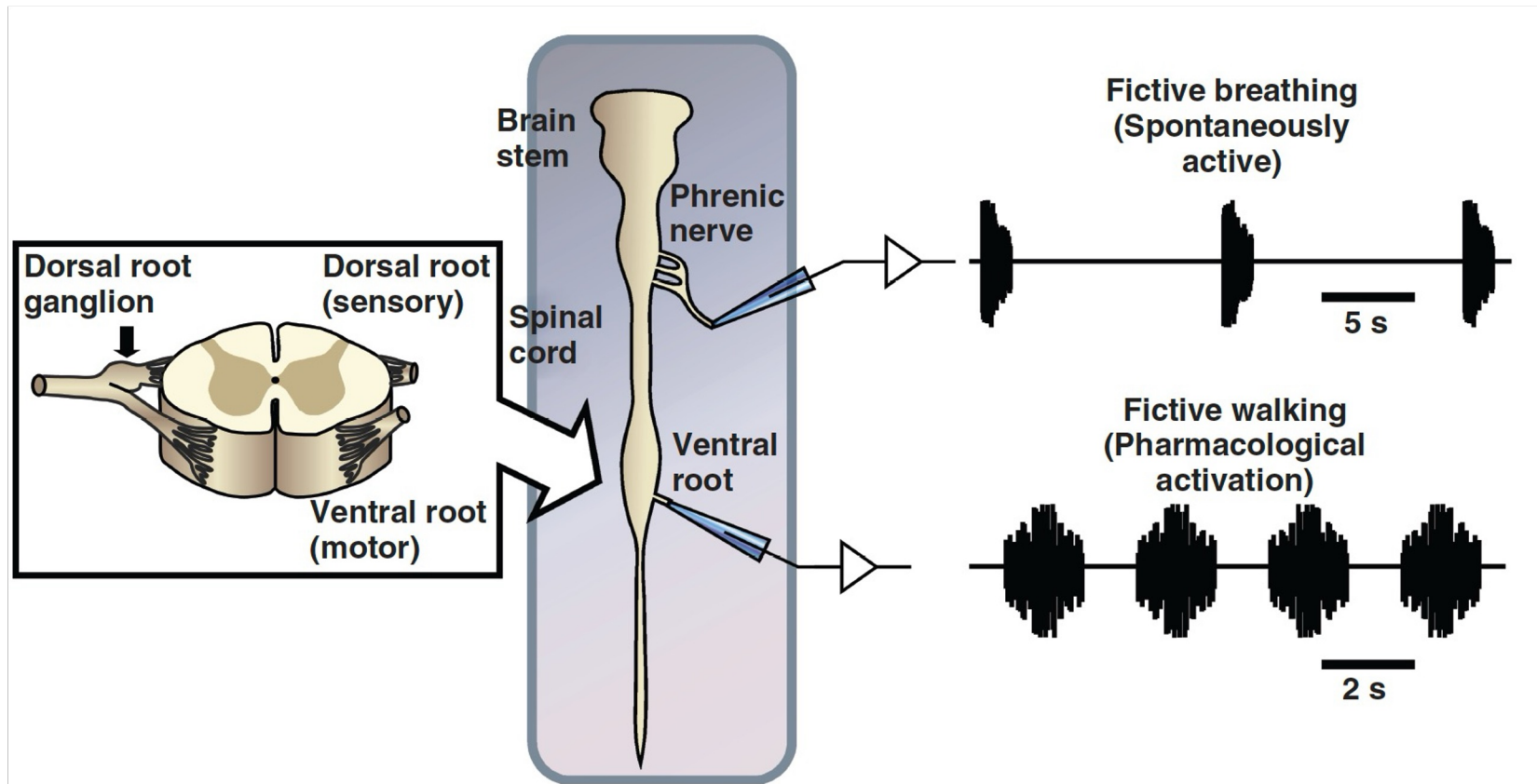
This model allows the body to perform repetitive actions without needing constant sensory or timing input from the body.



SN: sensory neuron; MN: motor neuron; eIN: excitatory interneuron; iIN: inhibitory interneuron

CPGs and fictive behaviors

Fictive behaviors are activity patterns observed in an isolated central nervous system that mimic the motor output seen in a fully intact animal, providing strong evidence for the presence of a CPG.



In an isolated rodent brainstem and spinal cord, rhythmic motor activity can be recorded in the **phrenic nerve**, which would normally drive **diaphragm movements for breathing**.

Similarly, with specific pharmacological activation, this setup can exhibit **walking-like** motor activity, observable in recordings from the **lumbar spinal cord's ventral roots**.

Spatiotemporal organization of motor output

Typically, a single muscle receives innervation from multiple spinal segments, and each spinal segment contributes to the innervation of multiple muscles.

	STER	SPLE	BIC	TRIC	DELTA	DELTP	TRAPS	TRAPI	LD	EST1	EST9	ESL2	RA	OE	OI	ILIO	GM	Gmed	TFL	SART	ADDL	ADDM	RF	Vlat	Vmed	BF	ST	LG	MG	SOL	PERL	PERB	FDB	EDL	TA
C3	X	X					X	X																											
C4		X					X	X		X																									
C5		X	X		X	X				X																									
C6		X	X	x	X	X			X	X																									
C7		X		X					X	X																									
C8		X		X					X	X																									
T1				x						X																									
T2										X	X																								
T3										X	X																								
T4										X	X																								
T5										X	X				x																				
T6										X		X		x																					
T7										X	X	X	X	X	x																				
T8										X	X	X	X	X	x																				
T9										X	X	X	X	X	X																				
T10												X	X	X	X																				
T11												X	X	X	X																				
T12												X	x	x	X																				
L1												X			x	X	x																		
L2																				X	X	x	X	X	X										
L3																				X	X	x	X	X	X										
L4																					X	x	X	X	X										
L5																						x	X	X	X										
S1																						x													
S2																																			

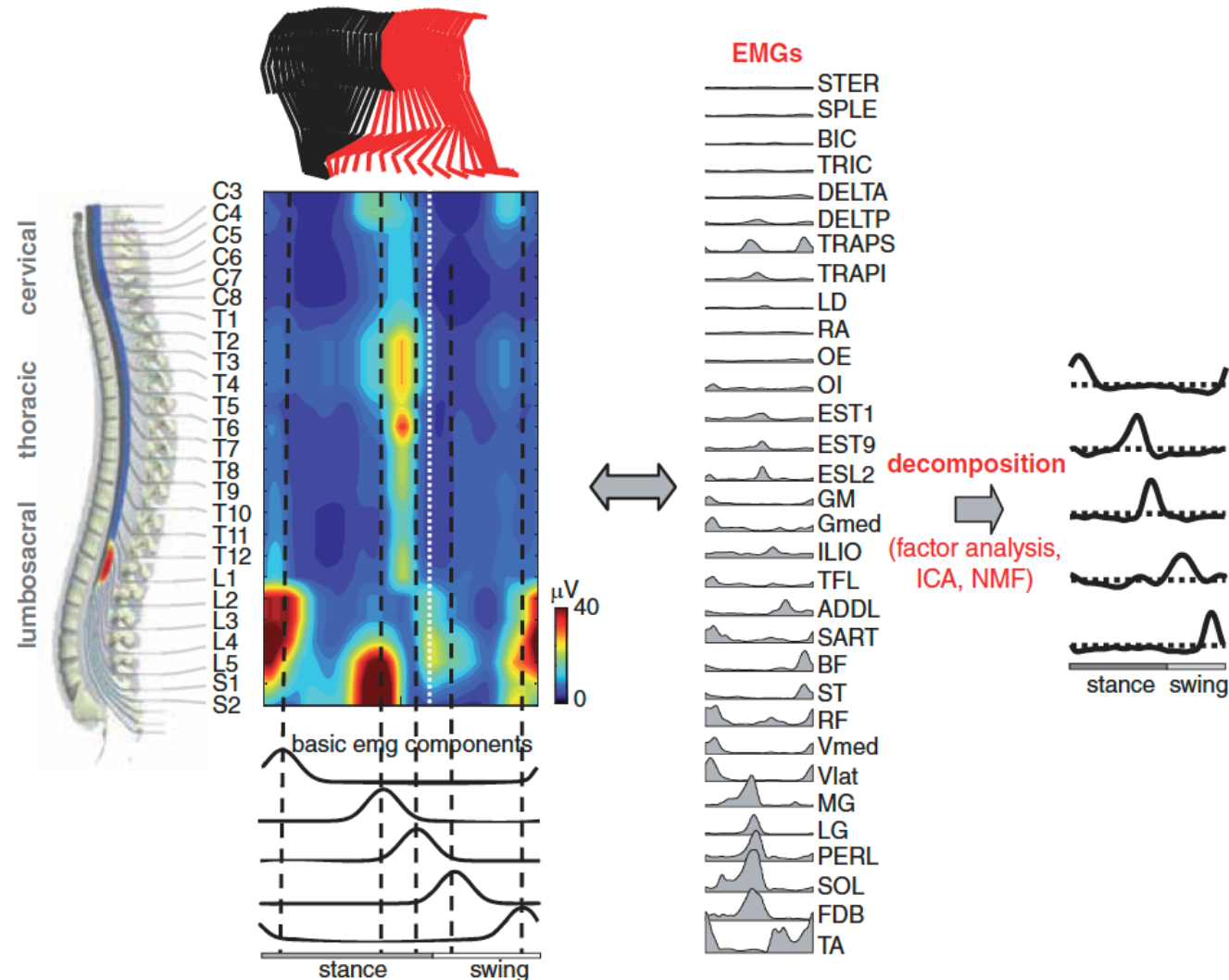
Results are from Kendall et al 1993.

Spatiotemporal organization of motor output

- By recording EMG activity from 32 leg and trunk muscles, we can map muscle activity onto the spinal cord, showing the approximate position of motor neuron pools active during each phase of walking.
- EMG data from each muscle is summed for each spinal segment based on segmental innervation charts, giving a view of MN pool activation.

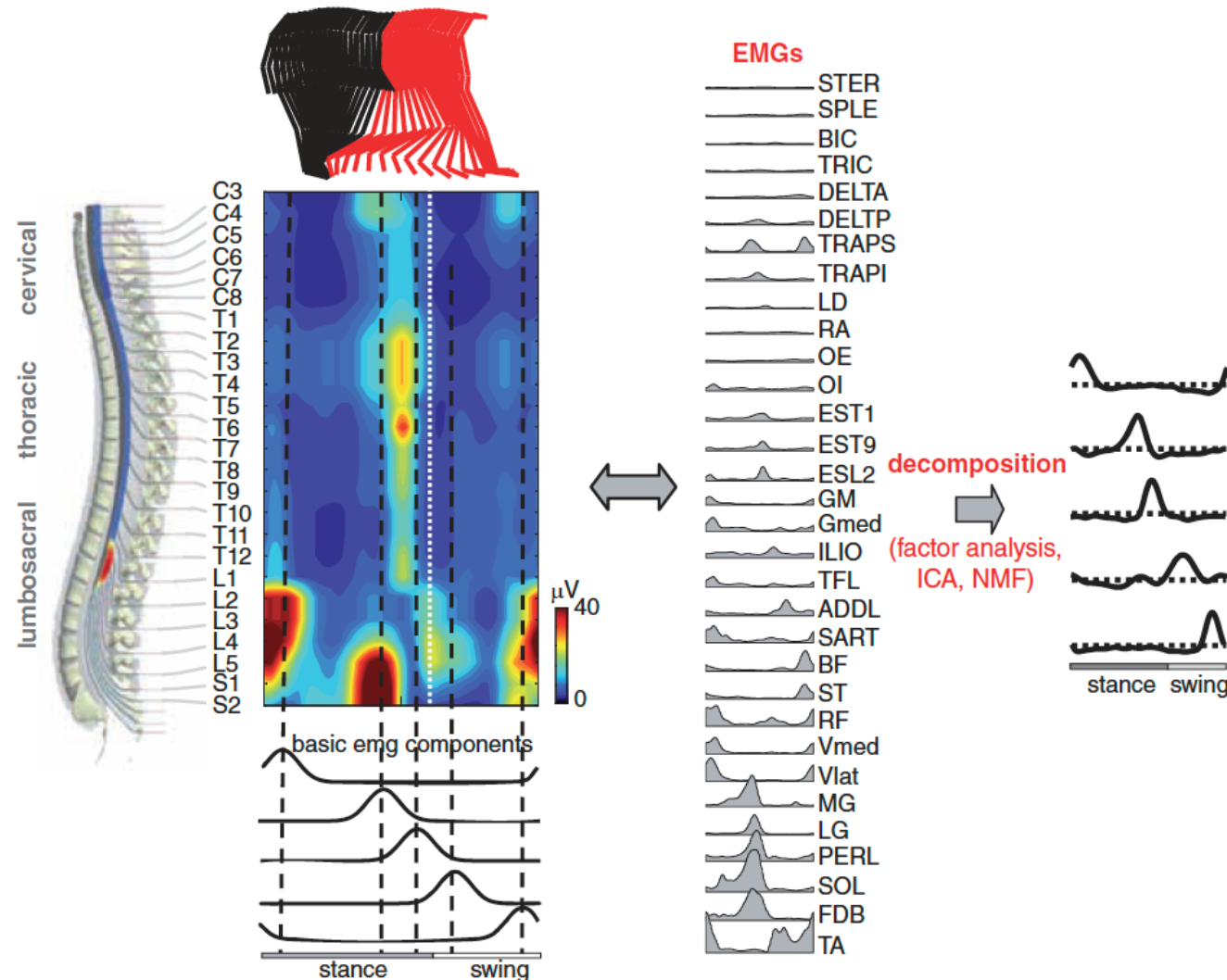
$$S_j = \frac{\sum_{i=1}^{n_j} k_{ij} \cdot EMG_i}{n_j}$$

- n_j : number of EMG signals corresponding to the j^{th} segment
- k_{ij} is a multiplicative coefficient for the i^{th} muscle (equals 1 if marked with "X" and 0.5 if marked with "x")



Spatiotemporal organization of motor output

- Assumes that rectified EMG activity provides an indirect measure of MN firing rates, which generally correlates with motor unit firing.
- While it doesn't reveal the CPG structure directly, this method illustrates how CPG output is distributed across MN pools.
- Allows analysis of MN activation intensity throughout different phases of the gait cycle, offering insight into motor control during walking.



Muscle synergies

- To perform a movement of a joint, the CNS coordinates numerous muscles, each requiring the recruitment of multiple motor units (*motor unit = motor neuron + innervated muscle fibers*) to generate sufficient force for movement.
- To execute a specific movement, the CNS must convert neural information about the limb's direction, amplitude, and velocity – generated by neural activity in cortical and subcortical areas – into commands that activate the appropriate muscles.
- This is an ill-posed problem: the musculoskeletal system has far more degrees of freedom than the movement requires, meaning there are many possible combinations of muscle actions that can achieve the same result. This redundancy raises the question of how the CNS manages and simplifies control over such a versatile system.
- To simplify control, the CNS organizes muscles into **muscle synergies**, grouping **muscles with similar functions to streamline movement**. By activating a smaller number of these synergies, rather than each individual muscle, the CNS achieves efficient motor control, activating groups of muscles that work together for specific movements.

Muscle synergies

- By analyzing surface EMG (sEMG) signals with signal processing techniques and iterative numerical algorithms, researchers can estimate the subdivisions that the CNS might use to streamline muscle control for specific motor tasks.
- Muscle synergies aim to break down the sEMG signal from each muscle's activity into two distinct components. This is a mathematical challenge addressed through Non-Negative Matrix Factorization (NNMF). Here's how it works:
- It's assumed that m muscles generate t time points of electrical potential, which are arranged into an $m \times t$ matrix. The NNMF algorithm decomposes this matrix into two smaller matrices:
 - **W** : the muscle weights matrix ($m \times n$), where n represents the number of synergies needed to reconstruct the original EMG signal.
 - **C** : the activation coefficients matrix ($n \times t$), with n matching the columns of **W** .
- This decomposition helps reveal the synergies that simplify muscle control.

Muscle synergies

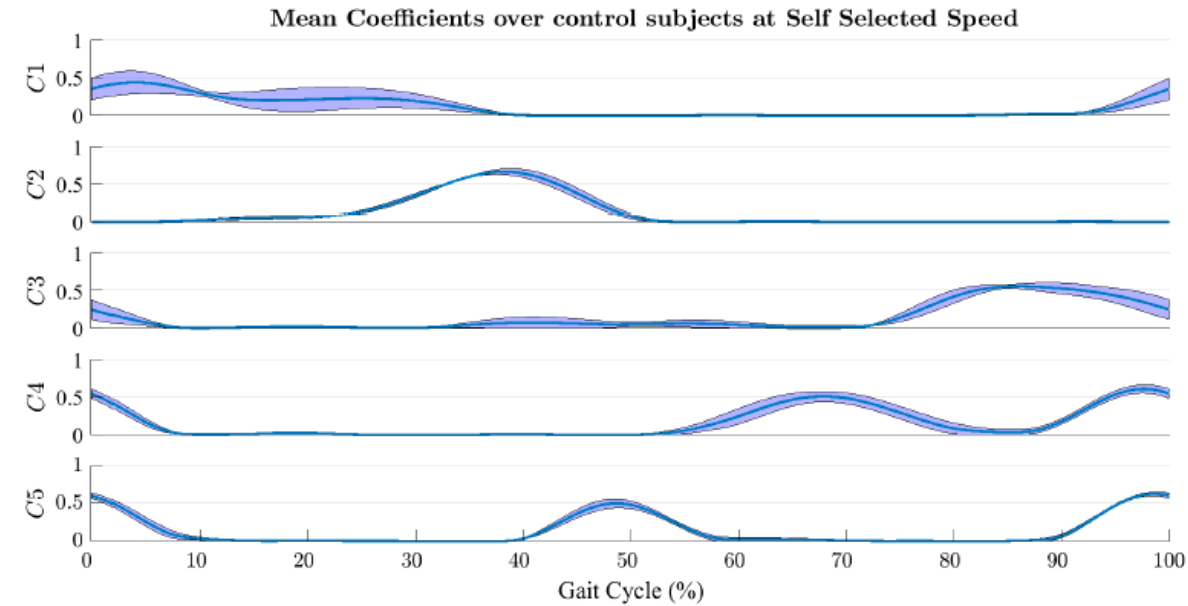
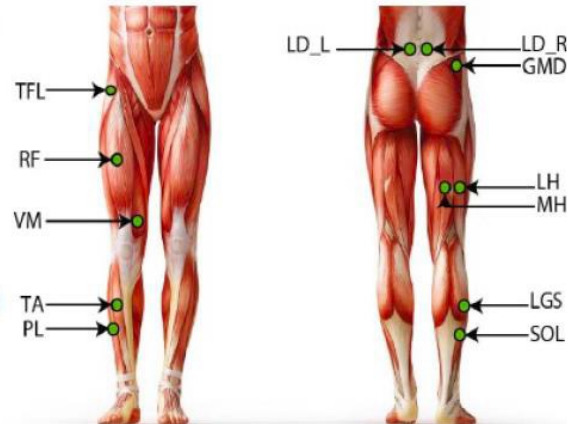
- The weights provide a value between 0 and 1 that indicates each muscle's level of activity within a synergy, independent of time. The activation coefficients vary over time and illustrate the temporal pattern of each muscle synergy's activity throughout the gait cycle. The product of matrices **W** and **C** reconstructs the original sEMG signal patterns, with minimal error.

$$sEMG(t) = \sum_{k=1}^K W_k C_k(t) + e$$

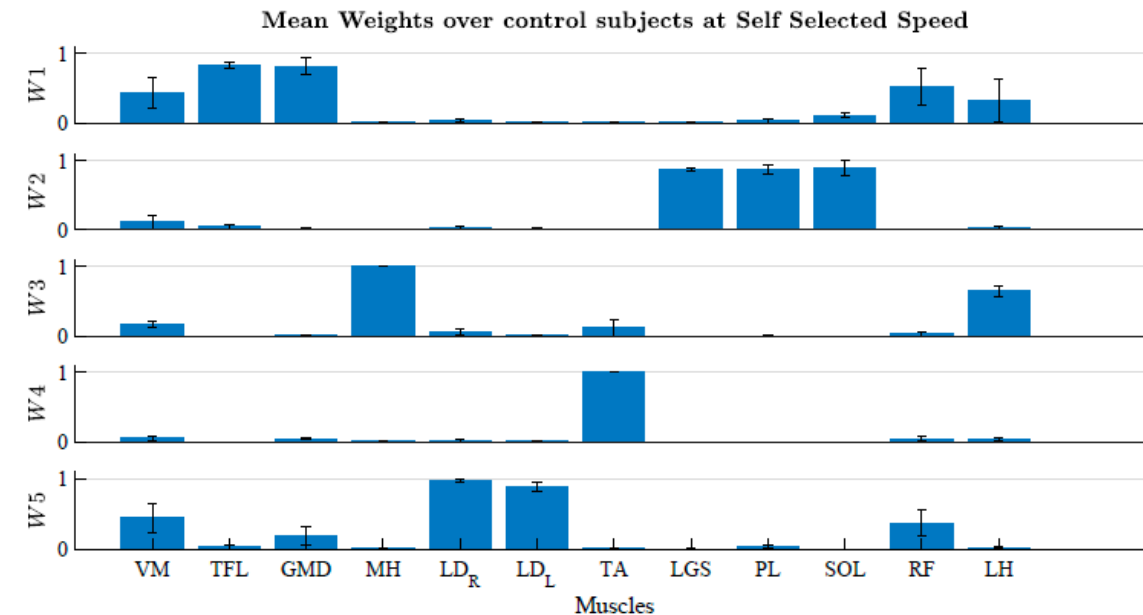
- Longissimus dorsii (@ L4 level) right & left (LD_R, LD_L)
- Gluteus medius (GMD)
- Tensor fasciae latae (TFL)

- Lateral Hamstrings (LH)
- Medial hamstrings (MH)
- Rectus Femoris (RF)
- Vastus Medialis (VM)

- Gastrocnemius Lat. (LGS)
- Soleus (SOL)
- Tibial Anterior (TA)
- Peroneus Longus (PL)



(a)



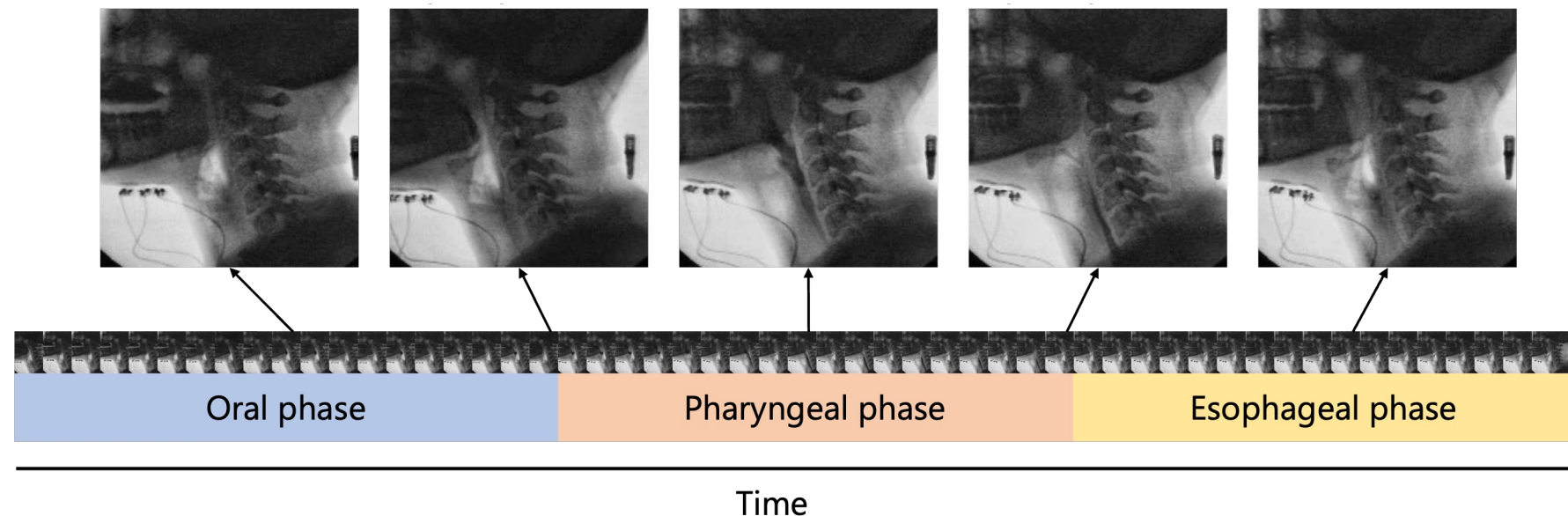
(b)

CPGs for swallowing

Swallowing serves two crucial purposes: to facilitate the digestive process (alimentary function) and to safeguard the upper respiratory tract. This intricate process involves the coordinated activation and inhibition of over 25 pairs of muscles. The initiation of the motor sequence occurs through the stimulation of the internal branch of the superior laryngeal nerve (SLN).

The act of swallowing unfolds in three distinct phases:

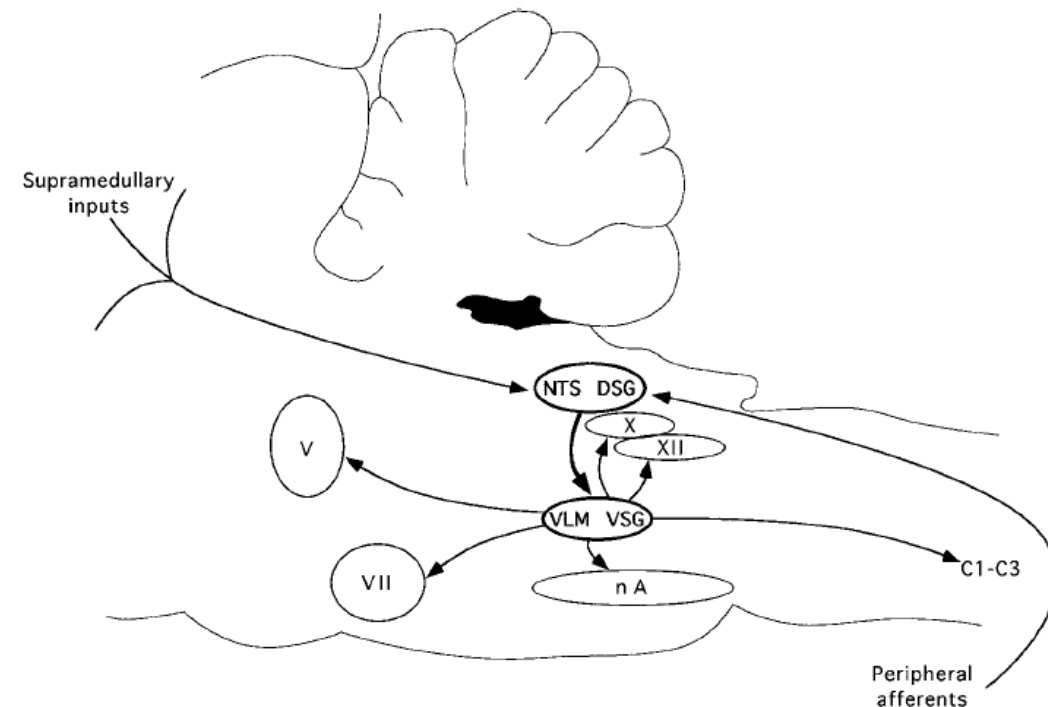
1. The Oral Preparatory Phase (voluntary).
2. The Pharyngeal Phase (involuntary).
3. The Esophageal Phase (involuntary), which includes primary peristalsis of the esophagus.



CPGs for swallowing

Once the pharyngeal phase is underway, the process becomes irreversible. The CPG located in the medulla oblongata orchestrates the sequential and rhythmic patterns of swallowing. This CPG can be subdivided into three systems:

1. The Afferent System, encompassing central and peripheral inputs to the center.
2. The Efferent System, representing outputs from the center, specifically motoneuron pools involved in swallowing.
3. The Organizing System, forming an inter-neuronal network that programs the motor patterns.



CPGs for swallowing

The swallowing CPG involves two main groups of neurons in the medulla oblongata:

Dorsal swallowing group (DSG): Located in the nucleus tractus solitarii (NTS) and nearby reticular formation.

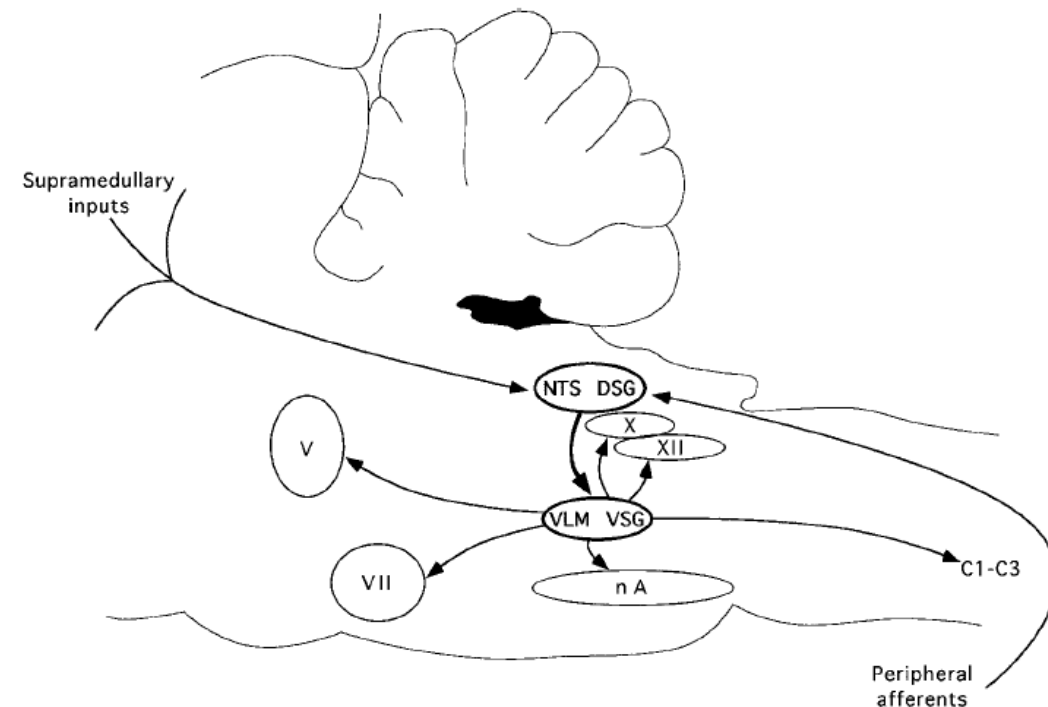
- Contains generator neurons that initiate, shape, and time the sequential or rhythmic swallowing pattern.

Ventral swallowing group (VSG): Situated in the ventrolateral medulla (VLM) next to the nucleus ambiguus.

- Contains switching neurons that distribute the swallowing drive to various motoneuron pools involved in swallowing.

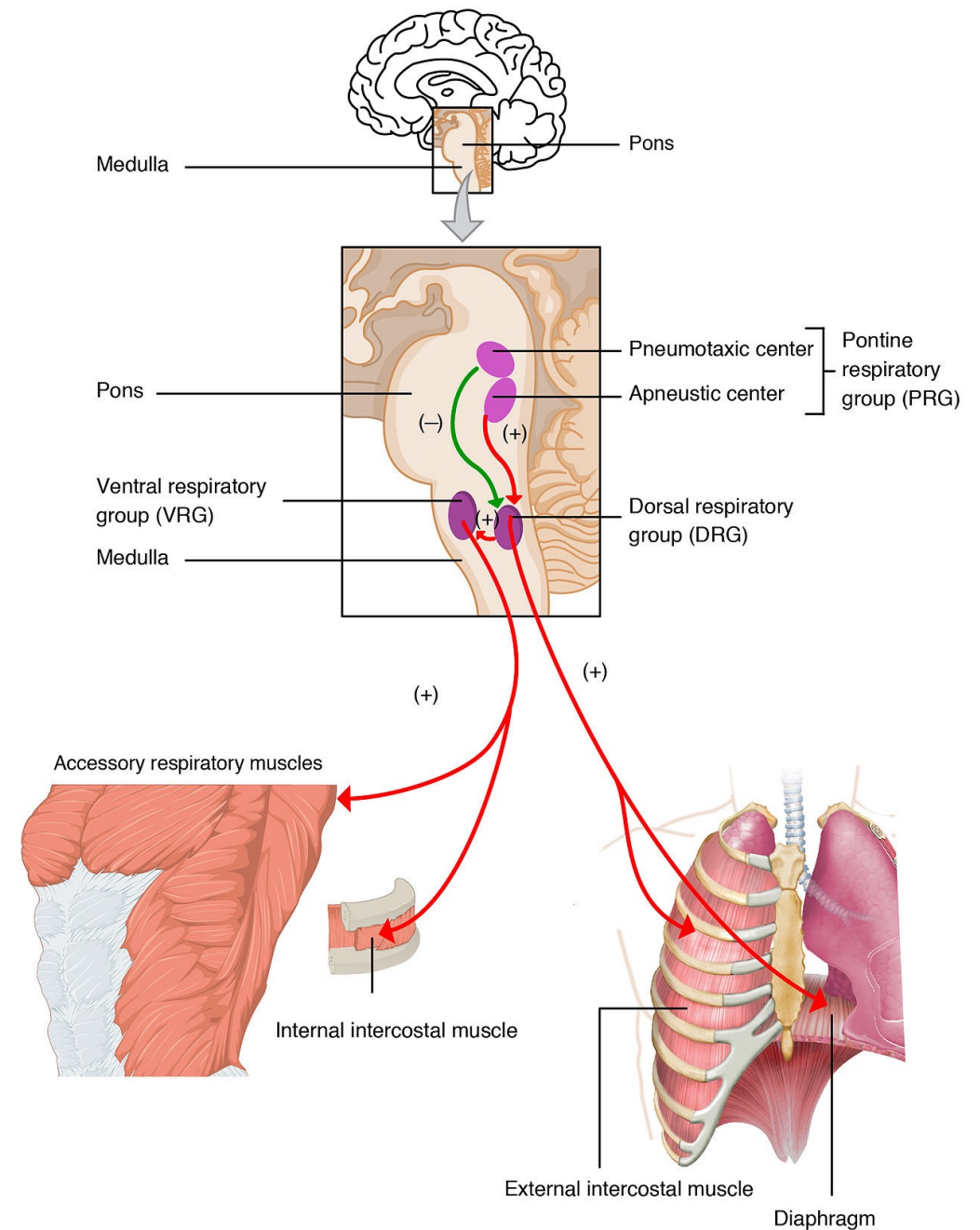
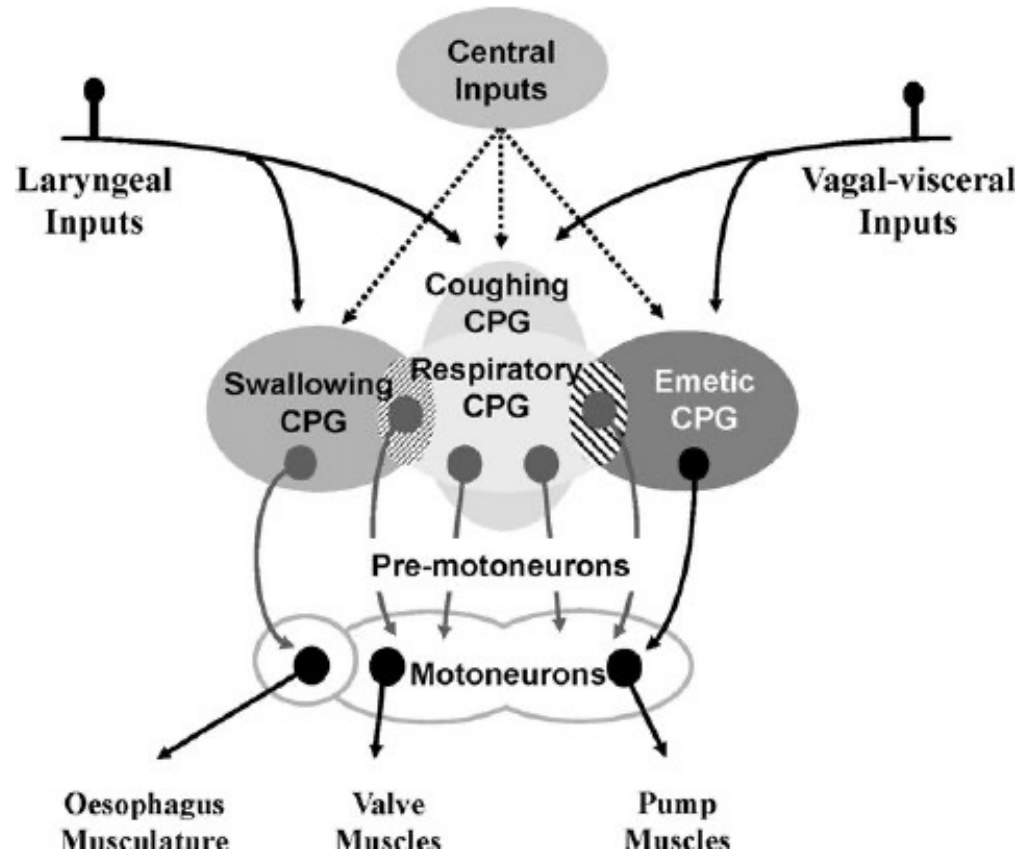
The pathway comprises:

- Peripheral afferent fibers, neurons in the DSG and VSG, and motoneurons.
- Forms an oligosynaptic loop that plays a role in swallowing and related reflexes.



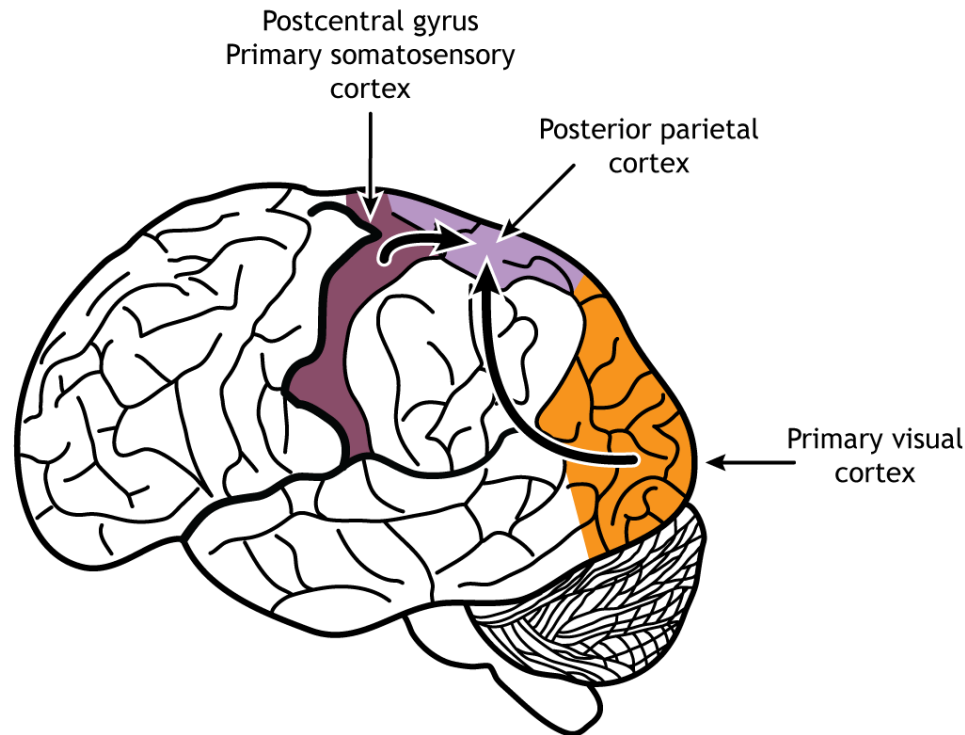
CPGs for breathing

Swallowing neurons are in the same medullary sites where are situated the neurons that belong to the CPGs involved in respiration and cardiovascular regulation.



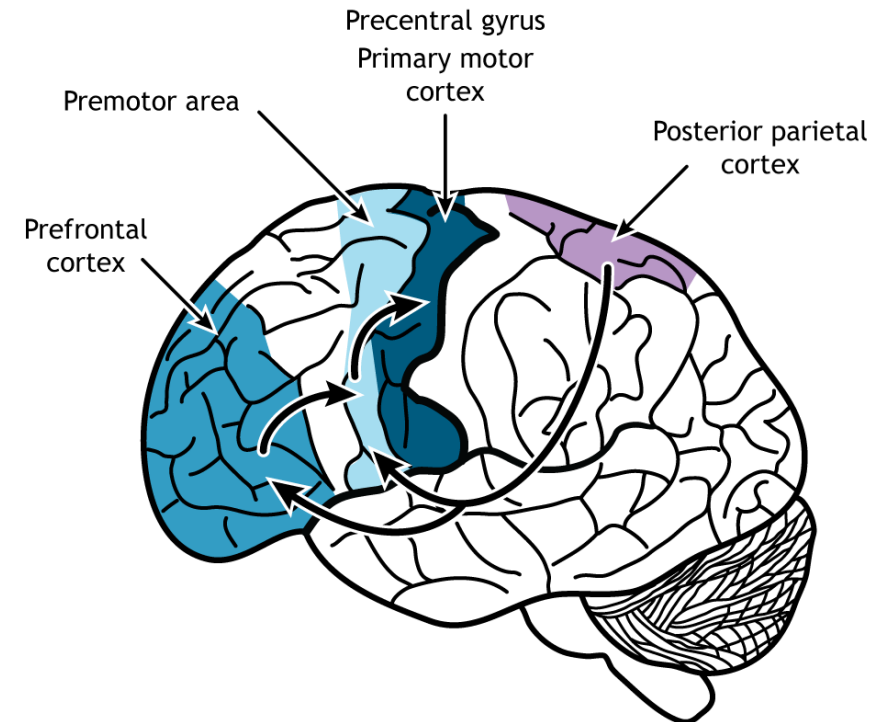
Planning of movement

Much of the cortex is dedicated to planning voluntary movement.



Information from various sensory systems is processed in the posterior parietal lobe.

Projections originate from both the primary somatosensory cortex and the primary visual cortex.



Sensory information from the posterior parietal lobe is processed in the prefrontal cortex and the premotor area, both situated in the frontal cortex.

These regions then relay information to the primary motor cortex located in the precentral gyrus.

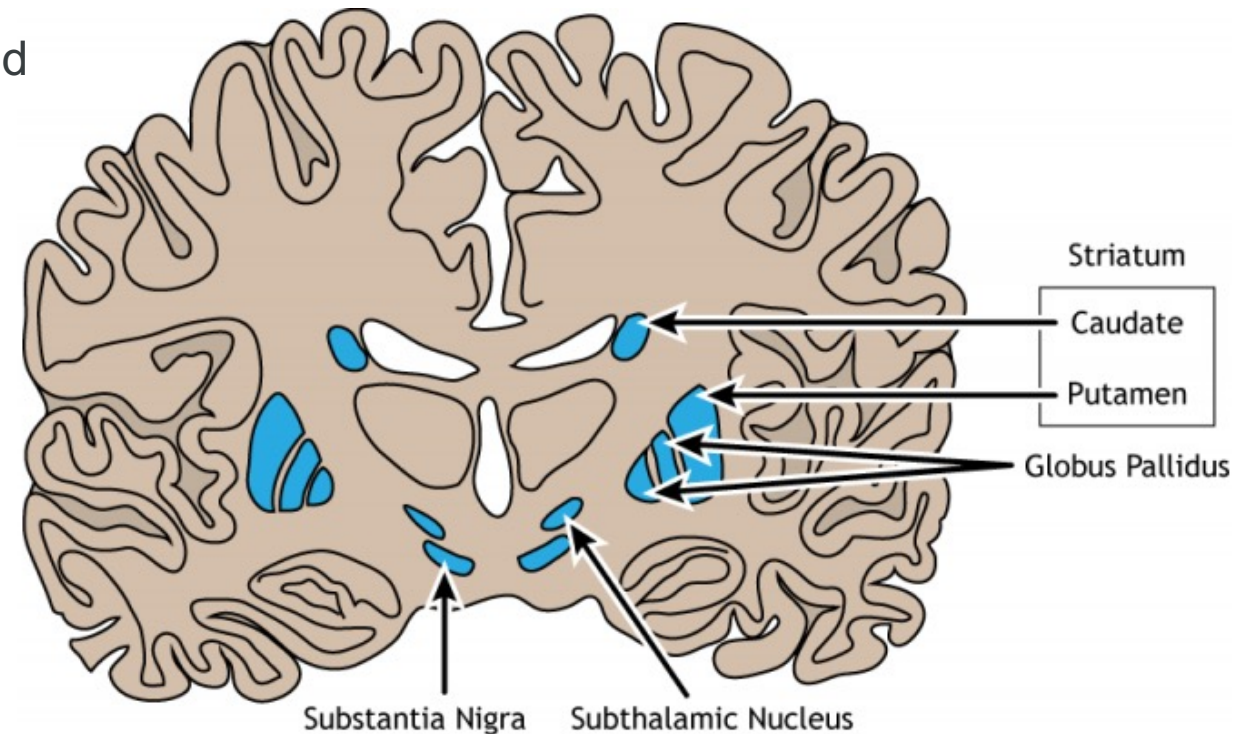
Basal ganglia

The basal ganglia are a group of subcortical nuclei (i.e., groups of neurons that lie below the cerebral cortex)

The basal ganglia are comprised of the:

- Striatum, which consists of the caudate nucleus and the putamen;
- Globus pallidus;
- Subthalamic nucleus;
- Substantia nigra.

The basal ganglia are primarily associated with motor control. For voluntary motor behavior, the basal ganglia are involved in the initiation or suppression of behavior and can regulate movement through modulating activity in the thalamus and cortex.



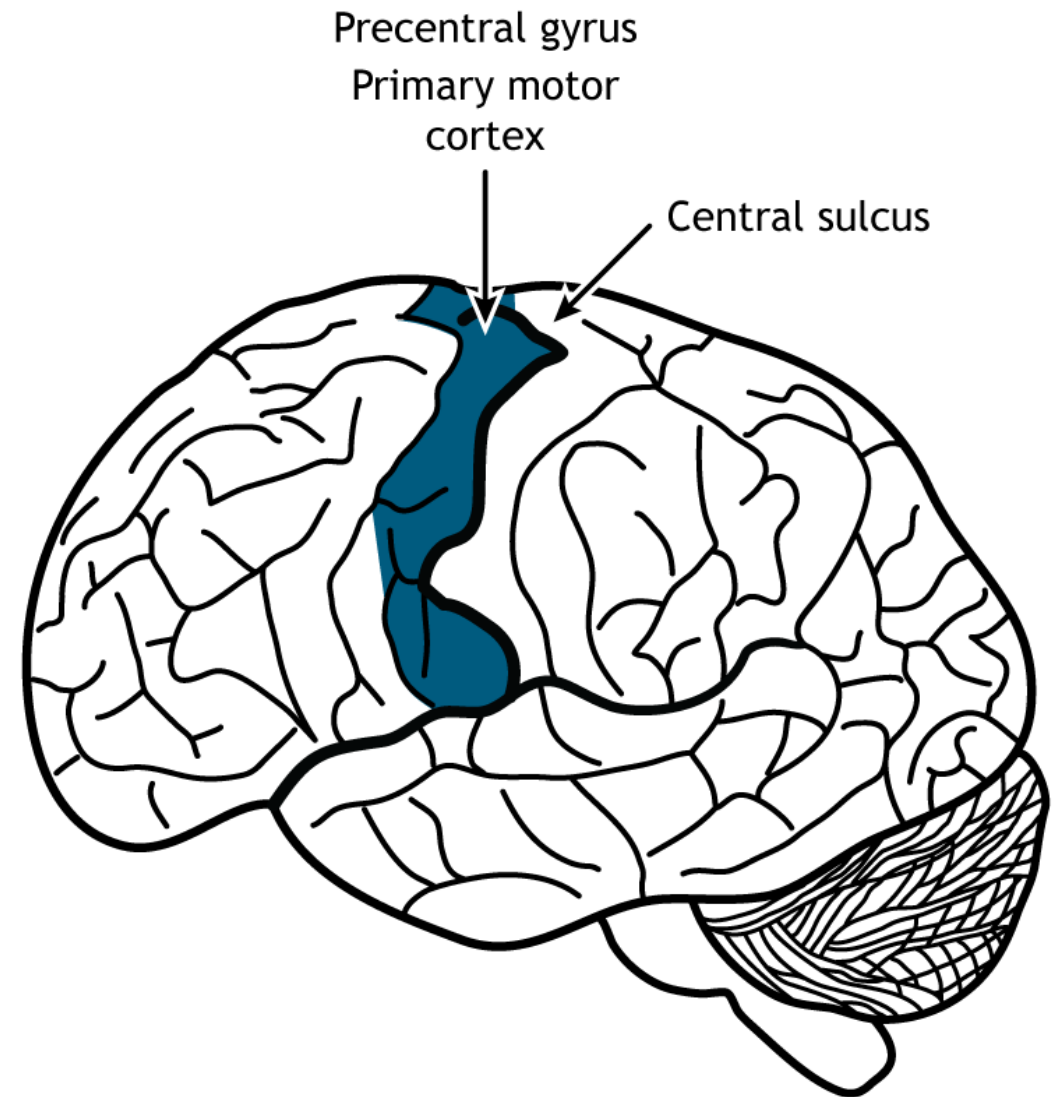
In addition to motor control, the basal ganglia also communicate with non-motor regions of the cerebral cortex and play a role in other behaviors such as emotional and cognitive processing.

Execution of movement

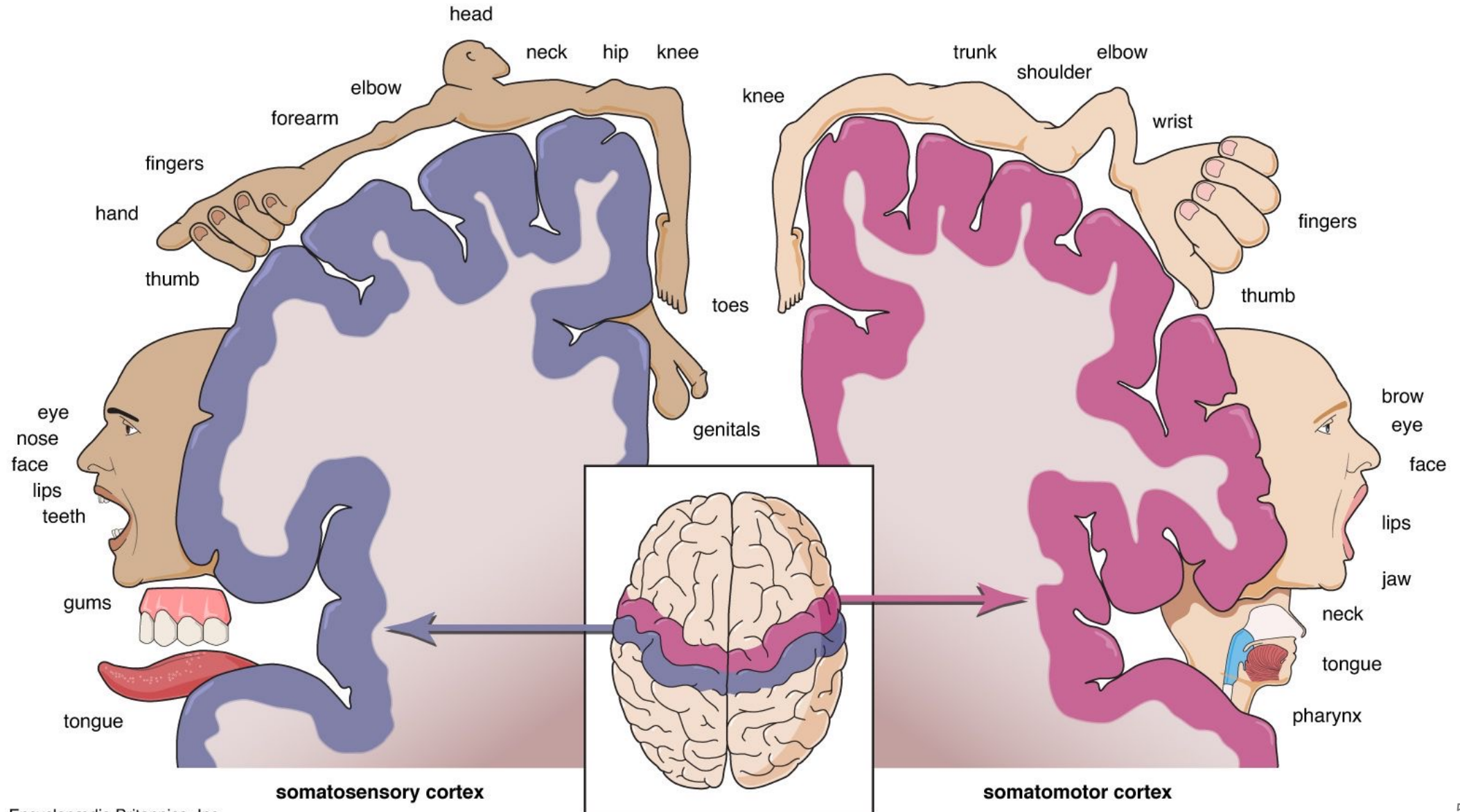
Once the plan for movement has been created, the primary motor cortex is responsible for the execution of that action.

The primary motor cortex lies just anterior to the primary somatosensory cortex in the precentral gyrus located in the frontal lobe.

This results in activation of an upper motor neuron causing excitation or inhibition in different neurons at once, indicating that the primary motor cortex is responsible for movements and not simply activation of one muscle.



Execution of movement



Execution of movement

The motor cortex controls movement using population coding mechanisms.

UMN tuning:

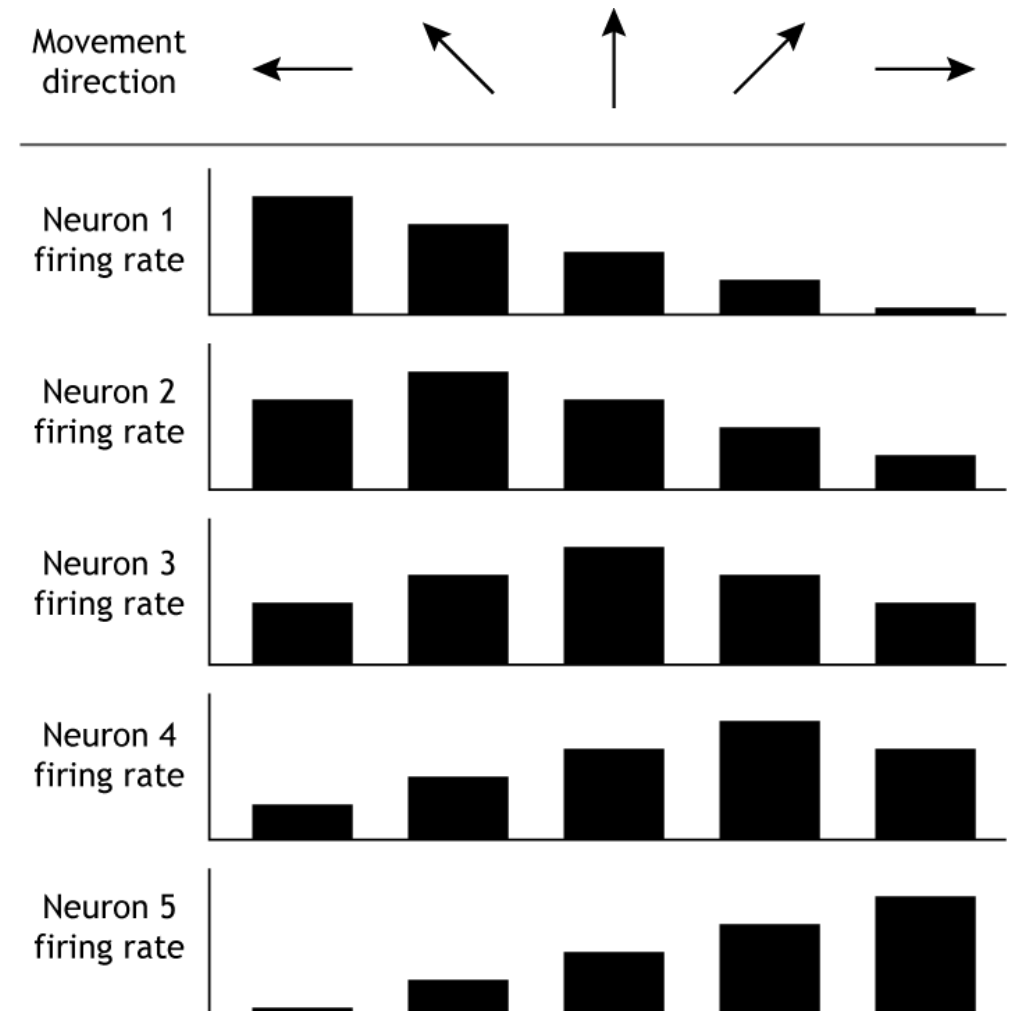
- UMNs are tuned to specific movements in certain directions.
- Their firing rate is highest for movements in a preferred direction, but they also respond to adjacent directions.

Example with monkeys:

- When a monkey is trained to move its hand to the left, neurons tuned for leftward movement become active right before and during the movement.
- Neurons tuned to other directions also fire, but at lower rates.

Collective firing rates:

- The direction of movement is not determined by the firing rate of a single neuron.
- Instead, it is the combined firing rates of a group of neurons that indicate the movement direction.



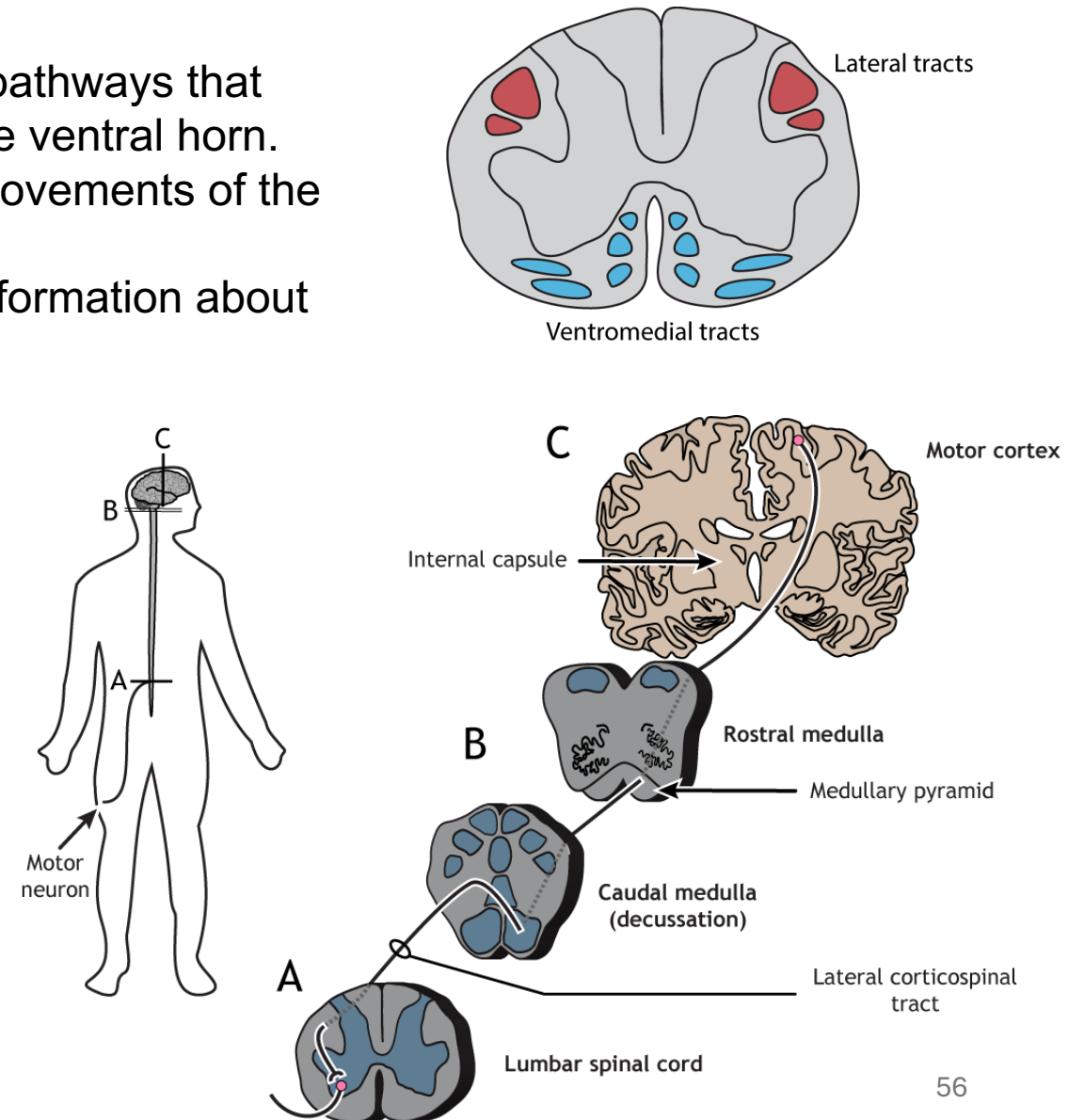
Execution of movement

Descending tracts in the spinal cord: there are multiple pathways that transmit information from the brain to motor neurons in the ventral horn.

- **Lateral tracts:** carry information related to voluntary movements of the arms and legs.
- **Ventromedial pathways:** responsible for conveying information about posture and balance.

Corticospinal tract (the largest of the lateral pathways)

- Transmits information directly from the motor and premotor cortices to motor neurons in the spinal cord.
- Axons travel through the brainstem and cross the midline at the base of the medulla.
- The right cortex processes information for the left side of the body, and vice versa.
- In the spinal cord, axons move through the lateral column and synapse in the ventral horn on motor neurons, primarily innervating distal muscles.



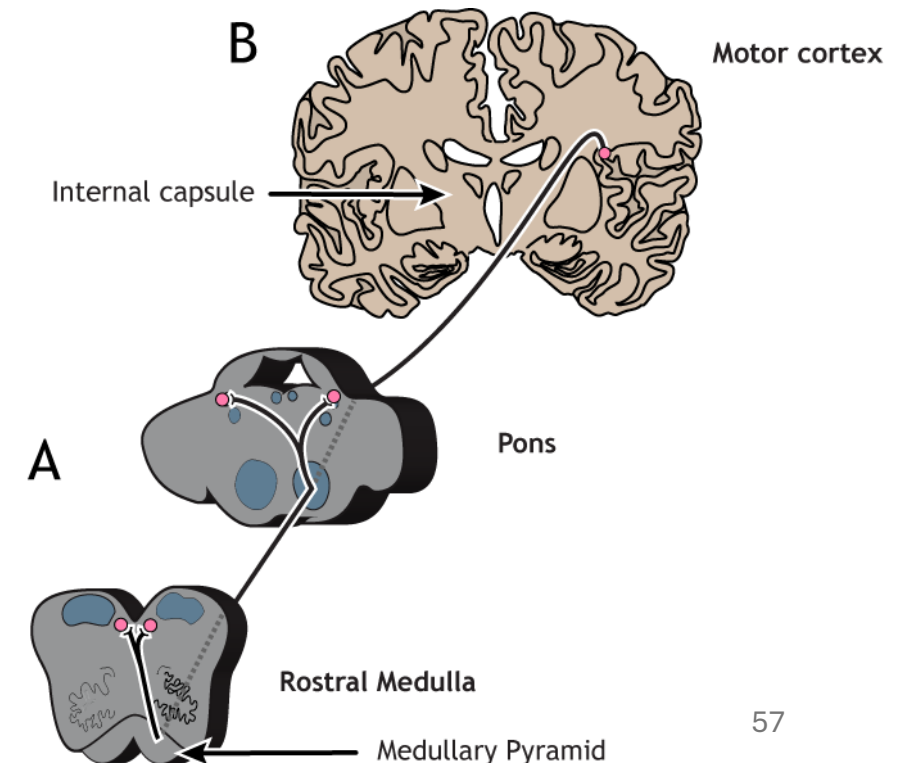
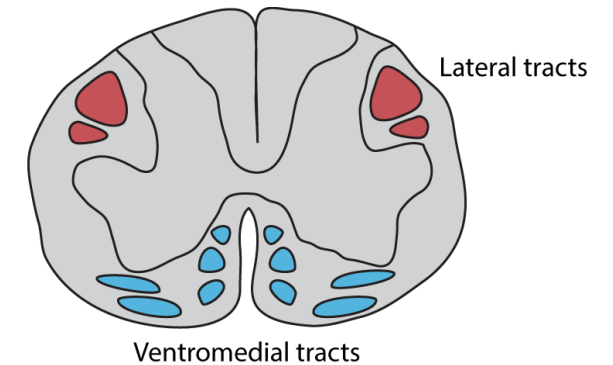
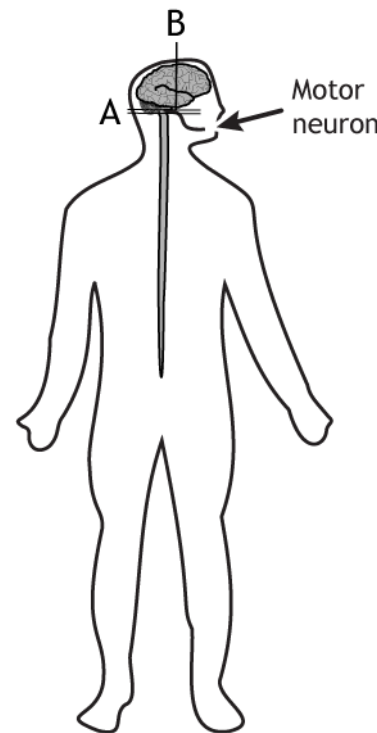
Execution of movement

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Corticobulbar Tract:

- Another lateral tract that conveys motor information to cranial nerves controlling facial movements.
- Travels ipsilaterally from the cortex into the brainstem.
- Branches off at the appropriate cranial nerve level in the pons or medulla.
- Innervates cranial nerve neurons bilaterally.



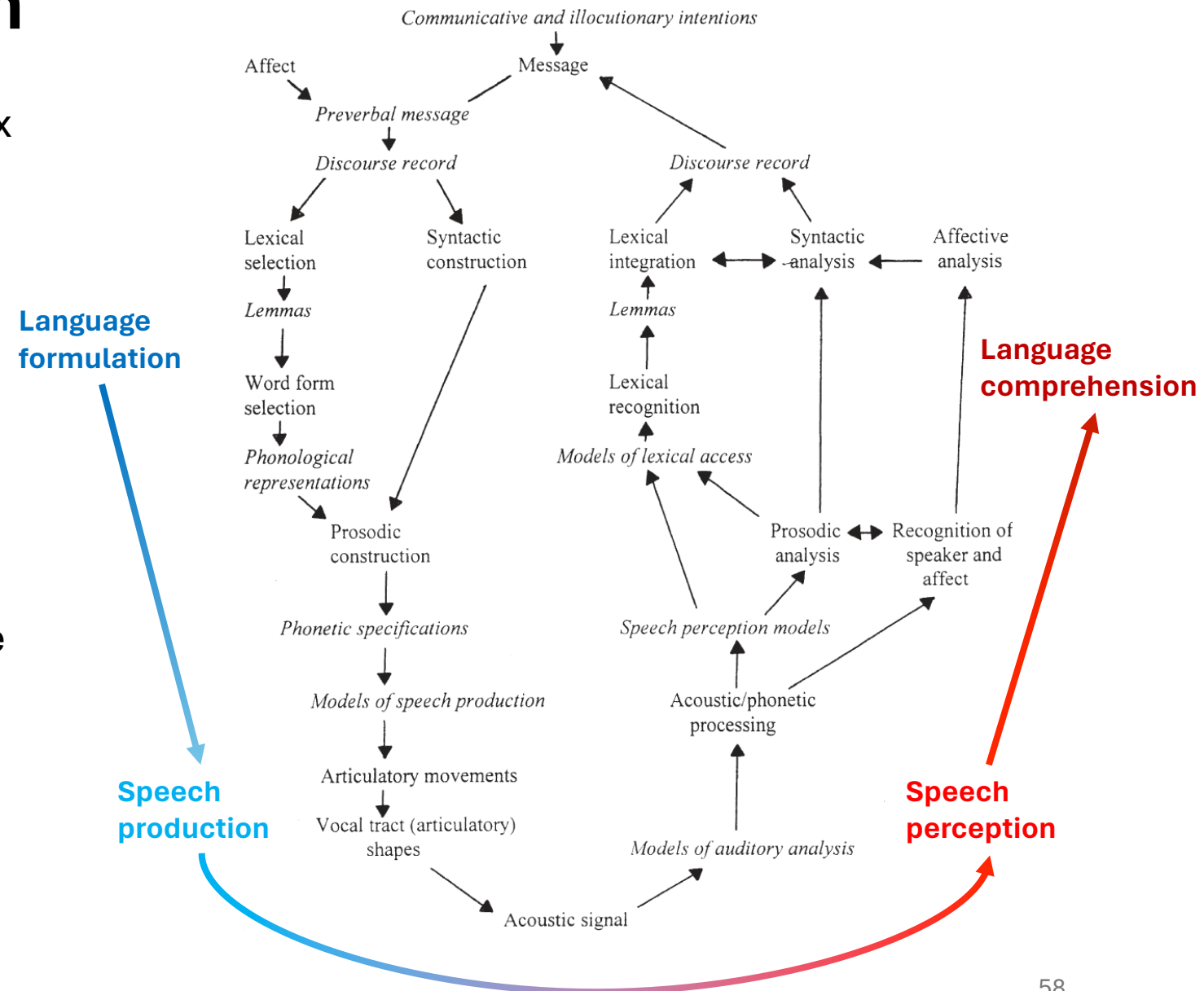
Motor control of speech

Speech motor control involves the complex systems and strategies that govern how speech is produced.

The process begins with a phonologic representation - essentially, a planned sequence of phonemes that forms the foundation of spoken language.

This phonological input is then transformed into precise articulatory movements, which generate an acoustic signal that conveys the intended message to listeners.

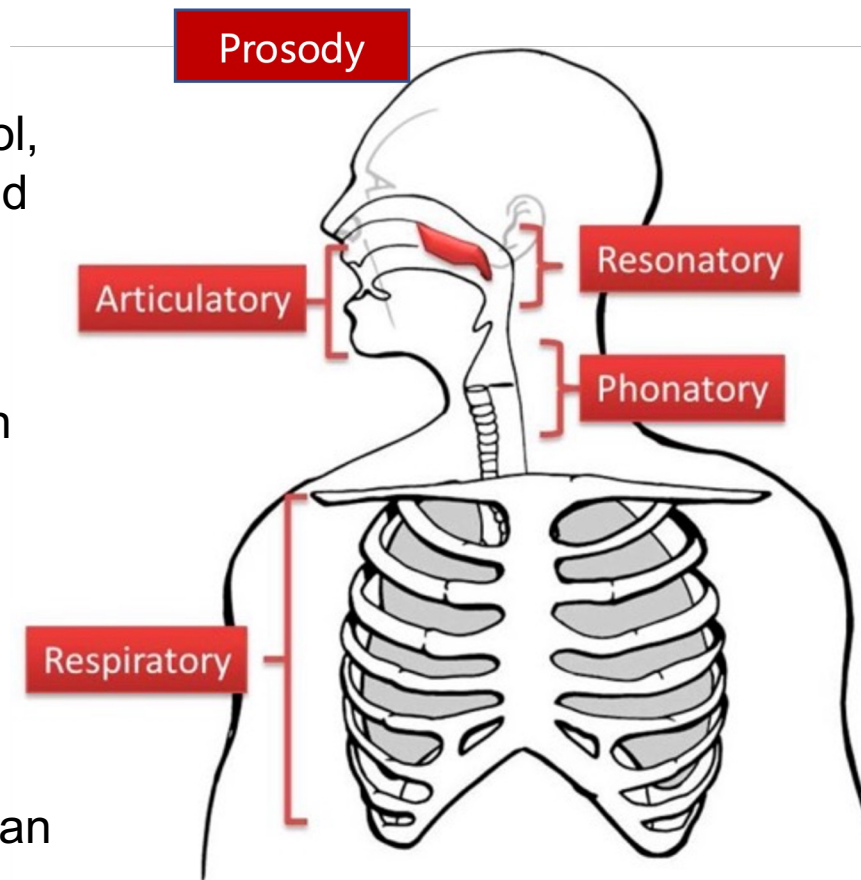
In this way, speech motor control serves as an essential link between language formulation and the creation of the audible speech signal.



Motor control of speech

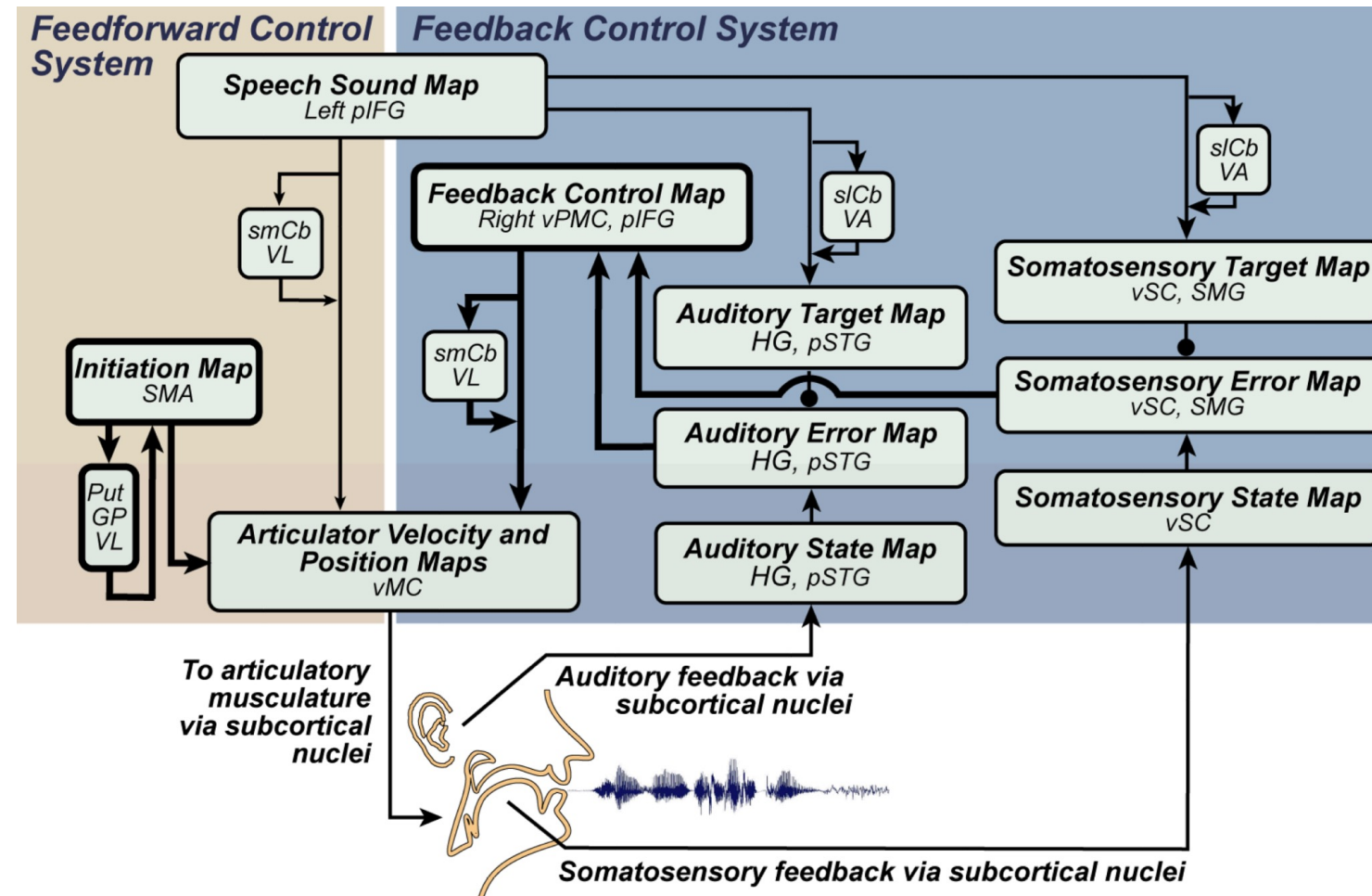
Speech production is one of the most complex human activities:

- It requires the coordination of multiple systems, including cognitive planning, phonological processing, sensory feedback, respiratory control, and precise motor execution, all working together to create coherent and intelligible speech.
- It uses more motor fibers than any other human mechanical activity. Despite its complexity, it generates an acoustic signal that conveys both linguistic meaning and emotional tone to listeners.
- There are no universally accepted animal models for spoken language, because no other species exhibits vocal behaviors that embrace the complexity and power of human speech.
- The cognitive, sensory, and motor requirements of speech production can be affected by normal aging processes and by various diseases that become more prevalent as people grow older.



Neural control of speech

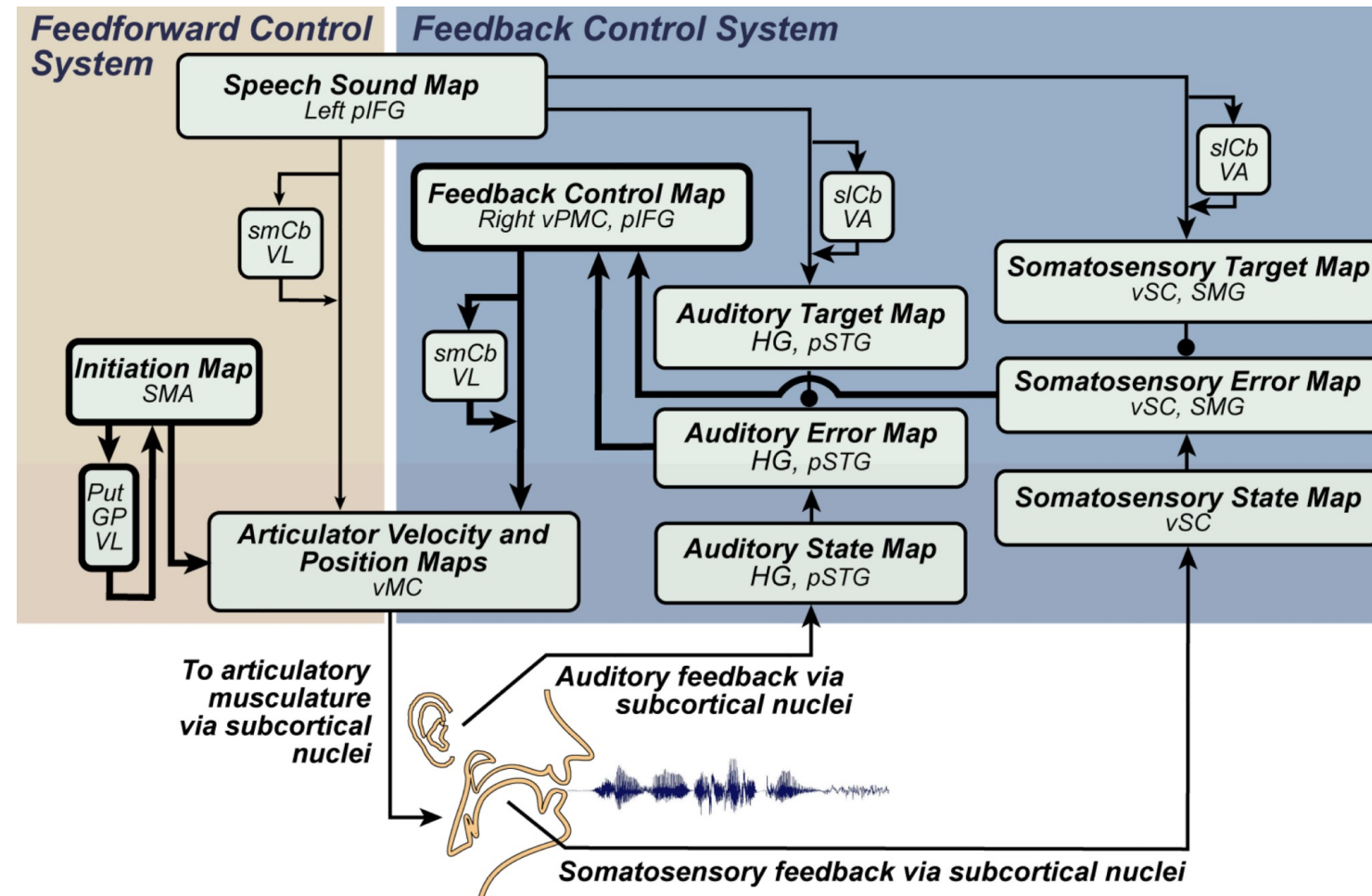
- The **Directions Into Velocities of Articulators (DIVA)** model is an adaptive neural network that describes the sensorimotor interactions involved in articulator control during speech production.
- The model combines **feedforward** and **feedback** control systems to learn how to control a simulated vocal tract.
- Sensory inputs are converted into velocity adjustments for speech articulation. These feedback-based velocity commands integrate with feedforward commands in the articulator position map, allowing for refined speech production.



GP = globus pallidus; **HG** = Heschl's gyrus; **pIFG** = posterior inferior frontal gyrus; **pSTG** = posterior superior temporal gyrus; **Put** = putamen; **sICB** = superior lateral cerebellum; **smCB** = superior medial cerebellum; **SMA** = supplementary motor area; **SMG** = supramarginal gyrus; **VA** = ventral anterior nucleus of the cerebellum; **VL** = ventral lateral nucleus of the thalamus; **vMC** = ventral motor cortex; **vPMC** = ventral premotor cortex; **vSC** = ventral somatosensory cortex.

Neural control of speech

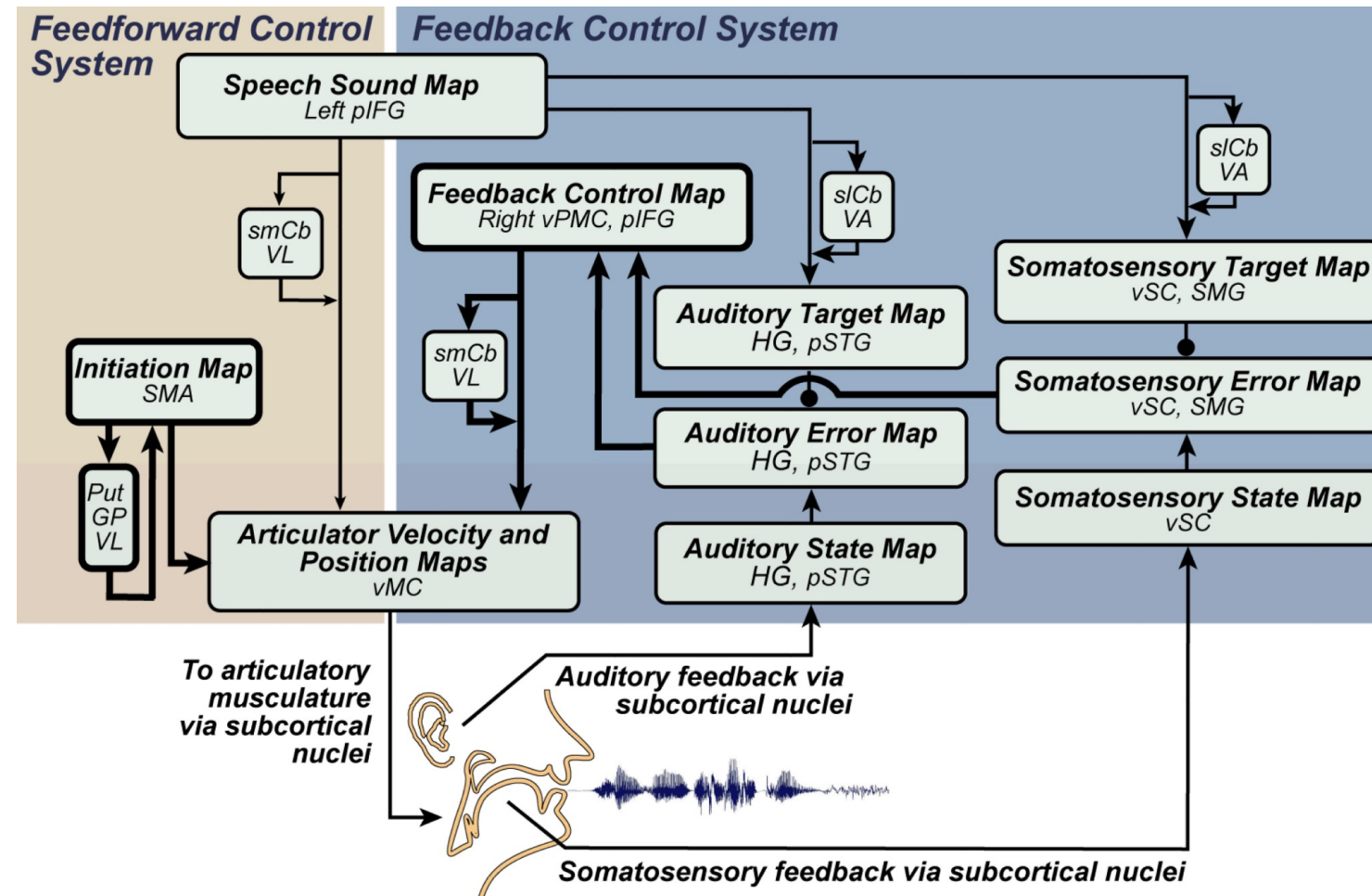
- **Blocks:** groups of neurons that, when activated, collectively "represent" a distinct component of speech (e.g., phoneme, syllable, motor command).
- **Arrows:** mappings (i.e., transformations from one neural representation to another) that occur through connections where cell activations in one neural map are processed and sent to another map via synapses.
- **Learning phase (babbling):** random movements of the speech organs generate sensory feedback (touch, position, and sound) used to learn the synaptic connections, between neural representations.
- Once babbling is complete, the model can then rapidly learn to produce new sounds based on audio examples



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Feedforward control

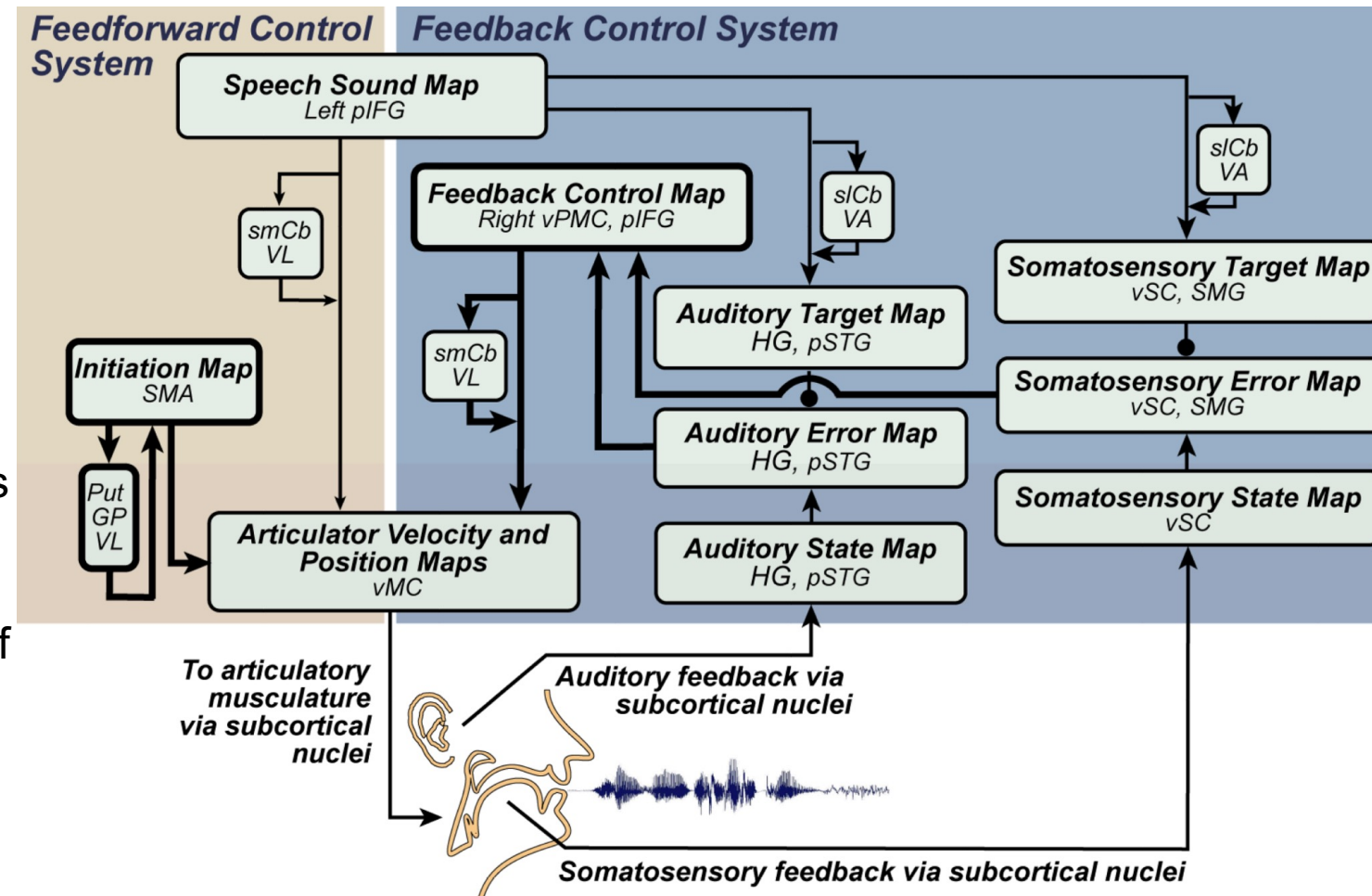
- Each unique sound that a speaker hears is stored as a specific cell in the **Speech sound map** (located in the left premotor cortex).
- **Movement control**: commands from the speech sound map are sent to the motor cortex, directing coordinated movements of speech muscles like the lips, jaw, and tongue to produce sounds.
- The **left side** of the brain is primarily responsible for speech, explaining why damage here often leads to speech difficulties.



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Feedback control

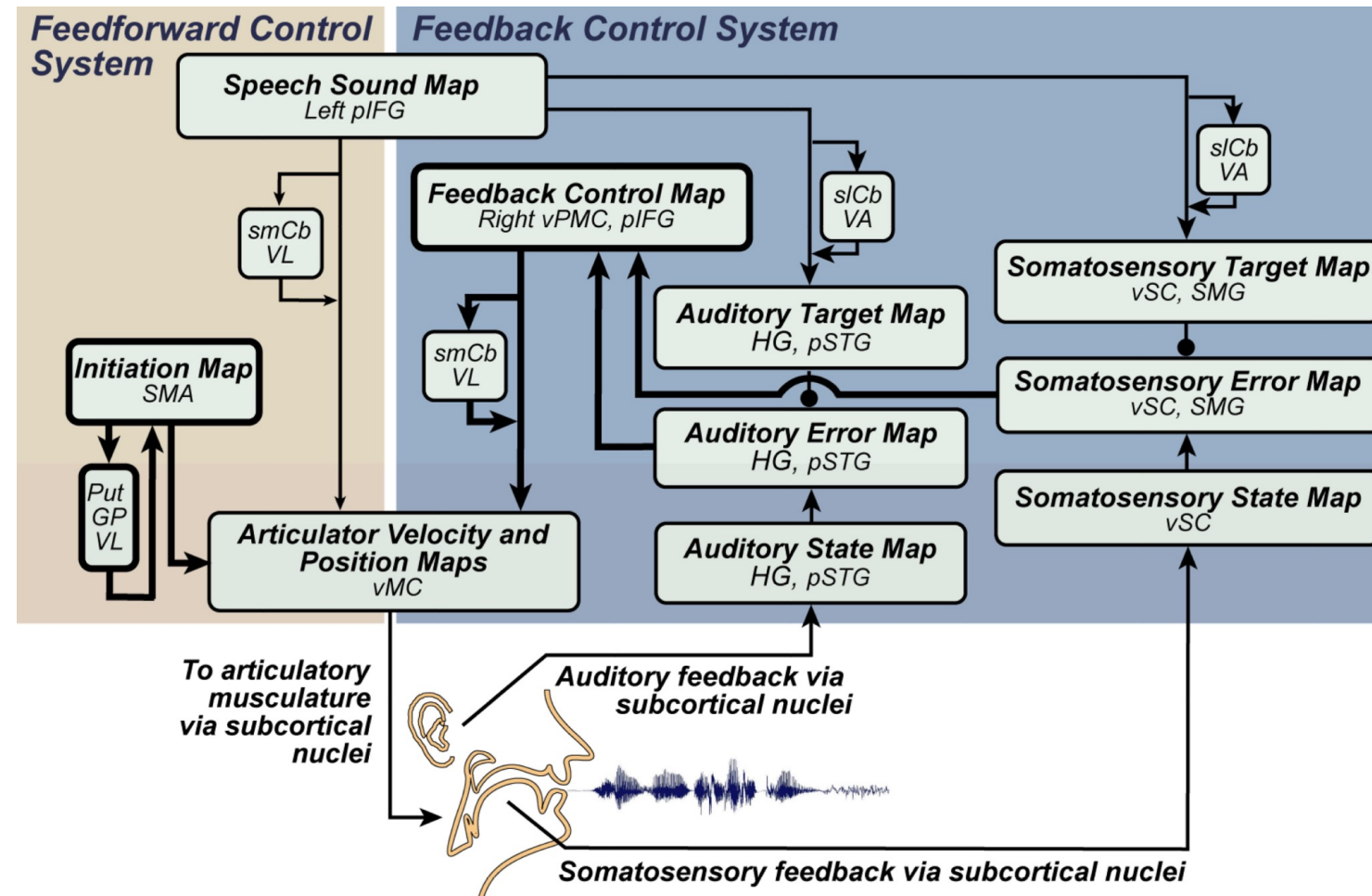
- The speech sound map also sends projections to auditory and somatosensory target maps. This dual projection allows encoding sensory expectations linked to the active speech sound.
- Auditory targets** are given by three pairs of inputs to the auditory target map that describe the upper and lower bounds for the 1st, 2nd, and 3rd formant frequencies of the speech sound being produced.
- Somatosensory targets** consist of a vector that describes the expected proprioceptive and tactile feedback for the sound being produced.



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Feedback control

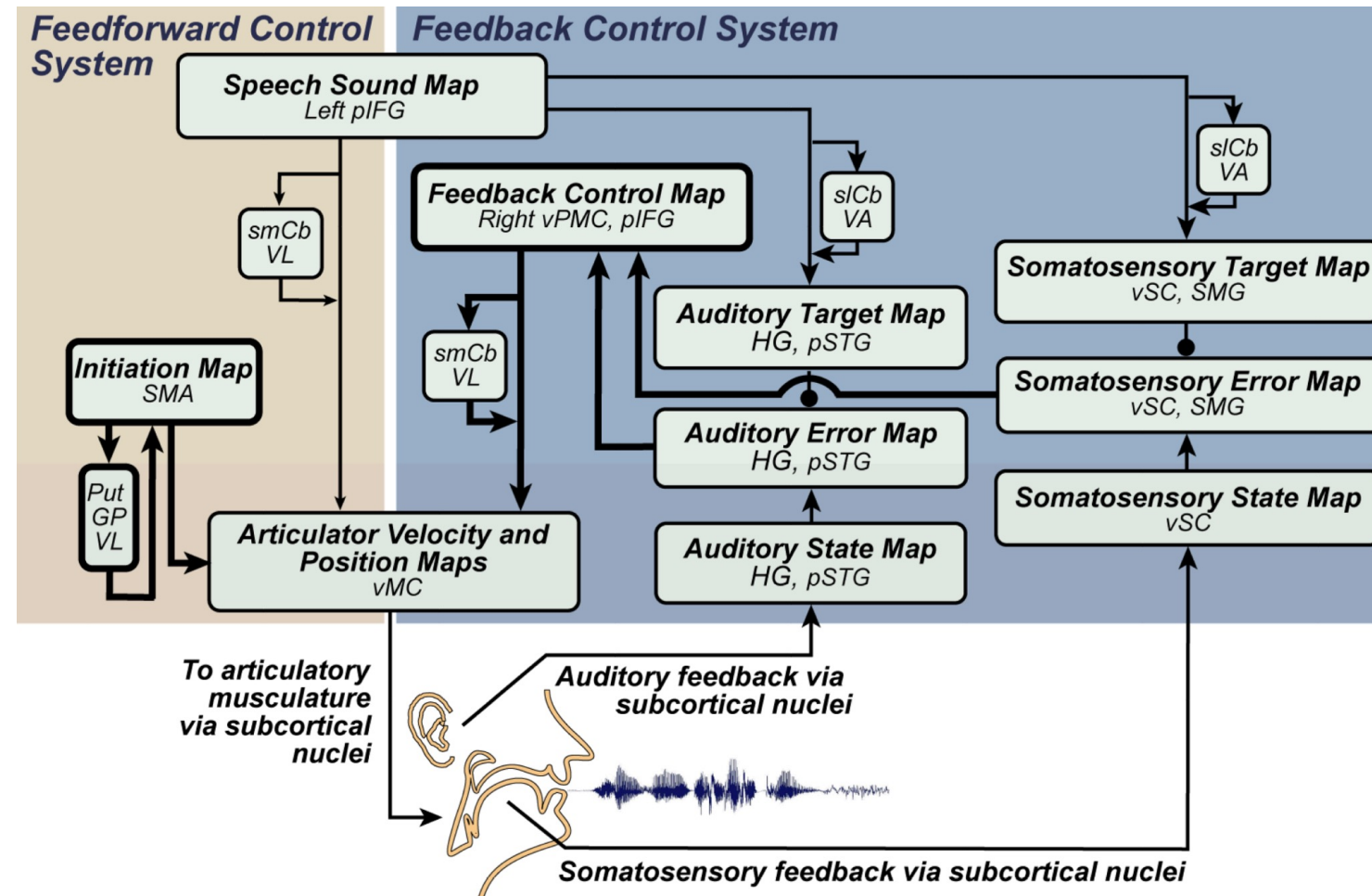
- The auditory and somatosensory target maps send inhibitory inputs to auditory and somatosensory error maps, respectively (hypothesized to be in two locations along the posterior superior temporal gyrus).
- Sensory error maps receive excitatory signals from sensory state maps in the auditory and somatosensory cortex, which relay an estimate of the current sensory state.



GP = globus pallidus; **HG** = Heschl's gyrus; **pIFg** = posterior inferior frontal gyrus; **pSTg** = posterior superior temporal gyrus; **Put** = putamen; **slCB** = superior lateral cerebellum; **smCB** = superior medial cerebellum; **SMA** = supplementary motor area; **SMG** = supramarginal gyrus; **VA** = ventral anterior nucleus of the cerebellum; **VL** = ventral lateral nucleus of the thalamus; **vMC** = ventral motor cortex; **vPMC** = ventral premotor cortex; **vSC** = ventral somatosensory cortex.

Feedback control

- This setup allows error maps to reflect the difference between expected and actual sensory feedback during speech.
- When sensory feedback deviates from the target region, an error signal is directed to the feedback control map in the right frontal/ventral premotor cortex.
- This feedback control map then translates the auditory and somatosensory errors into corrective motor commands, sending these commands to the articulator velocity map in the motor cortex on both sides.



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1. Background on nervous system injuries and disorders

1.1 Overview of the nervous system

- Central and peripheral nervous system
- Neuronal pathways: sensory and motor circuits

1.2 Principles of motor control

- Motor control of human movement
- Motor control of speech

1.3 Injuries and disorders of the nervous system

- Types and causes of injuries
- Neurodegenerative diseases
- Overview of neurological deficits related to motor and speech control

Types of neurological disorders

Based on the pathology

- **Neurodegenerative disorders:** progressive deterioration of neurons, often leading to worsening cognitive and/or motor symptoms.
- **Acute or traumatic:** Sudden onset, often due to an injury or acute event.
- **Infectious:** Caused by bacteria, viruses, fungi, or parasites that invade the nervous system.
- **Autoimmune and Inflammatory Disorders:** Involving immune system attacks on nervous tissue.
- **Genetic and Congenital Disorders:** Arise from inherited or developmental abnormalities.

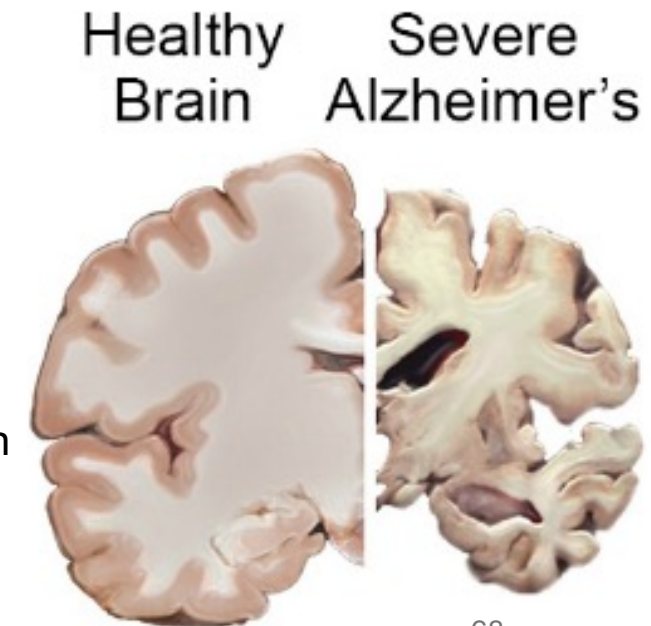
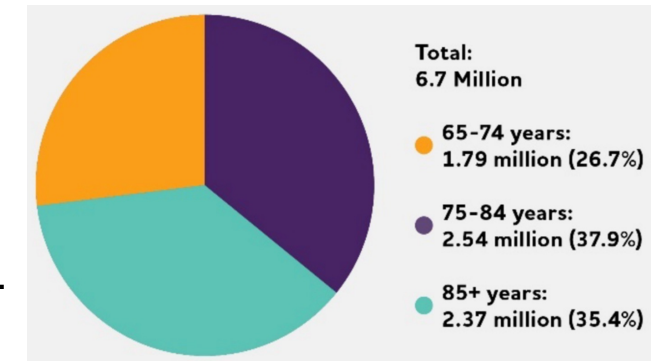
Based on Primary Site of Impact

- **CNS:** disorders primarily affecting the brain or spinal cord.
- **PNS:** disorders affecting nerves outside the brain and spinal cord.
- **Autonomic Nervous System:** disorders affecting involuntary bodily functions.
- **Motor Neurons:** disorders primarily targeting motor neurons, impacting voluntary muscle movement.

Alzheimer's disease (AD)

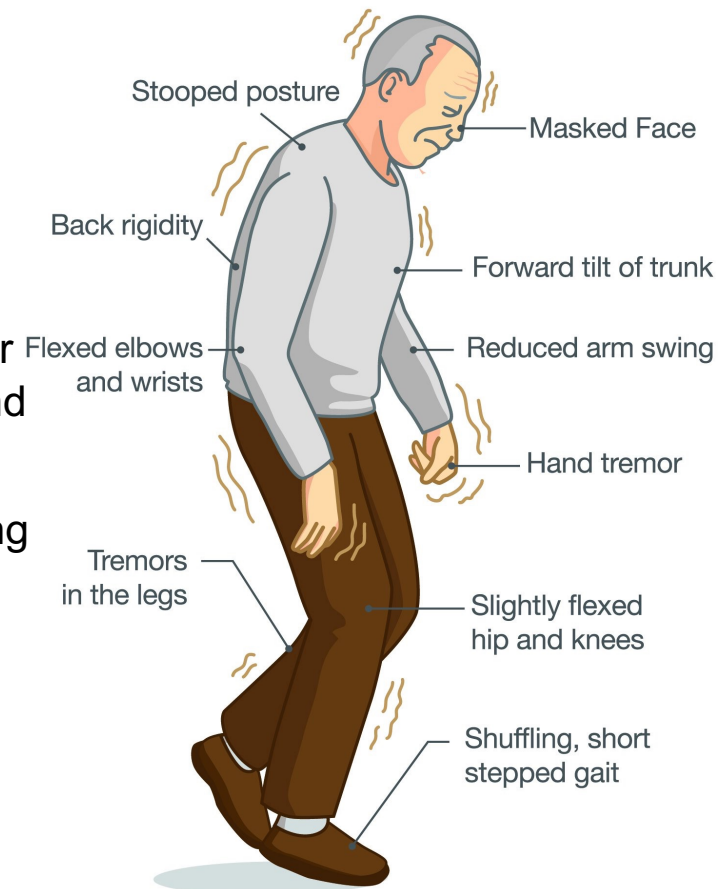
- **Type:** neurodegenerative disorder that progressively damages brain cells.
- **Description and causes:** AD leads to progressive memory loss and cognitive decline, primarily due to abnormal protein build-up in the brain, with causes that include genetic factors and, to some extent, lifestyle and environmental influences.
- **Affected systems:** brain areas involved in memory, cognition, and, later, motor functions.
- **Key symptoms:** memory loss, confusion, mood swings, and difficulty with language and problem-solving, with symptoms progressively worsening over time.
- **Diagnosis:** typically confirmed through a combination of cognitive assessments, imaging tests like MRI and PET scans, and sometimes blood or genetic tests.
- **Treatment and management:** While there is no cure, treatments such as medications (e.g., cholinesterase inhibitors) and cognitive therapies can help manage symptoms and slow progression in some cases.
- **Prognosis:** Alzheimer's disease is a progressively worsening condition, with an average life expectancy of 4 to 8 years after diagnosis, though some may live longer depending on the stage of diagnosis and care.

- Stats: 6.7 million people (U.S., 2023)
- Age of onset: >65 years old
- Higher prevalence in women



Parkinson's disease (PD)

- **Type:** progressive neurodegenerative disorder, affecting motor control.
 - **Description and causes:** loss of dopaminergic neurons in the brain. The exact causes remain unknown in most cases.
 - **Affected systems:** brain, specifically the basal ganglia and substantia nigra regions, which regulate movement.
 - **Key symptoms:** tremor, muscle rigidity, bradykinesia, and postural instability. Non-motor symptoms (mood changes, sleep disturbances, and cognitive issues) may also occur and gradually worsen over time.
 - **Diagnosis:** motor symptoms and neurological exams; imaging tests can aid in supporting the diagnosis but are not definitive on their own.
 - **Treatment and management:** medications like levodopa to restore dopamine levels, deep brain stimulation for advanced cases, and physical therapy to maintain mobility.
 - **Prognosis:** progressive, with symptoms worsening over time; life expectancy can be near-normal with early and effective symptom management, although advanced stages can significantly impact quality of life.
- Stats: ~1 million people (U.S., 2022)
 - Age of onset: >60 years old
 - Higher prevalence in men

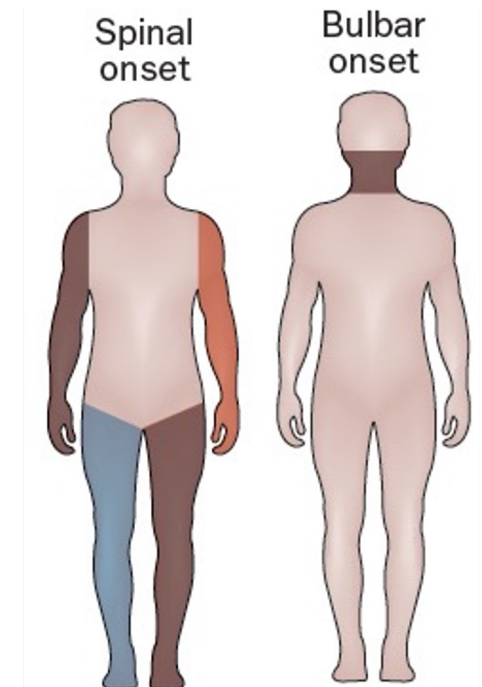


Huntington's disease (HD)

- Stats: 3-7 per 100,000 persons
 - Age of onset: 30-50 years old
 - Affects men and women equally
-
- **Type:** genetic neurodegenerative disorder caused by an inherited gene mutation.
 - **Description and causes:** mutation in the HTT gene, leading to abnormal protein production that damages neurons, primarily in the brain.
 - **Affected systems:** brain, especially the basal ganglia and cerebral cortex, affecting movement control, cognition, and emotional regulation.
 - **Key symptoms:** involuntary movements (chorea), cognitive decline, and emotional changes. As the disease progresses, it leads to severe motor dysfunction, dementia, and psychiatric symptoms.
 - **Diagnosis:** confirmed through genetic testing for the HTT mutation, supported by neurological exams and sometimes brain imaging to assess neuronal loss.
 - **Treatment and management:** there is no cure; symptoms are managed with medications for movement issues, psychiatric treatments for mood and behavioral symptoms, and physical therapy to help maintain mobility.
 - **Prognosis:** HD is progressive and typically fatal within 15-20 years after onset; symptoms worsen over time, significantly reducing quality of life in later stages.

Amyotrophic lateral sclerosis (ALS)

- **Stats:** 2 to 5 people per 100,000 globally
- **Age of onset:** 40-70 years old
- **Higher prevalence in men**
- **Type:** neurodegenerative; affects motor neurons with loss of motor function.
- **Description and causes:** gradual degeneration of motor neurons. The cause is largely unknown, but about 5-10% of cases are hereditary due to genetic mutations.
- **Affected systems:** CNS and PNS, specifically targeting motor neurons in the brain and spinal cord. It generally spares cognitive functions.
- **Key symptoms:** muscle twitching, cramping, and weakness, often in the limbs. Difficulty speaking (dysarthria), swallowing (dysphagia), and breathing, with symptoms worsening over time.
- **Diagnosis:** combination of clinical examinations, EMG to assess muscle function, nerve conduction studies, and ruling out other conditions through imaging and lab tests.
- **Treatment and management:** There is no cure for ALS; treatments aim to slow progression and manage symptoms (Riluzole and edaravone). Physical therapy, speech therapy, and breathing support improve quality of life.
- **Prognosis:** ALS is progressive and ultimately fatal; average life expectancy of 2-5 years after diagnosis. Some individuals live longer, especially with supportive care and adaptive devices to maintain daily functions.

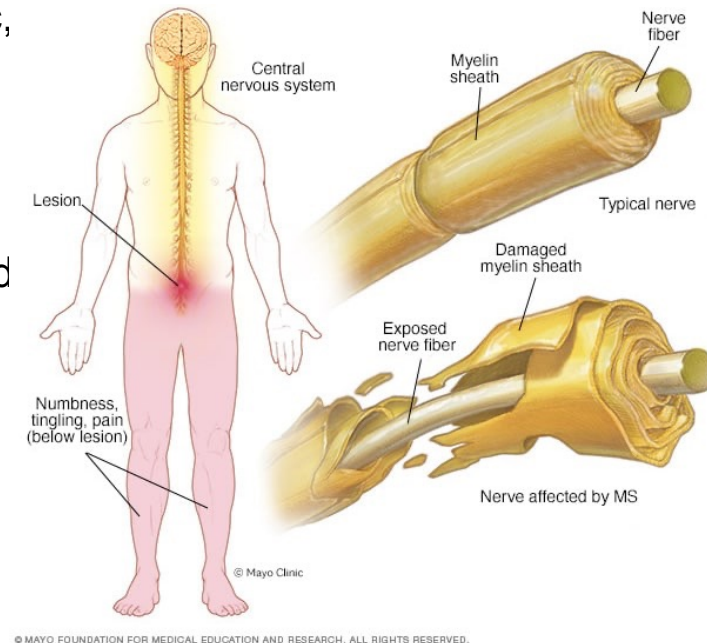


Stephen William Hawking 1942-2018

Multiple sclerosis (MS)

- **Type:** autoimmune, neurodegenerative disorder characterized by chronic inflammation that damages the CNS.
- **Description and causes:** the immune system attacks the myelin sheath of nerve fibers in the brain and spinal cord. The exact cause is unknown, though it may involve genetic, environmental, and immune-related factors.
- **Affected systems:** brain, spinal cord, and optic nerves. This affects sensory, motor, and cognitive functions.
- **Key symptoms:** fatigue, muscle weakness, vision problems, balance, coordination, and cognitive issues. Symptoms may be relapsing-remitting or progressively worsening.
- **Diagnosis:** MRI scans that detect lesions in the CNS, along with lumbar punctures and neurological exams to assess functional impairment.
- **Treatment and management:** Treatments aim to slow disease progression, manage relapses, and alleviate symptoms. Disease-modifying therapies, corticosteroids, physical therapy, and lifestyle management are commonly used.
- **Prognosis:** MS is a lifelong condition with a variable course. With treatment, many people manage the disease effectively and maintain a good quality of life.

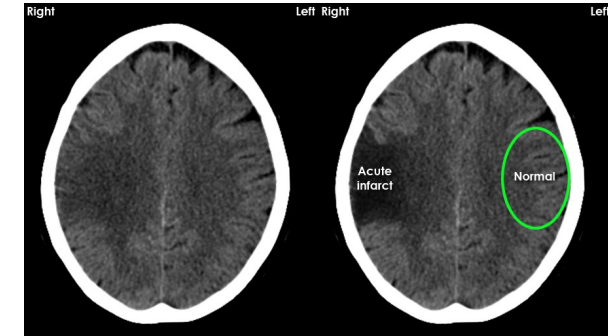
- Stats: ~2.5 million people worldwide
- Age of onset: 20-40 years old
- Higher prevalence in women



Stroke

- **Type:** acute; occurs when there is an interruption of blood flow to the brain.
- **Description and causes:** blood supply to part of the brain is interrupted or reduced, depriving brain tissue of oxygen and nutrients. Primary causes include ischemia (due to a blood clot or embolism) or hemorrhage (due to a ruptured blood vessel).
- **Affected systems:** brain (it depends on the location of the blood flow disruption). It can impair various functions (e.g., movement, speech, memory, and sensation).
- **Key symptoms:** sudden weakness or numbness on one side of the body, difficulty speaking or understanding speech, vision problems, loss of balance or coordination, and a severe headache.
- **Diagnosis:** imaging (e.g., CT scan or MRI) to identify the type and extent of the stroke.
- **Treatment and management:** blood flow restoration or surgery to repair blood vessels. Rehabilitation therapies (physical, occupational, and speech therapy) for recovery.
- **Prognosis:** Recovery from stroke varies widely; some individuals may experience significant recovery within months, while others may face long-term disabilities. Life expectancy can be affected by stroke severity and complications.

- Stats: >795,000 people/year (U.S.)
- Age of onset: any (higher risk after 55)
- Slightly higher prevalence in men



FACE
DROOPING



ARM
WEAKNESS



SPEECH
DIFFICULTY



TIME TO
CALL 911

Spinal cord injuries (SCI)

- **Type:** acute, traumatic neurological injury to the spinal cord.
- **Description and causes:** a damage to the spinal cord disrupts communication between the brain and the periphery. Primary causes: trauma from accidents or non-traumatic conditions (e.g., tumors, infections).
- **Affected systems:** spinal cord, resulting in loss of movement, sensation, and autonomic function below the level of injury.
- **Key symptoms:** partial or complete loss of motor and sensory function, tetraplegia or paraplegia, altered reflexes, pain or tingling sensations, and autonomic dysregulation.
- **Diagnosis:** physical examination, imaging studies (e.g., X-rays, CT scans, MRIs) to assess the extent and location of the injury.
- **Treatment and management:** spine stabilization, preventing further injury, and managing symptoms. This may include medications, surgery, rehabilitation therapies (physical, and occupational), and assistive devices for mobility.
- **Prognosis:** prognosis for spinal cord injuries varies widely, depending on the severity and location of the injury. Life expectancy can be affected by complications related to the injury, such as respiratory issues or infections.

- Stats: 15 million people (WHO, 2024)
- Mostly in young adults (16-30 years old)
- Higher prevalence in men

